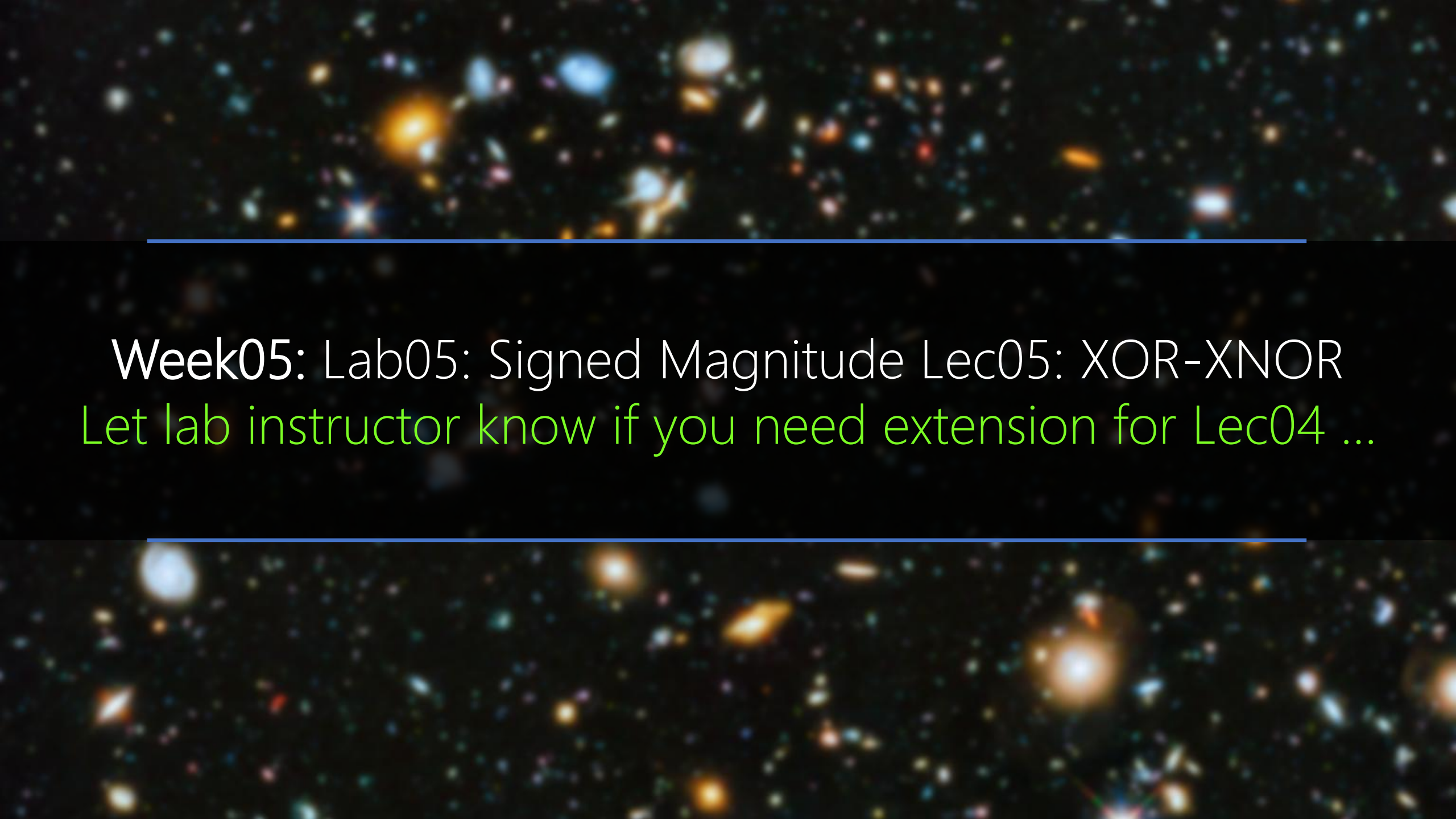


Week05: Lab05: Signed Magnitude Lec05: XOR-XNOR

Let lab instructor know if you need extension for Lec04 …

S2025: A Digital Odyssey

The Ultimate Trip.

Predicting Multiplication with GPT-2: Implicit vs. Explicit CoT

This demo showcases GPT-2's ability to directly predict the product of two large numbers without intermediate steps, using our stepwise internalization method. Compare the performance of implicit CoT (our method), no CoT, and explicit CoT. Implicit CoT offers accuracy and speed, while explicit CoT provides detailed reasoning but is slower.

First Number (up to 20 digits)

100

Second Number (up to 20 digits)

-200

Multiply!

Ground Truth Product

- 2 0 0 0 0

Implicit CoT Prediction (Ours)

No CoT Prediction

Explicit CoT Steps & Prediction

$$1 * 200 = 100$$
$$2 * 200 = 200$$
$$3 * 200 = 300$$
$$4 * 200 =$$

- Paper 1: Implicit Chain of Thought Reasoning via Knowledge Distillation
- Paper 2: From Explicit CoT to Implicit CoT: Learning to Internalize CoT Step by Step
- Code Repository
- Tweet Announcement

Predicting Multiplication with GPT-2: Implicit vs. Explicit CoT

This demo showcases GPT-2's ability to directly predict the product of two large numbers without intermediate steps, using our stepwise internalization method. Compare the performance of implicit CoT (our method), no CoT, and explicit CoT. Implicit CoT offers accuracy and speed, while explicit CoT provides detailed reasoning but is slower.

First Number (up to 20 digits)

100

Second Number (up to 20 digits)

-200

Multiply!

Ground Truth Product

- 2 0 0 0 0

Implicit CoT Prediction (Ours)

No CoT Prediction

Explicit CoT Steps & Prediction

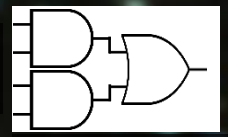
$$1 * 200 = 100$$
$$2 * 200 = 400$$
$$3 * 200 = 600$$
$$4 * 200 =$$

- Paper 1: Implicit Chain of Thought Reasoning via Knowledge Distillation
- Paper 2: From Explicit CoT to Implicit CoT: Learning to Internalize CoT Step by Step
- Code Repository
- Tweet Announcement

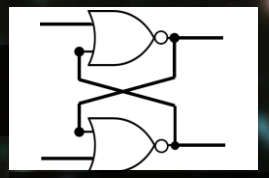
Number Systems $|(12)_{10} \rightarrow (1100)_2$

Logic Gates |

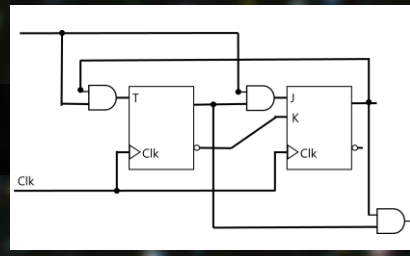
Combinational Logic |

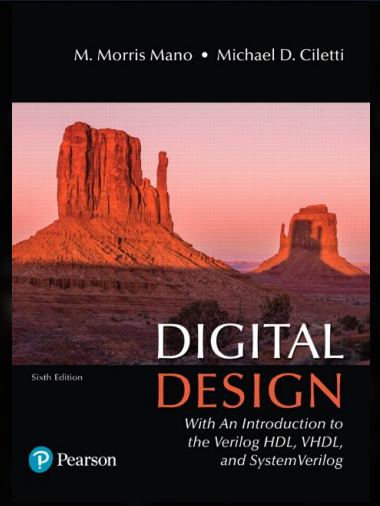


Flip-Flop |



Sequential Logic |

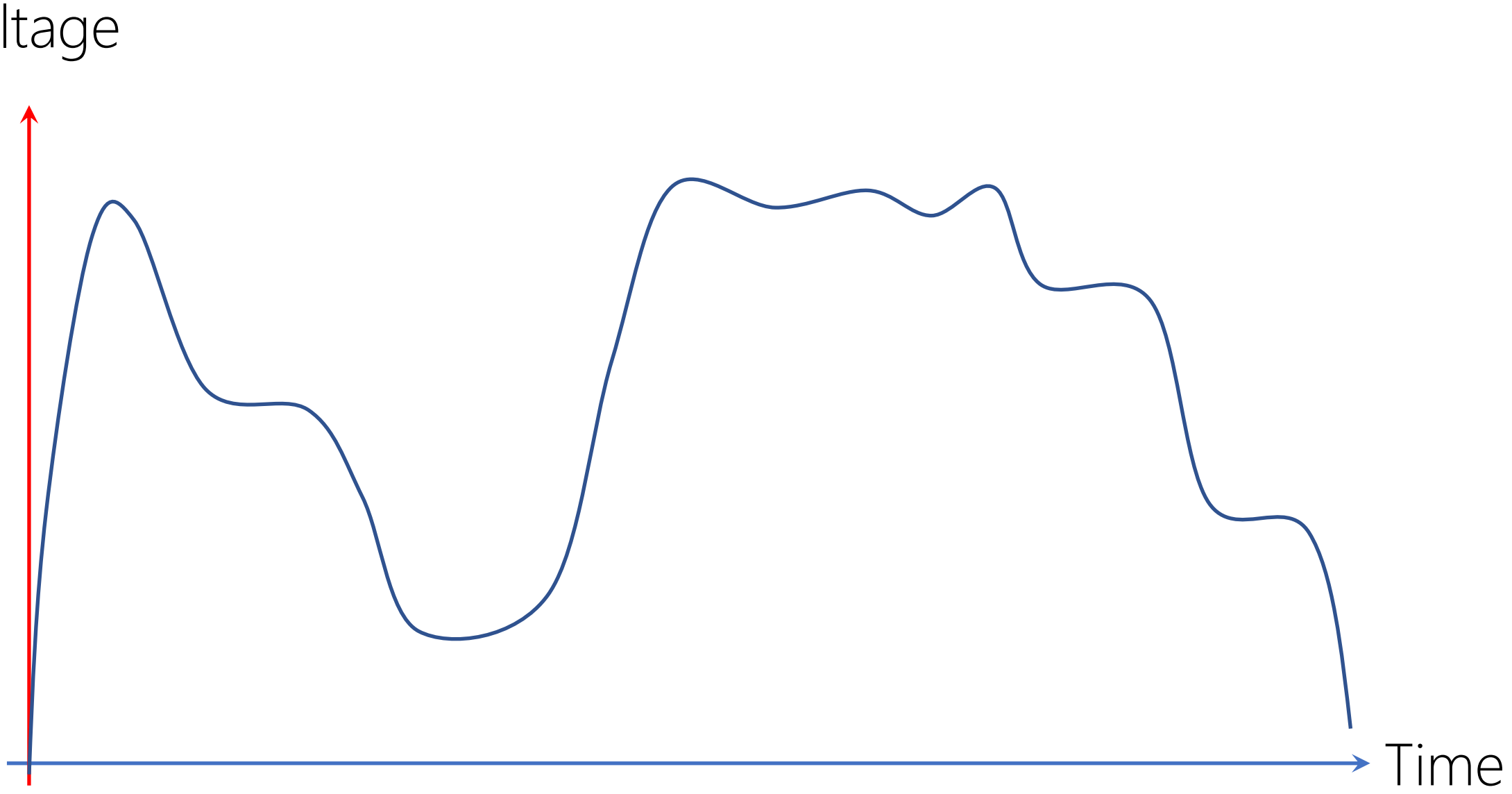




Chapter 2
Boolean Algebra and Logic Gates

ANALOG SYSTEMS

Continuous


Voltage
Time

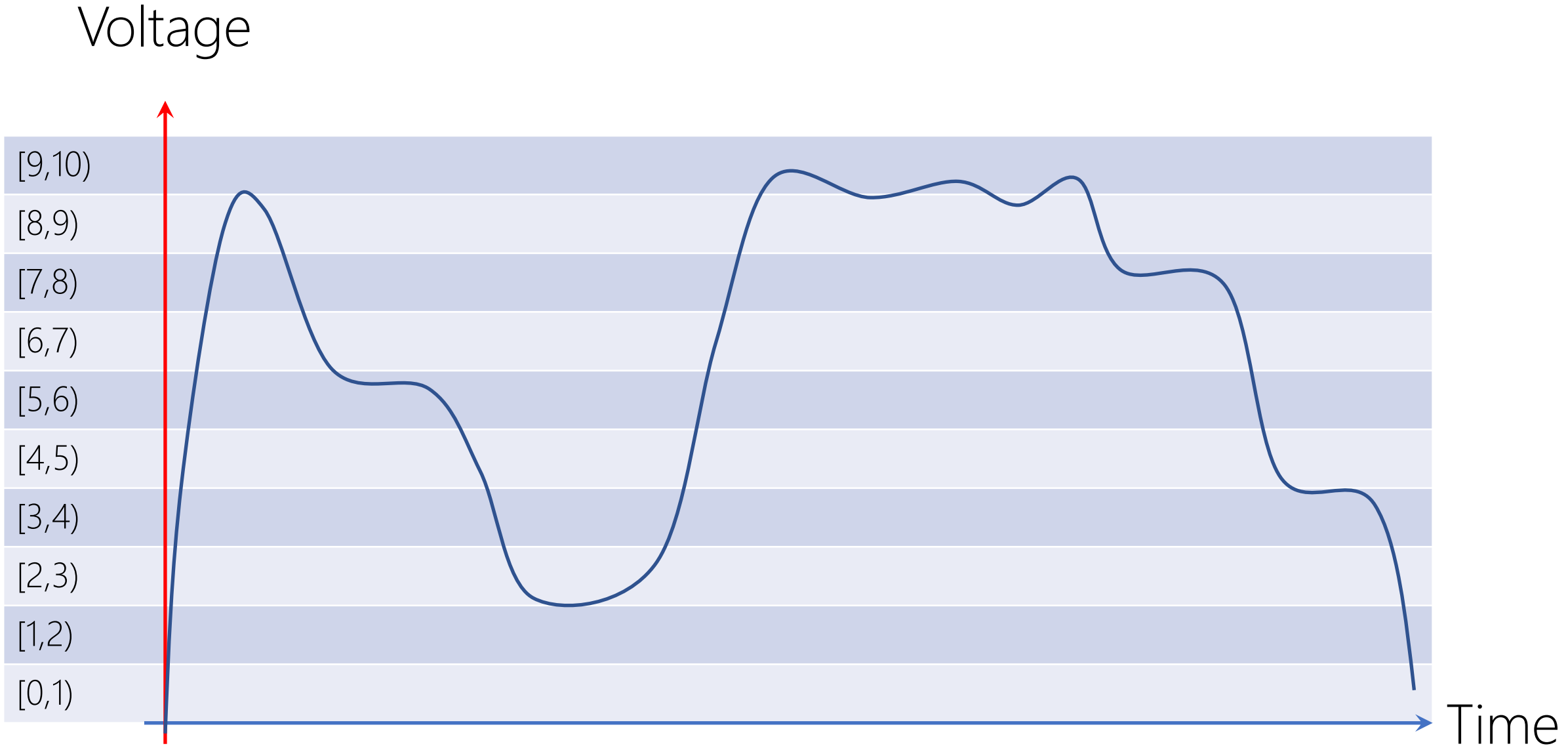
DIGITAL SYSTEMS

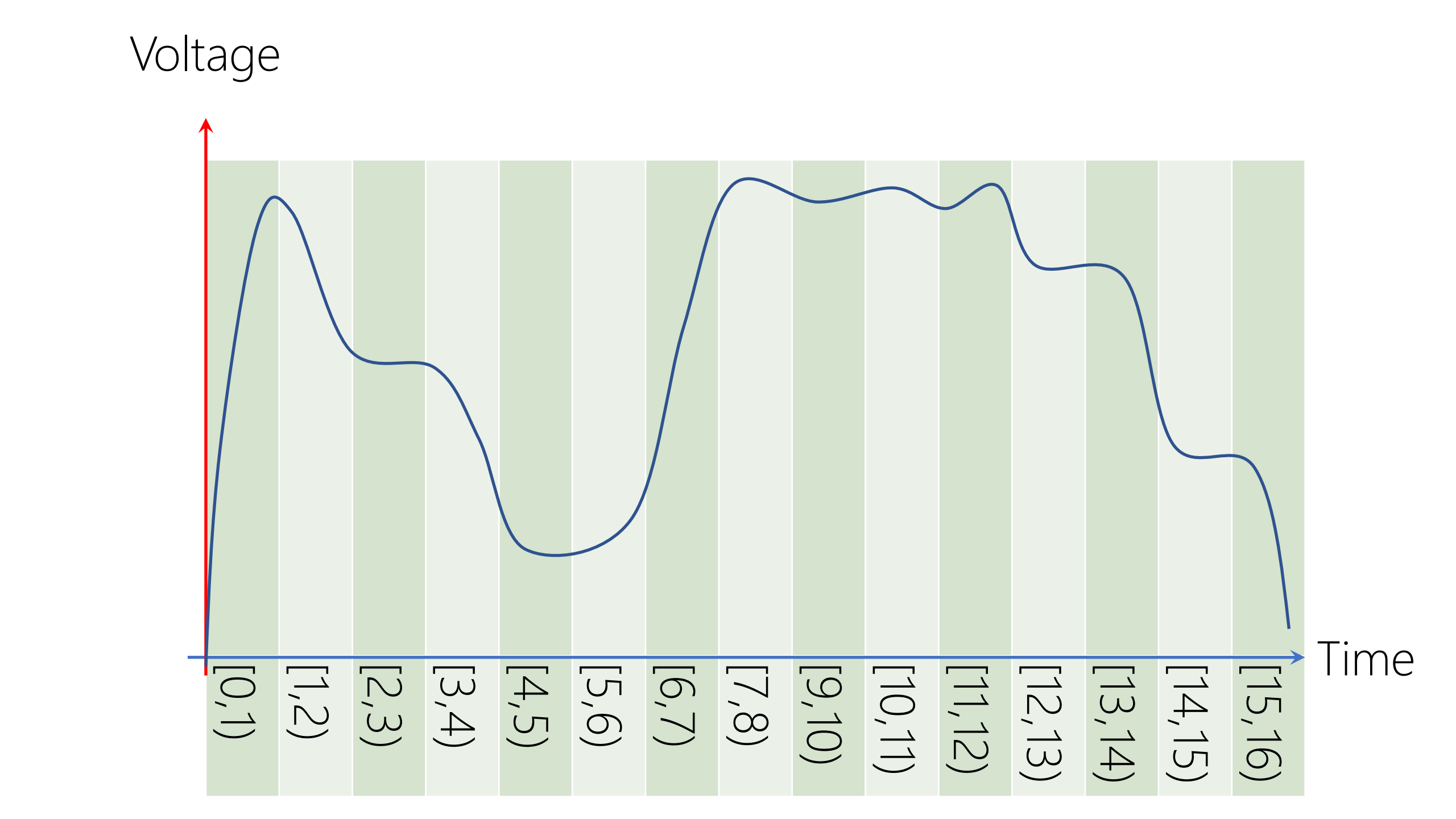
Discrete

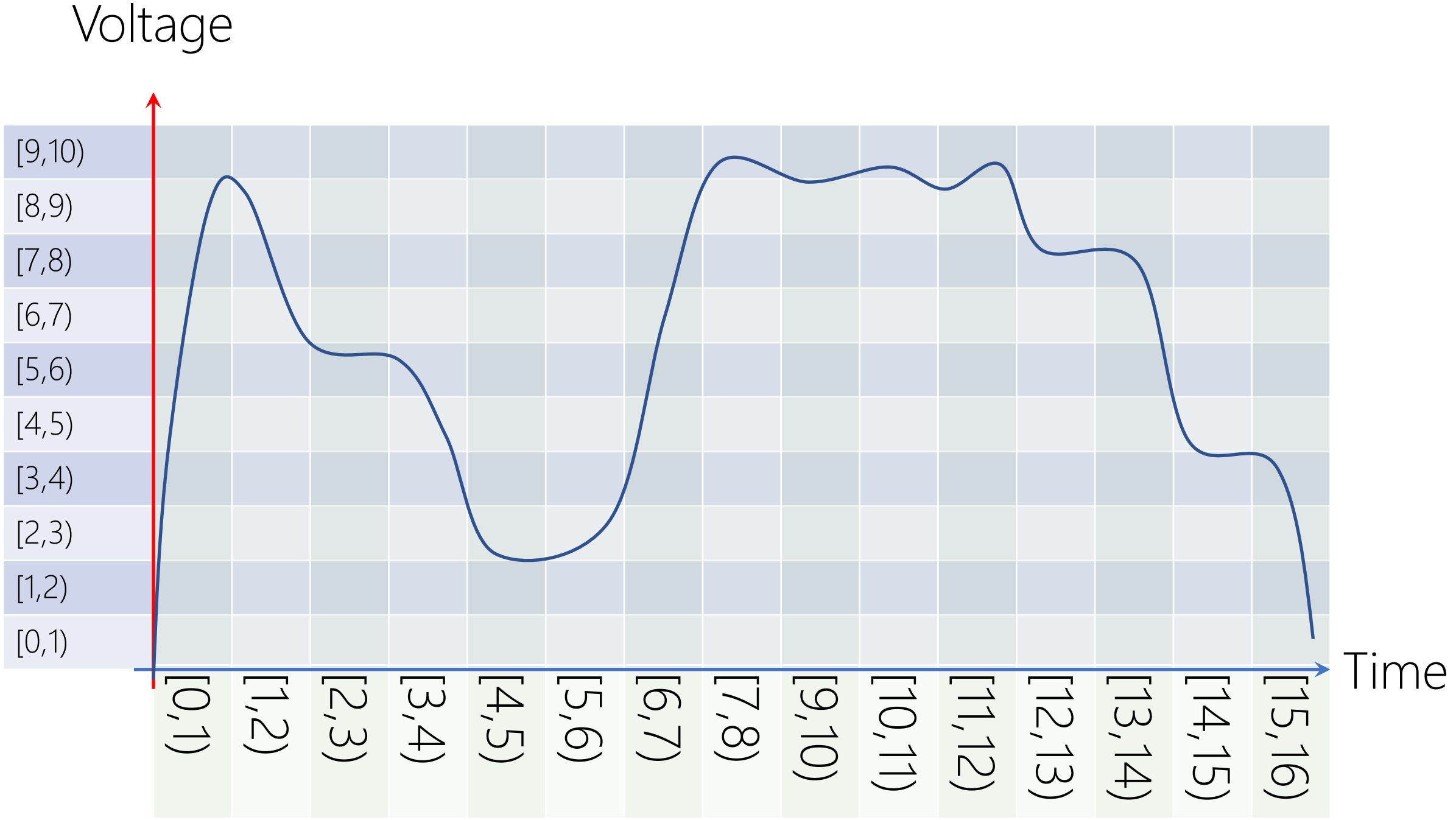
ELECTRICITY NUMBER SYSTEM

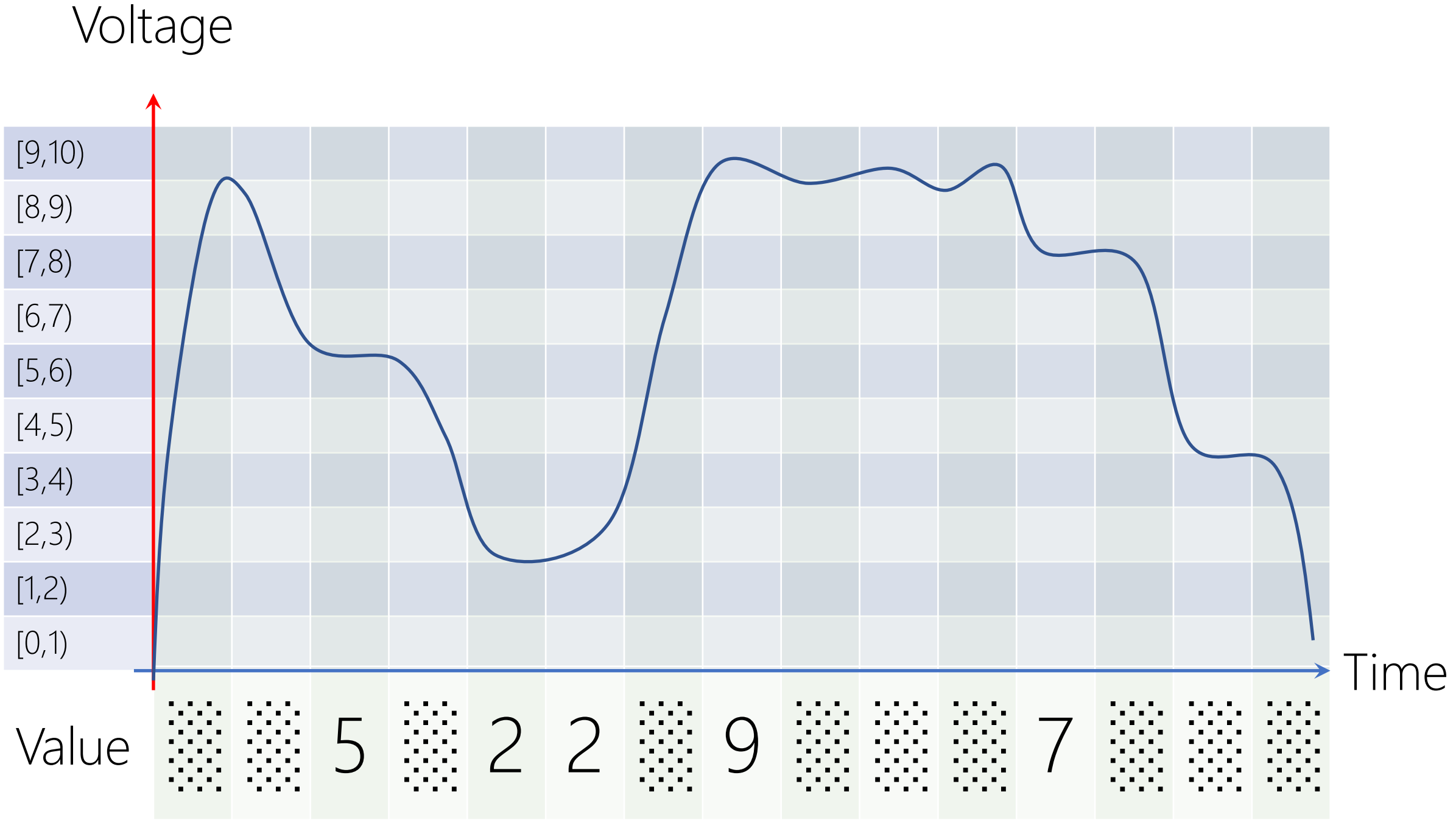
BASE-?

BASE-10

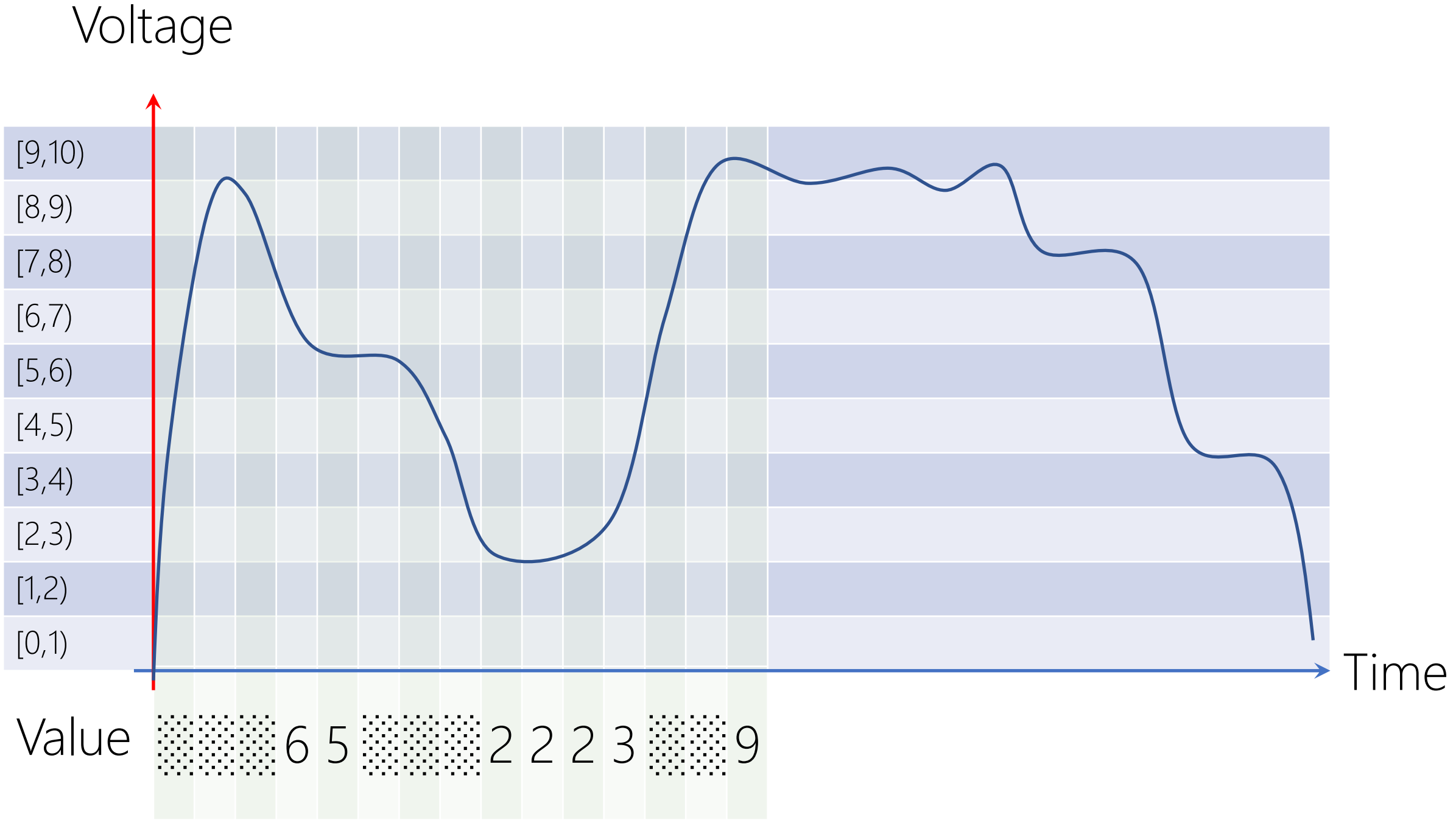

Voltage
[9,10)
[8,9)
[7,8)
[6,7)
[5,6)
[4,5)
[3,4)
[2,3)
[1,2)
[0,1)
Time

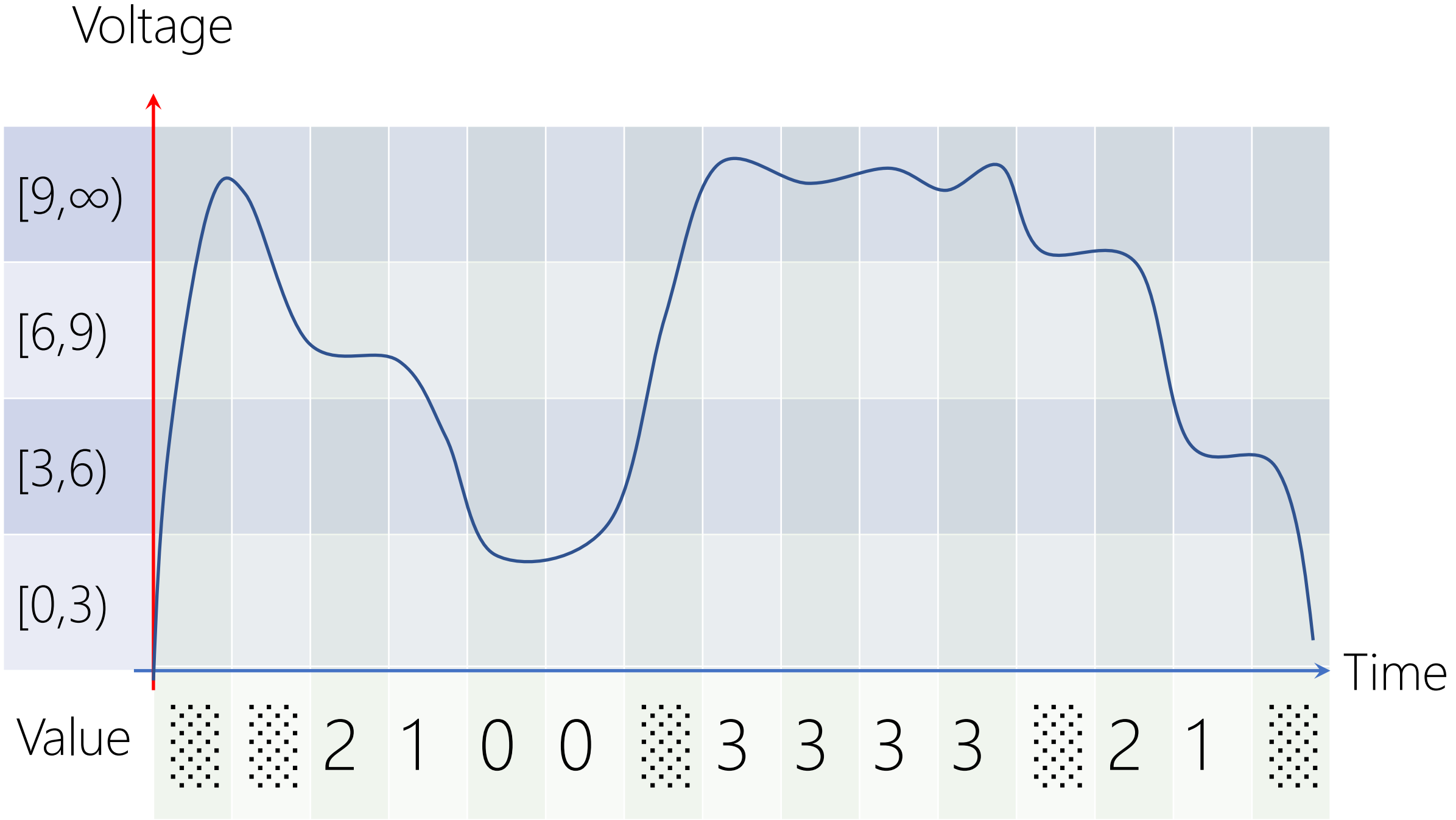

Voltage
Time
[0,1)
[1,2)
[2,3)
[3,4)
[4,5)
[5,6)
[6,7)
[7,8)
[9,10)
[10,11)
[11,12)
[12,13)
[13,14)
[14,15)
[15,16)

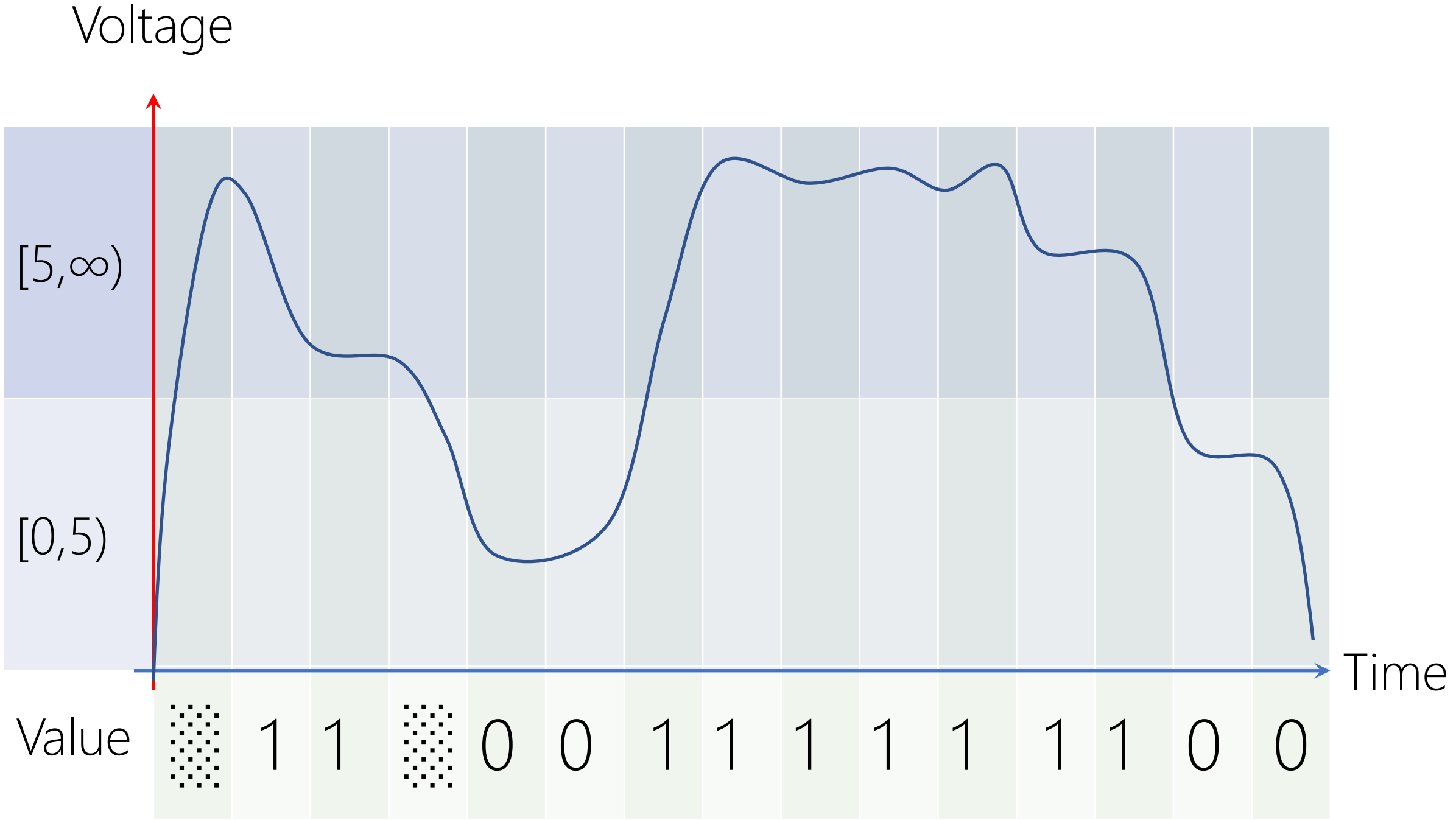

Voltage
[9,10)
[8,9)
[7,8)
[6,7)
[5,6)
[4,5)
[3,4)
[2,3)
[1,2)
[0,1)
Time
[0,1)
[1,2)
[2,3)
[3,4)
[4,5)
[5,6)
[6,7)
[7,8)
[9,10)
[10,11)
[11,12)
[12,13)
[13,14)
[14,15)
[15,16)


Voltage
[9,10)
[8,9)
[7,8)
[6,7)
[5,6)
[4,5)
[3,4)
[2,3)
[1,2)
[0,1)
Time
Value
5
2
2
9
7

GRANULARITY **for** ROBUSTNESS


Voltage
[9,10)
[8,9)
[7,8)
[6,7)
[5,6)
[4,5)
[3,4)
[2,3)
[1,2)
[0,1)
Time
Value
6 5
2 2 2 3
9


Voltage
[9,∞)
[6,9)
[3,6)
[0,3)
Time
Value
2 1 0 0 3 3 3 3 2 1


Voltage
[5,∞)
[0,5)
Time
Value
1 1 0 0 1 1 1 1 1 1 1 0 0

(1) RELIABILITY

Robust to Noise

Fundamentally Hardware/Engineering Problem

TERNARY COMPUTER

https://en.wikipedia.org/wiki/Ternary_computer

Balanced Trinary {−1,0,1}

Entirely from *Wood*!

Thomas Fowler 1840

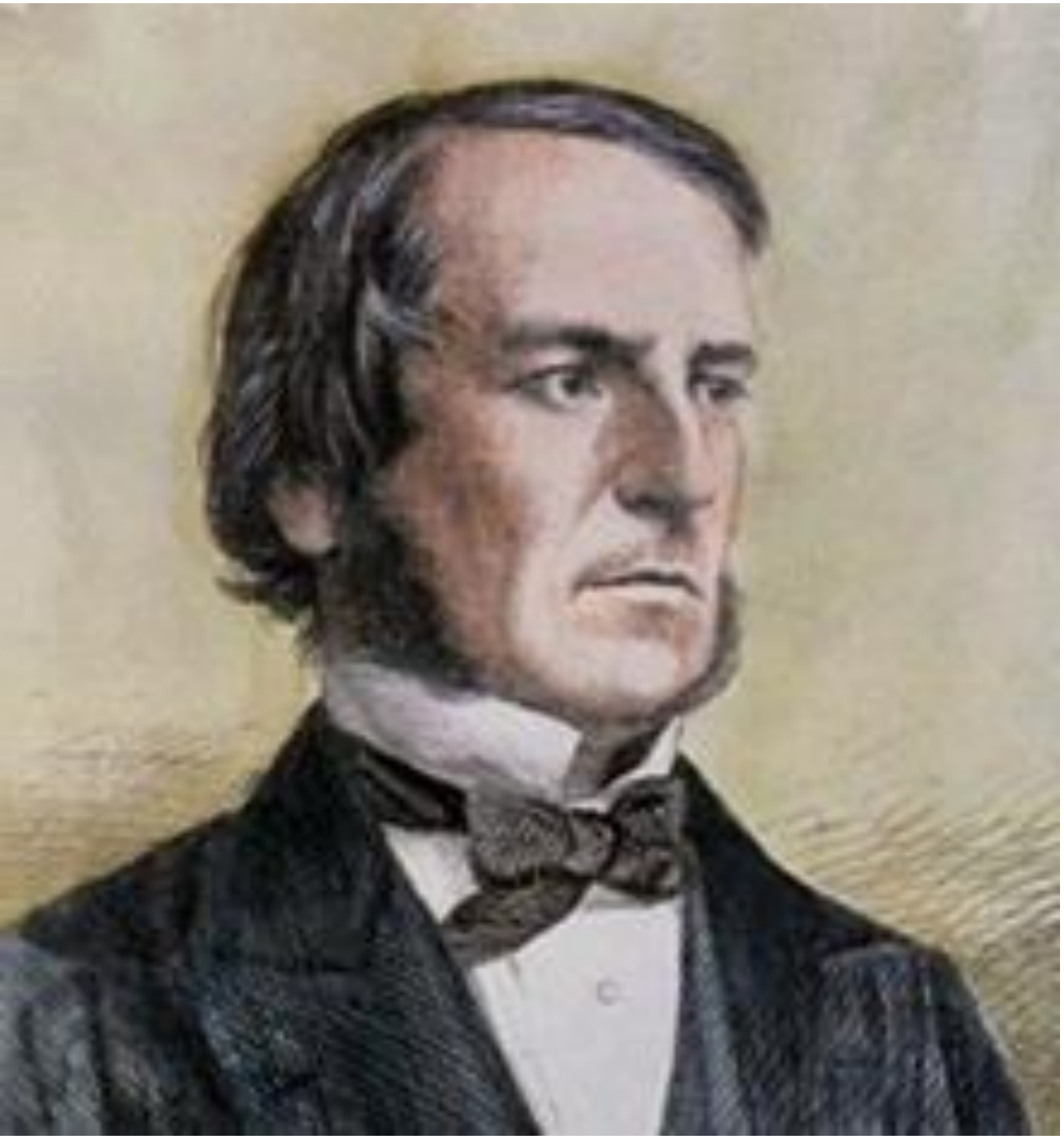
More History and Etymology → https://en.wikipedia.org/wiki/Computer

DECIMAL COMPUTER

https://en.wikipedia.org/wiki/Decimal_computer

They are not actually base-10! We'll cover them later.

(2) TRUE VS. FALSE



George Boole (/bu:l/)

Mathematician
Philosopher
Logician

The Laws of Thought (1854)

Boolean Algebra!

Claude Elwood Shannon

Mathematician
Electrical Engineer
Cryptographer

M.Sc. Thesis (1937)

A Symbolic Analysis of Relay and Switching Circuits
21 years old!

Switching Algebra!

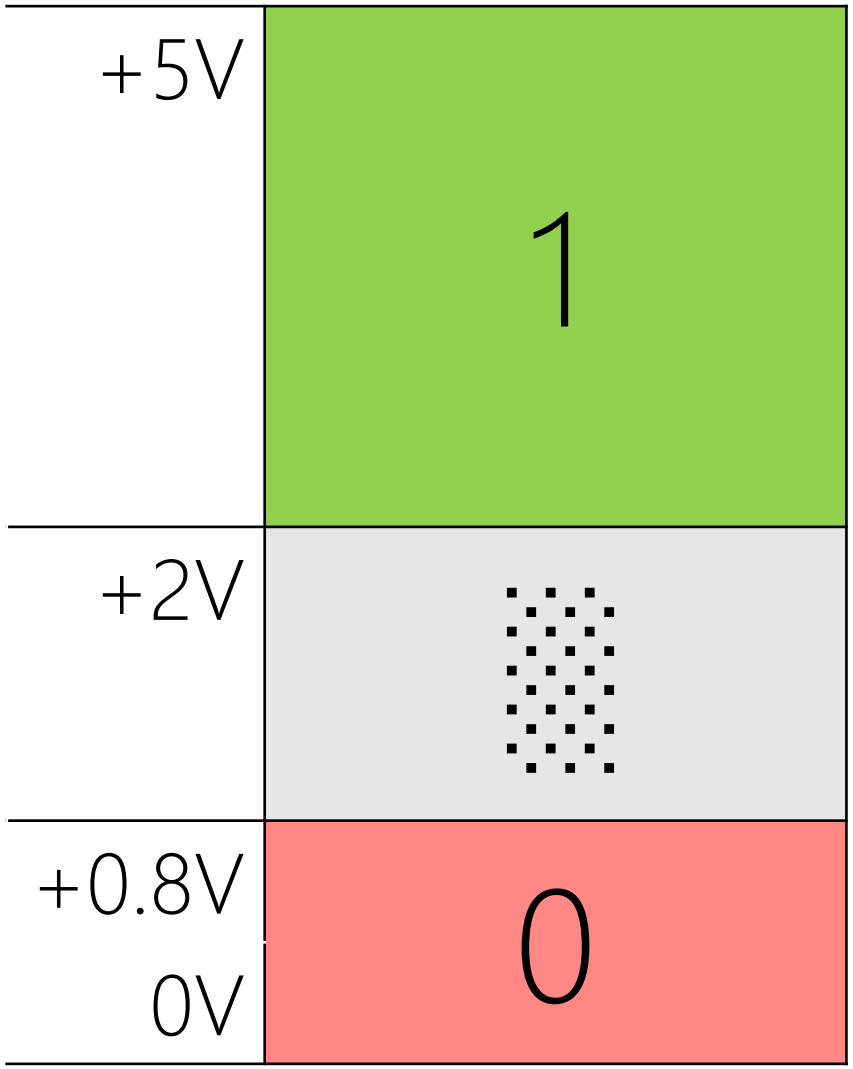
BINARY COMPUTER

More Reliability in Engineering
Deep Logic Foundation

2001: A Space Odyssey (1968), Stanley Kubrick, Arthur C. Clarke

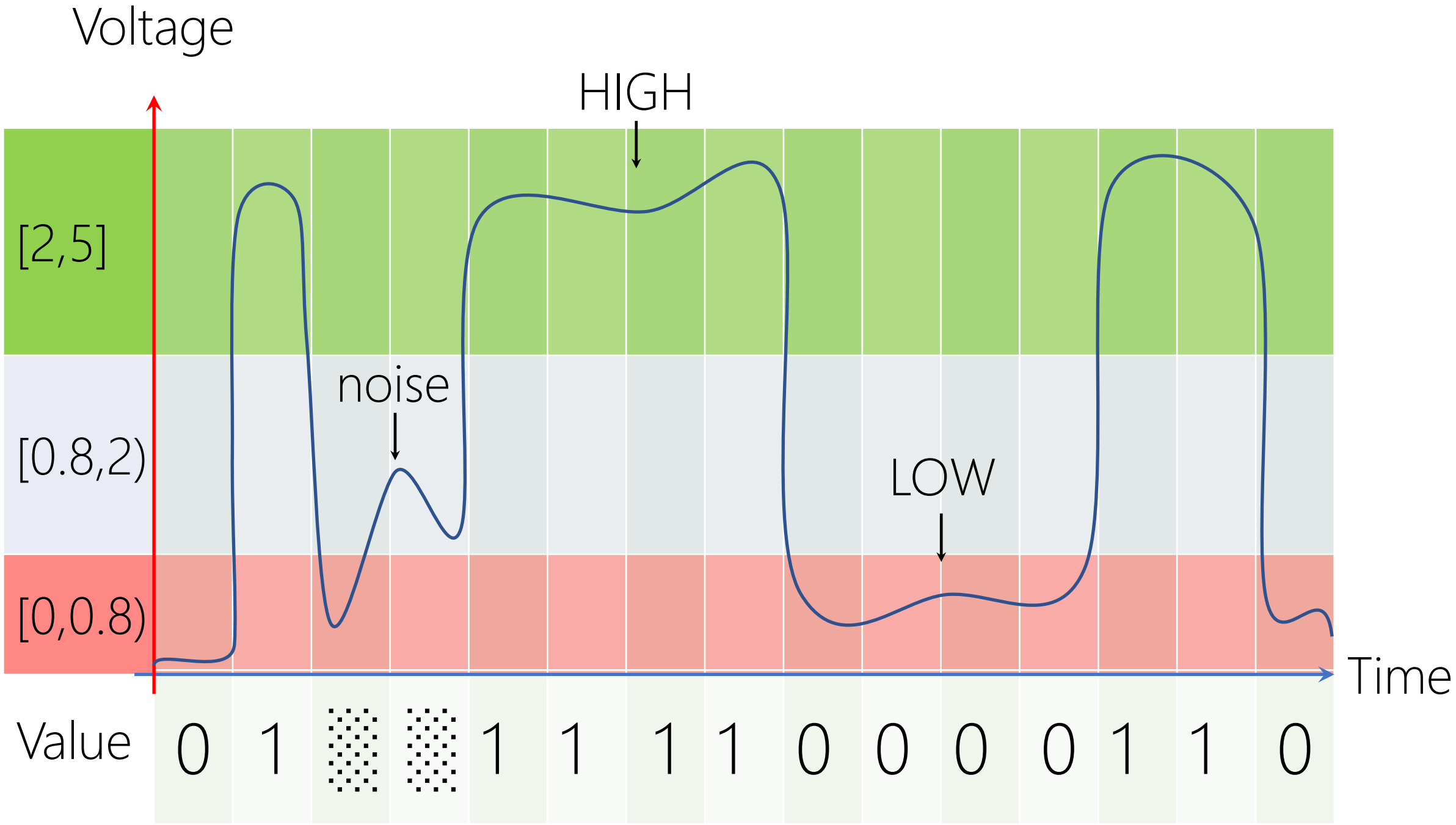
Ape Learning Scene: https://www.youtube.com/watch?v=VABNA_an2A0

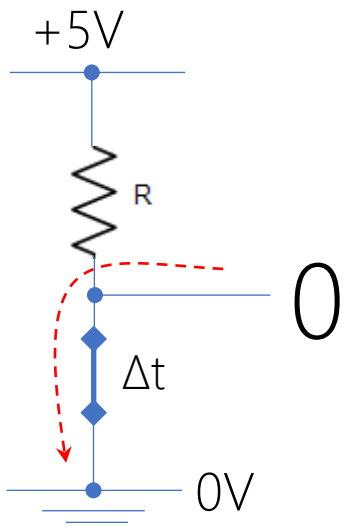
Bone/Satellite Match Cut: https://www.youtube.com/watch?v=avjdKTqiVvQ



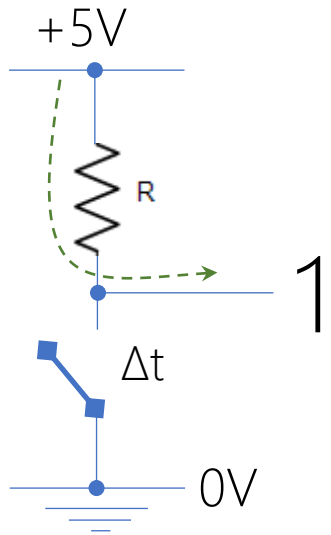
POSITIVE LOGIC

https://www.electronics-tutorials.ws/logic/logic_1.html

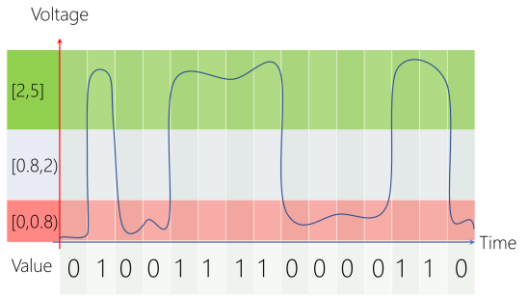

Voltage
HIGH
[2,5]
noise
[0.8,2)
LOW
[0,0.8)
Time
Value
0 1 1 1 1 0 0 0 0 1 1 0

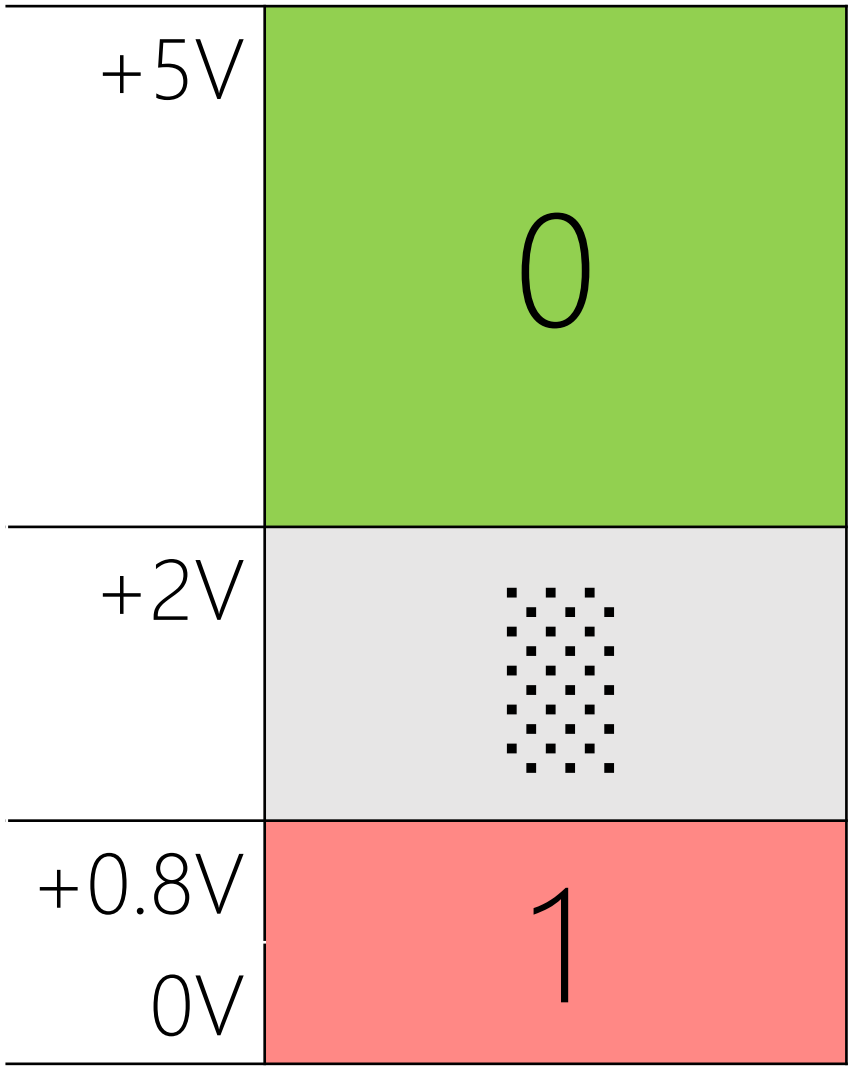


Earth (Ground)



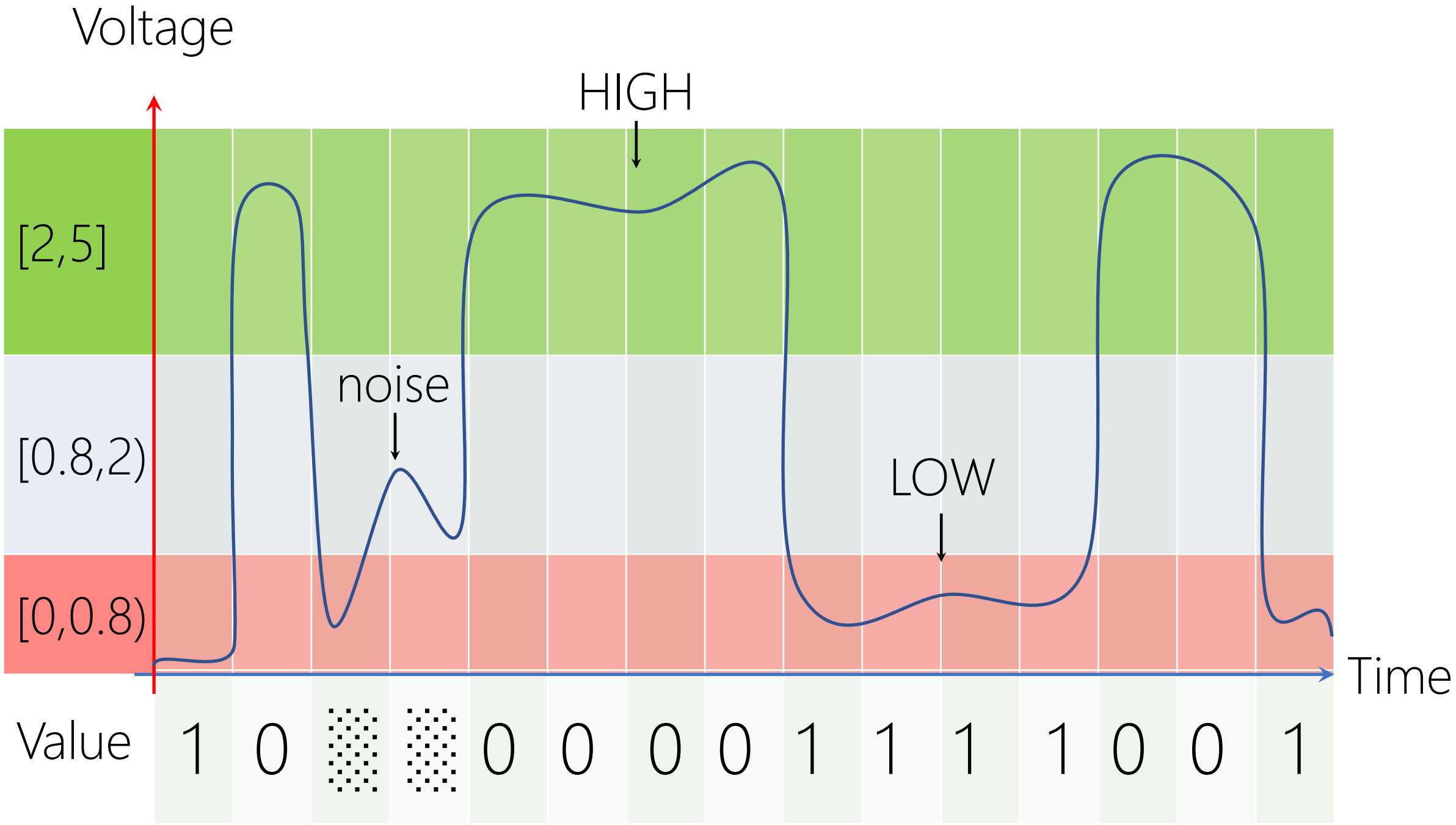
Earth (Ground)

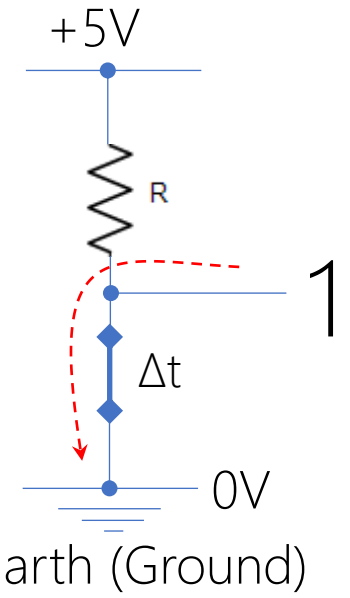




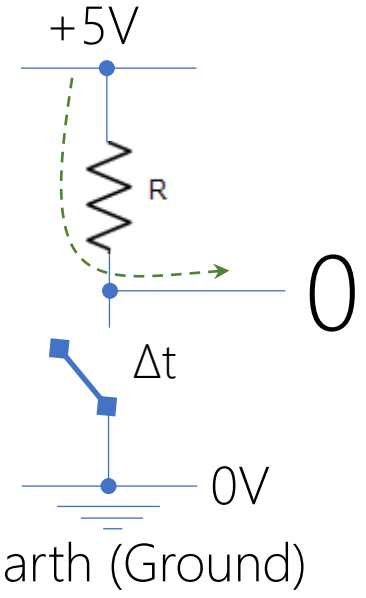
NEGATIVE LOGIC

https://www.electronics-tutorials.ws/logic/logic_1.html

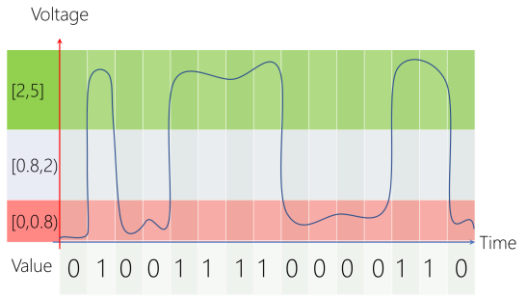

Voltage
HIGH
[2,5]
noise
[0.8,2)
LOW
[0,0.8)
Time
Value
1 0 0 0 0 1 1 1 1 0 0 1



Earth (Ground)



Earth (Ground)



DESIGN COMPUTER

Positive Logic
Button-Up Approach

DESIGN COMPUTER

Positive Logic
Button-Up Approach

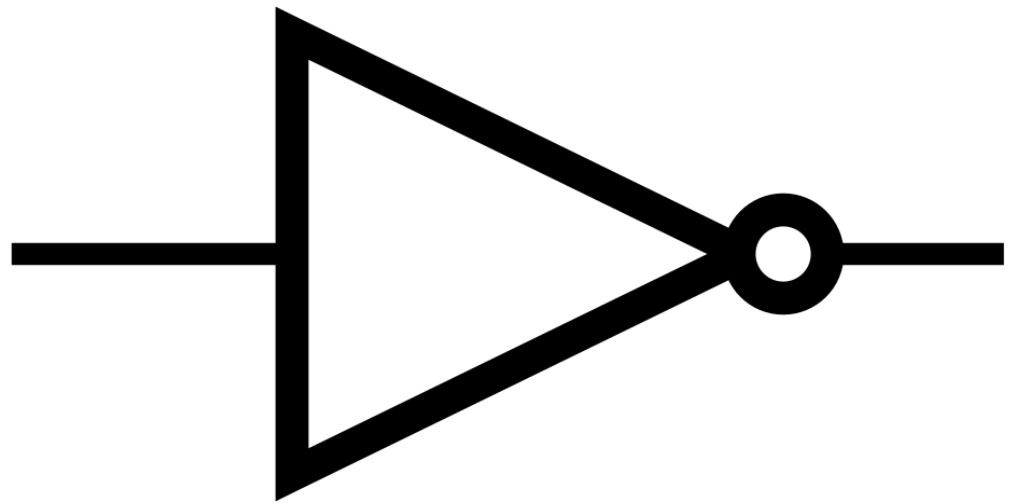
Finding simpler, but equivalent, computers reduces the overall cost!
Rely primarily on mathematical methods in Boolean algebra!

BUILD COMPUTER

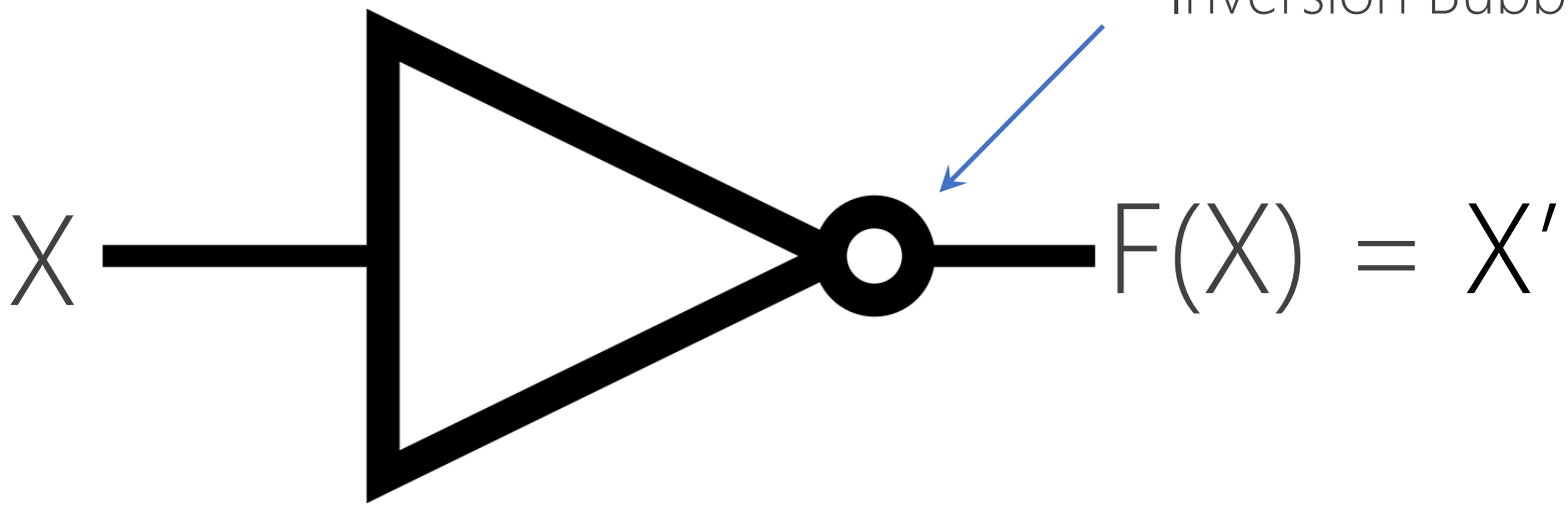
Electrical and Computer Engineering

LOGIC GATES

BOOLEAN GATES



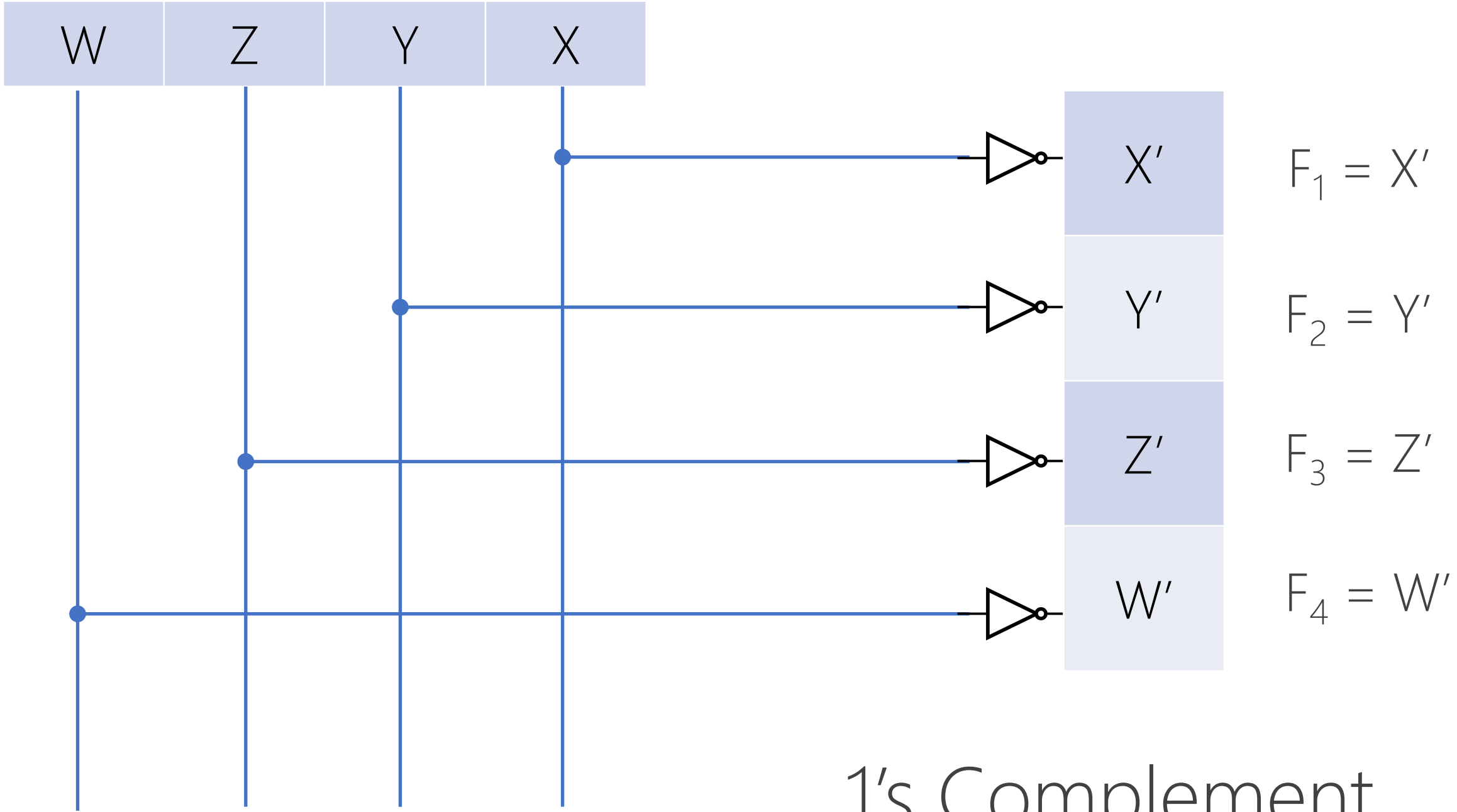
Truth Table



X	NOT X	Invertor X	X′	$\overline{X}$
0	1			
1	0			

Boolean Expression/Function: F(X) = X′

> *inverse of X gives F* <



1's Complement

BINARY VARIABLE

aka. Boolean variable

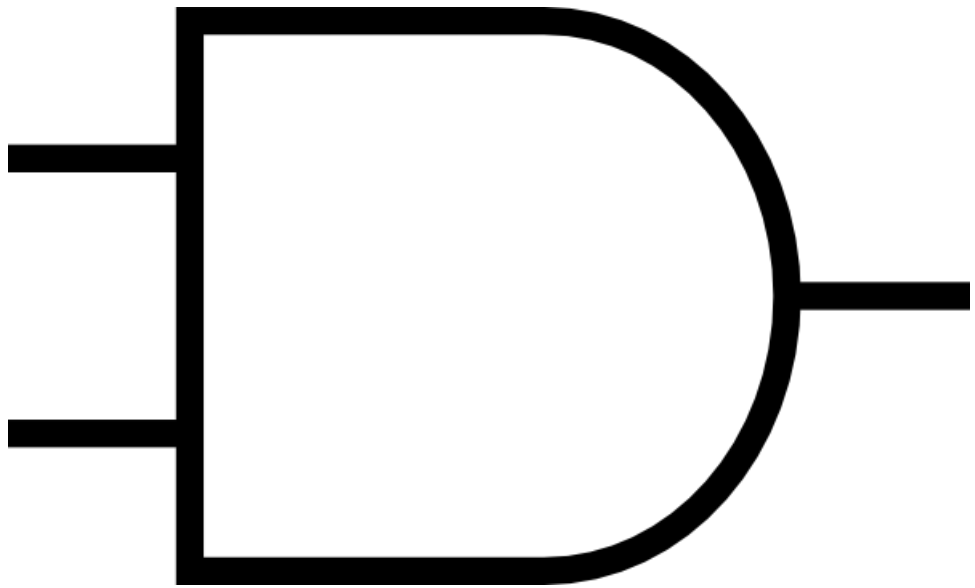
A variable that can have a value in {0,1}

X, Y, Z, W, …

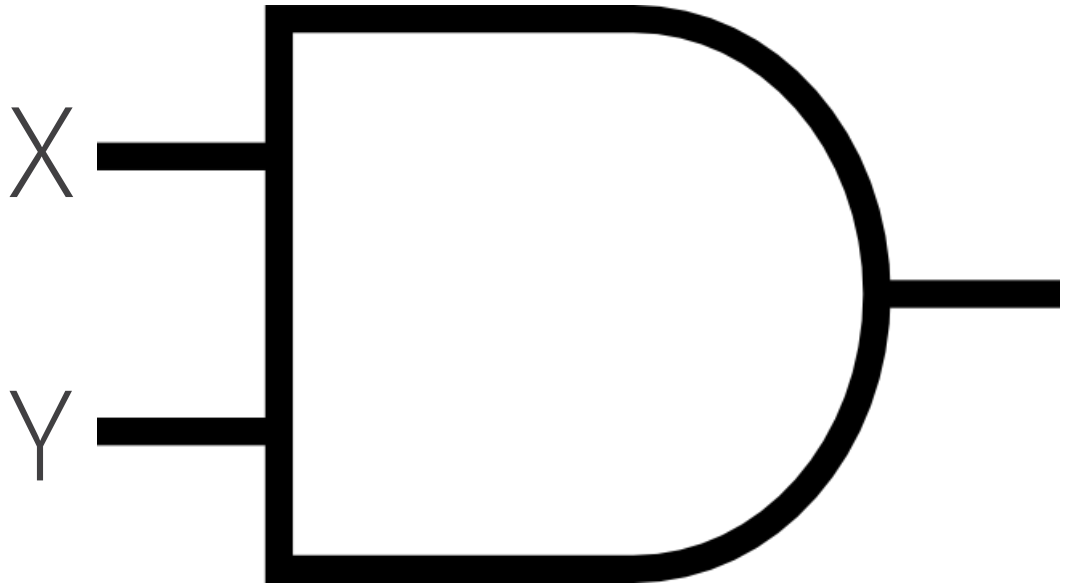
BOOLEAN FUNCTION

aka. Boolean Expression

A function that accept Boolean variable(s) and output a value in {0,1}, e.g., F(X) = X′

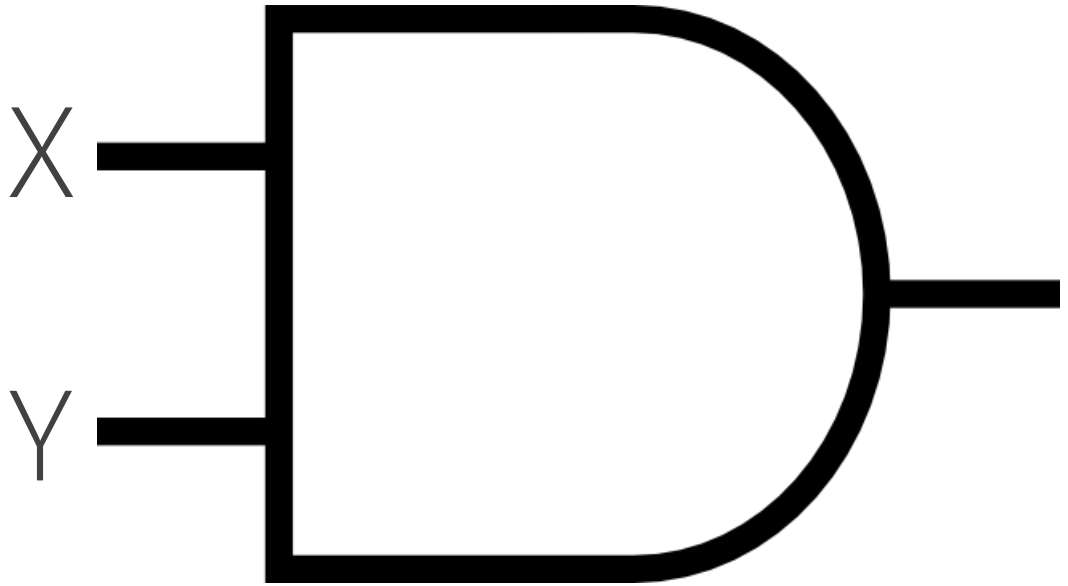


Truth Table



$F(Y, X) = YX$

Y	X	X AND Y	Y · X	Y * X
0	0		0	
0	1		0	
1	0		0	
1	1		1	



$F(Y,X) = YX = XY$

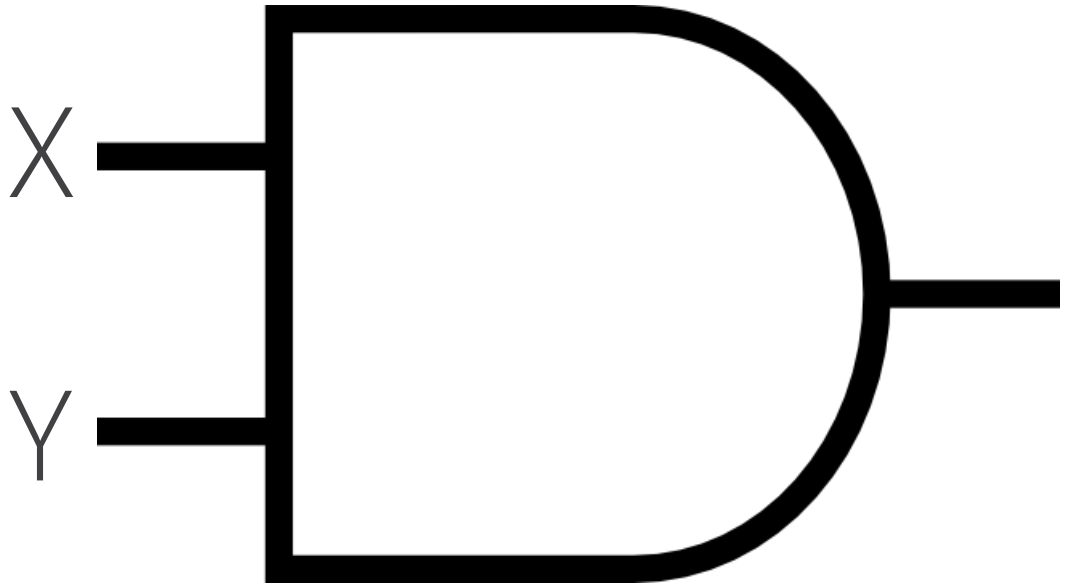
Commutative

X	Y	Y AND X	X · Y	X * Y
0	0		0	
0	1		0	
1	0		0	
1	1		1	

BOOLEAN FUNCTION

Equality F1 = F2

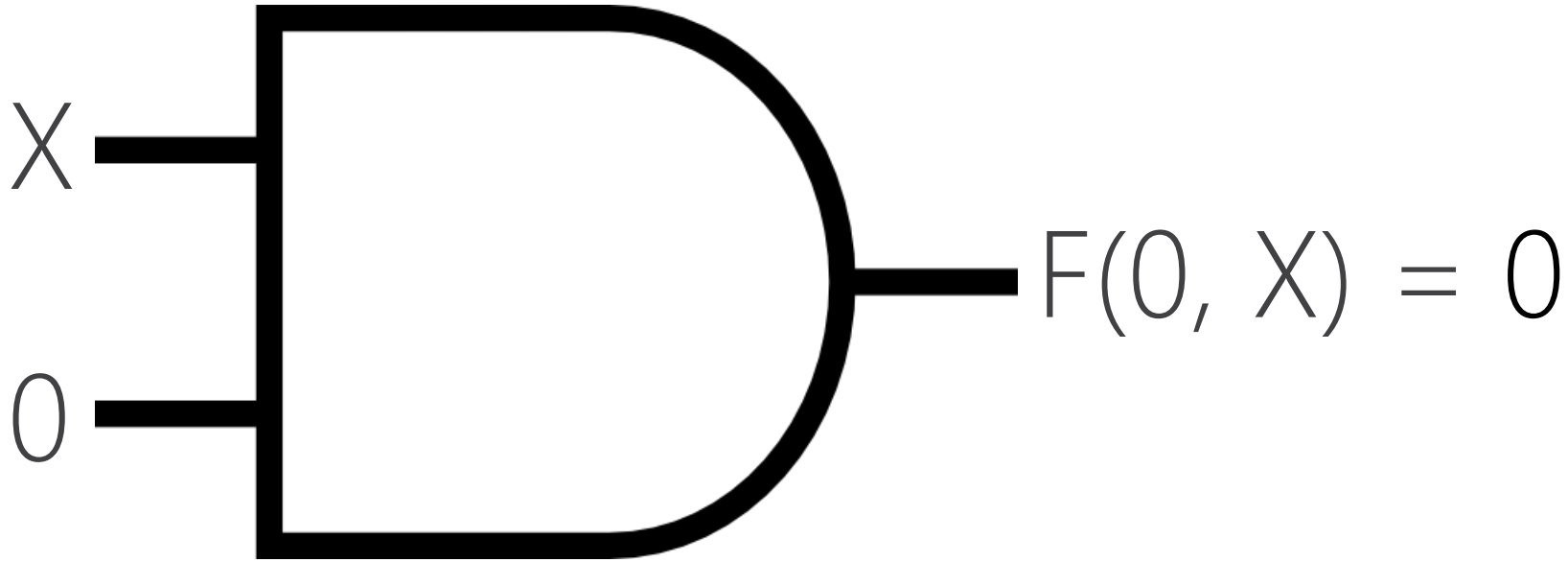
For the same input(s), F1 and F2 result in same output



$F(Y,X) = YX = XY$

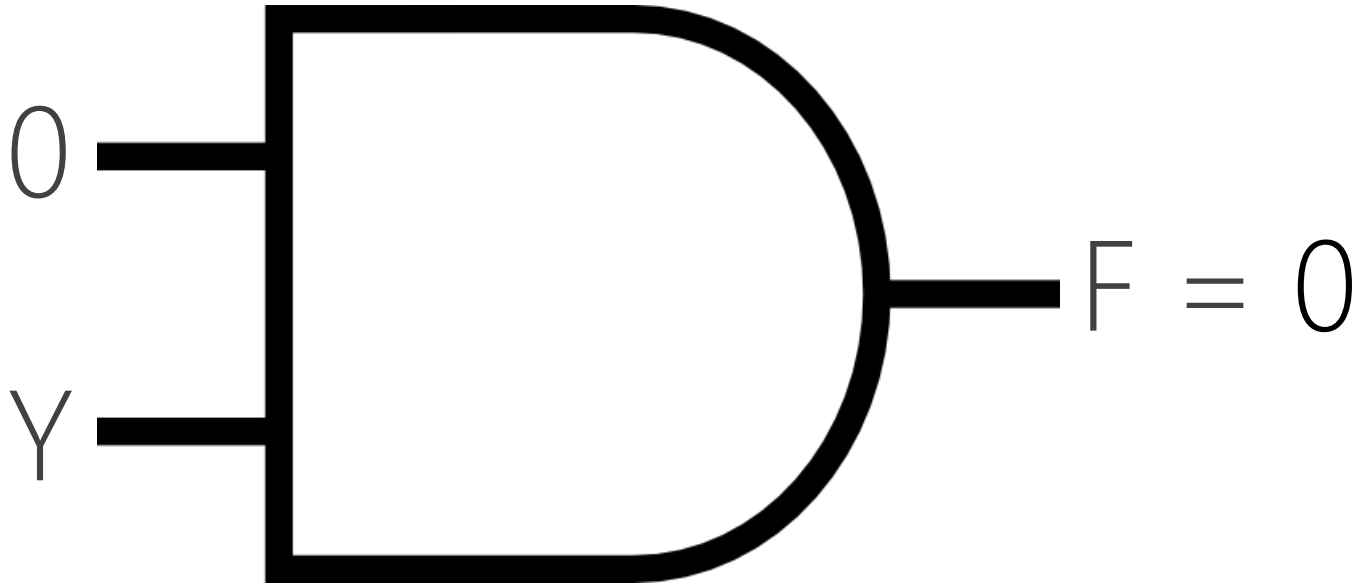
Commutative

X	Y	$X \cdot Y$	$Y \cdot X$	$Y \cdot X == X \cdot Y$
0	0	0	0	Yes
0	1	0	0	Yes
1	0	0	0	Yes
1	1	1	1	Yes



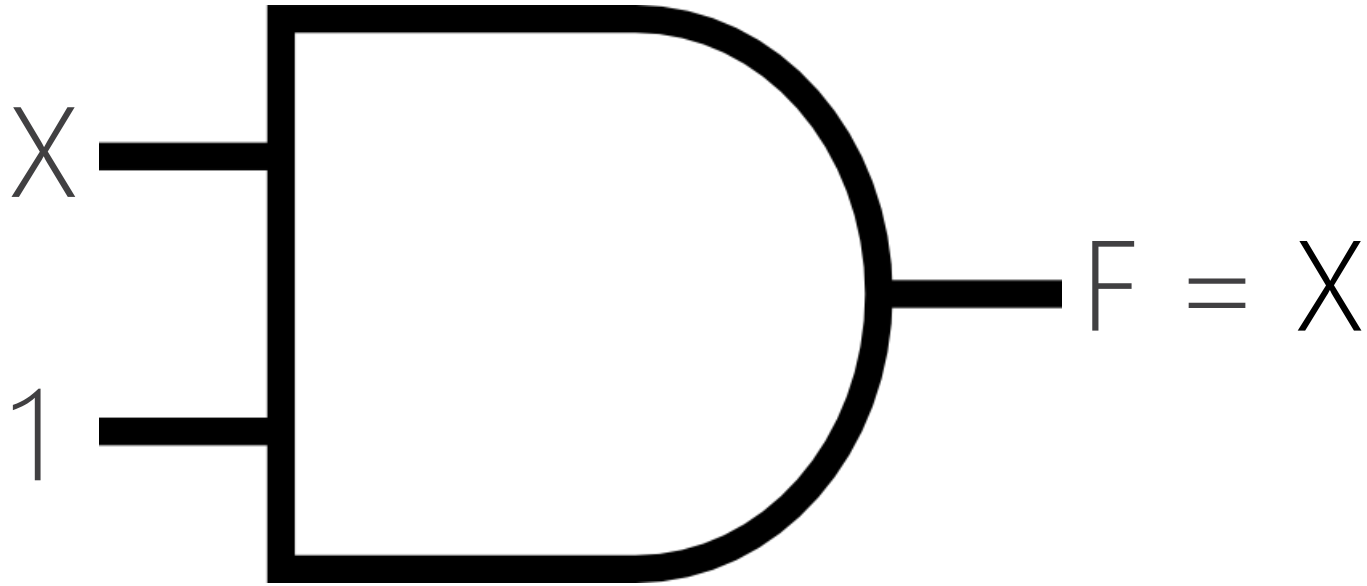
Y	X	YX
0	1	0
0	0	0

$$F(X,0) = X0 = 0$$



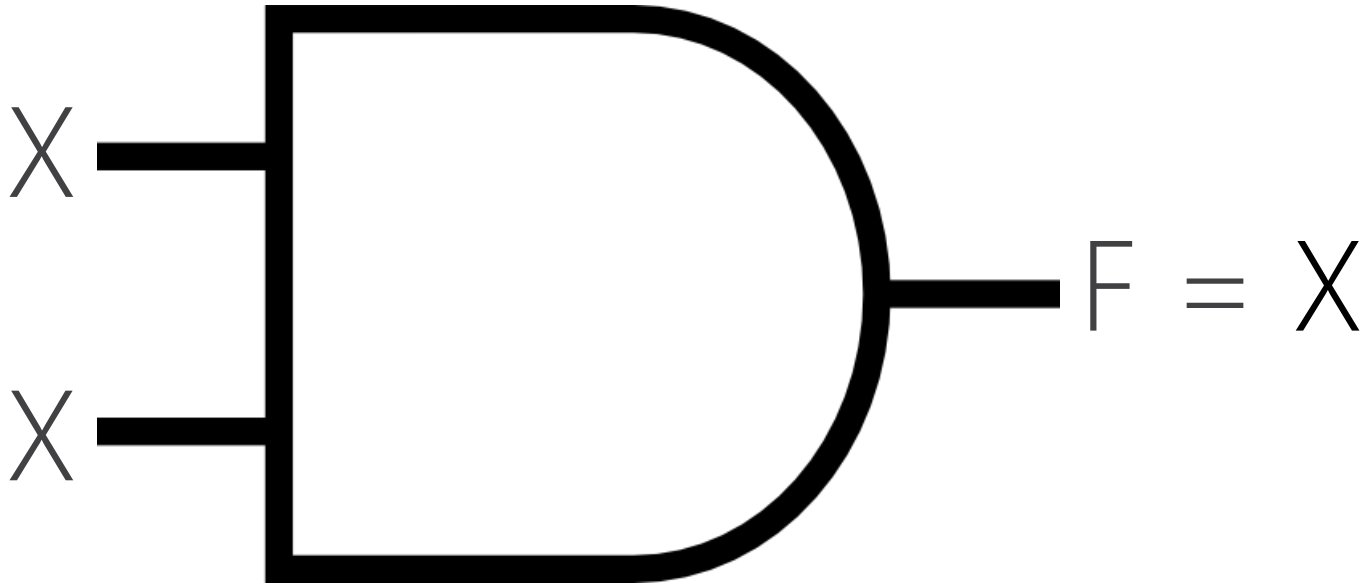
Y	X	YX
0	0	0
1	0	0

$$F = 0Y = 0$$



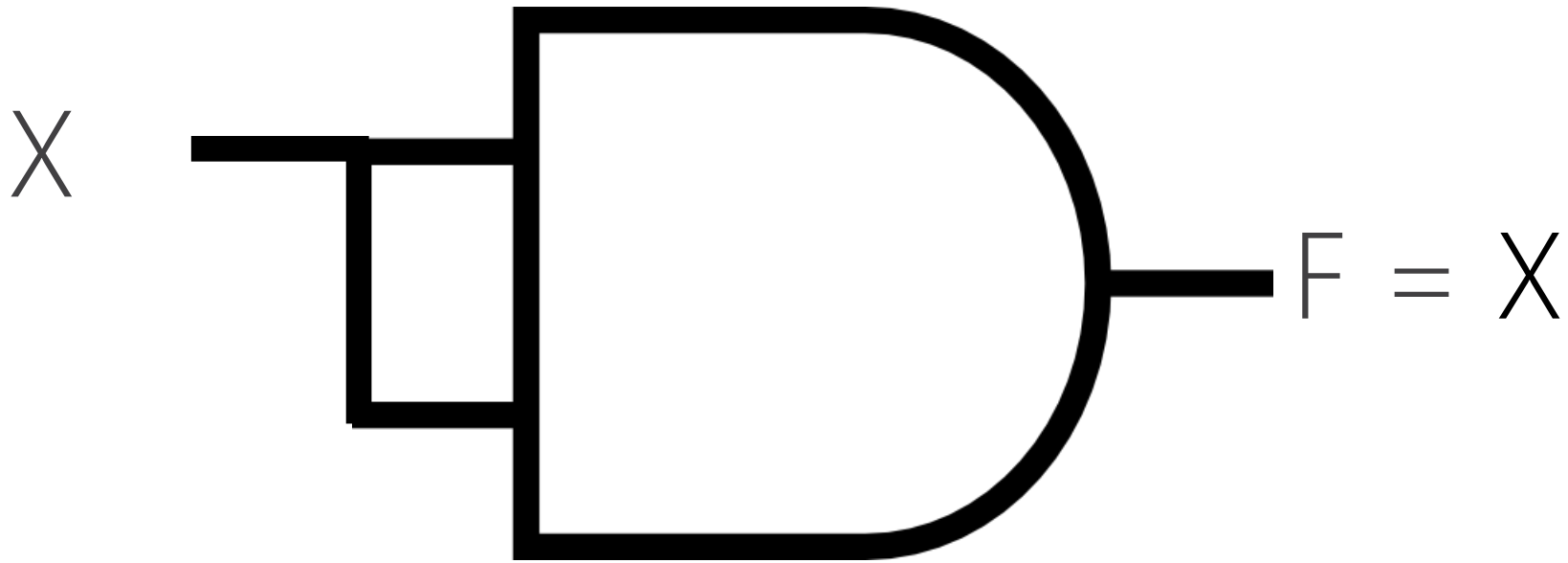
Y	X	YX
1	0	0
1	1	1

$$F = X1 = 1111X1111 = X$$



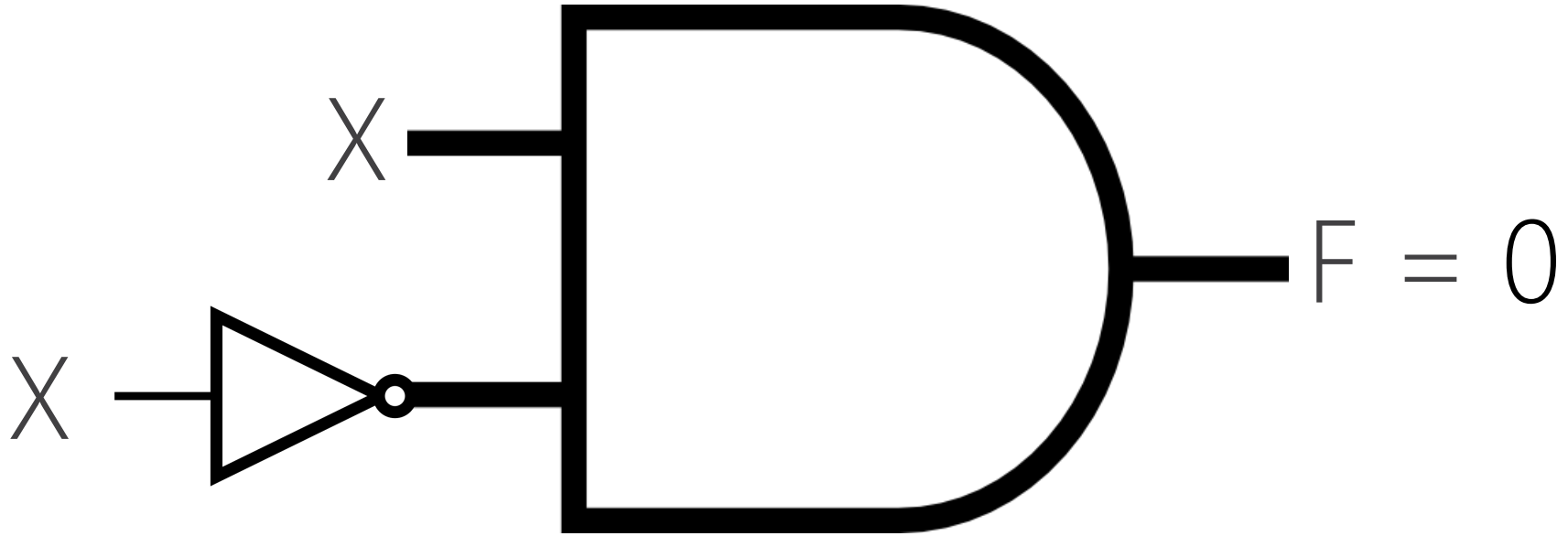
X	X	XX
0	0	0
1	1	1

$$F = XX = X$$



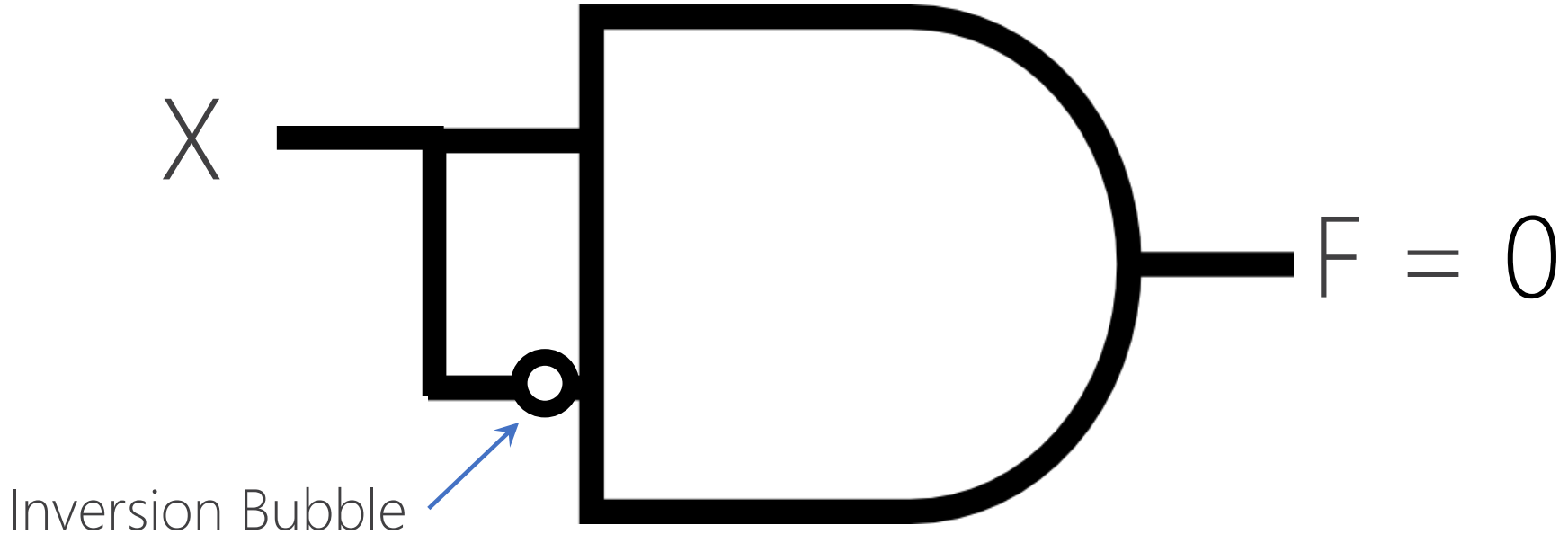
X	X	XX
0	0	0
1	1	1

$$F = XX = X$$



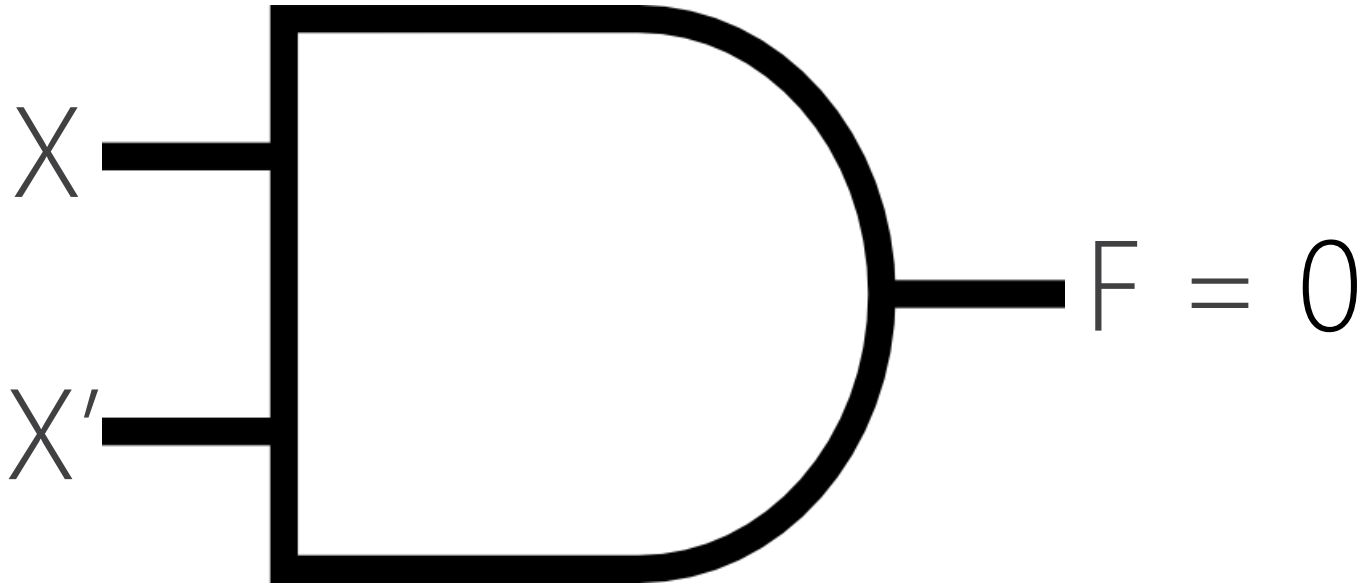
X′	X	X′X
1	0	0
0	1	0

$$F = XX' = 0$$



X′	X	X′X
1	0	0
0	1	0

$$F = XX' = 0$$



X′	X	X′X
1	0	0
0	1	0

$$F = XX' = 0$$

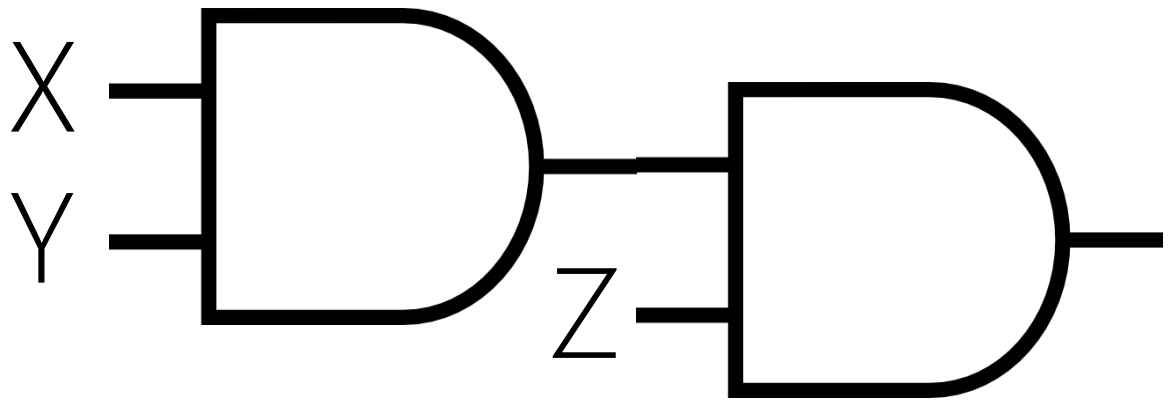
DESIGN

given the functionality, design the structure of a system

3-INPUT AND

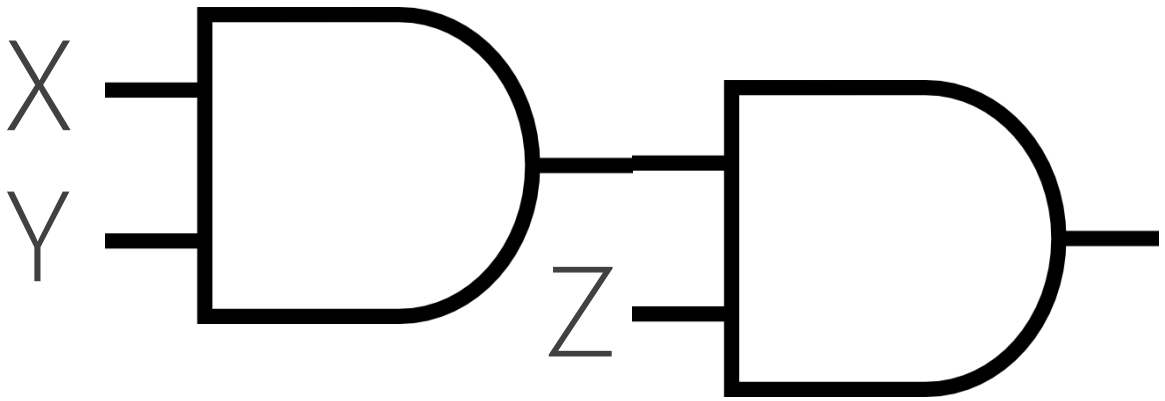
TRUTH TABLE
3-INPUT AND

Z	Y	X	ZYX
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



$$F = Z(YX)$$

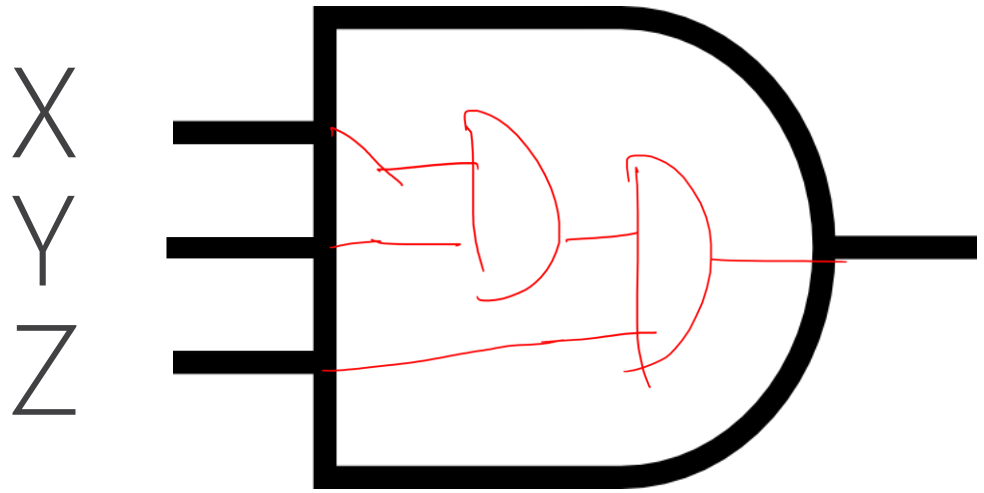
Z	Y	X	Z(YX)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



$$F = Z(YX) = Z(XY) = (ZX)Y = (XZ)Y = XZY$$

Associative

Z	Y	X	Z(YX)	Z(XY)	(ZX)Y	XZY
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	1	1	1	1



$$F = ZYX$$

Z	Y	X	ZYX
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

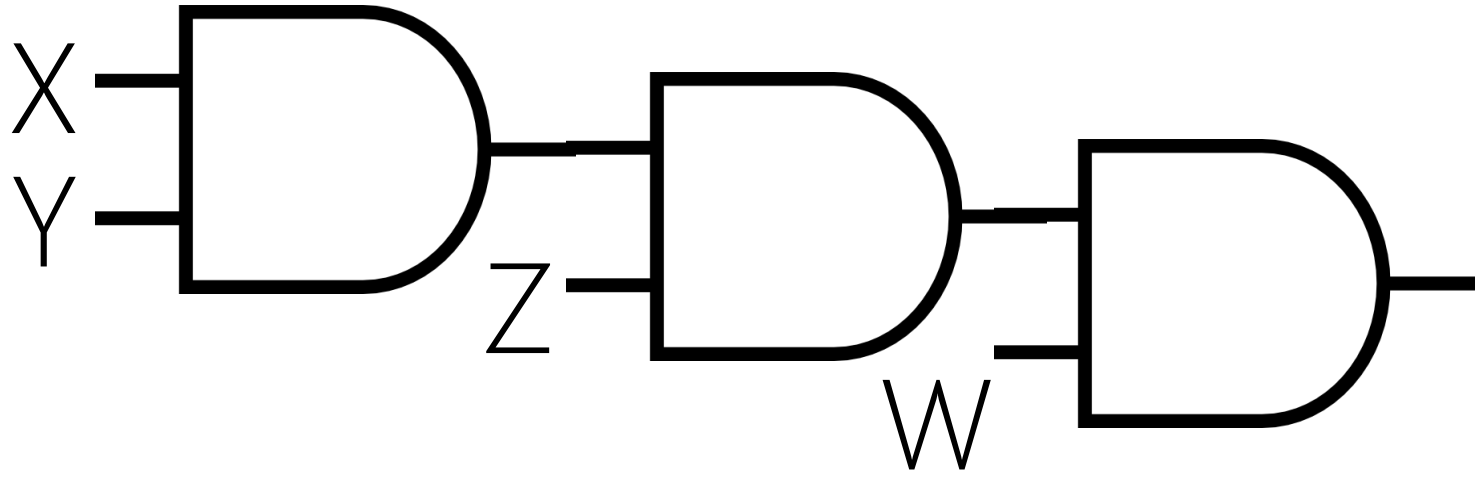
DESIGN

given the functionality, design the structure of a system

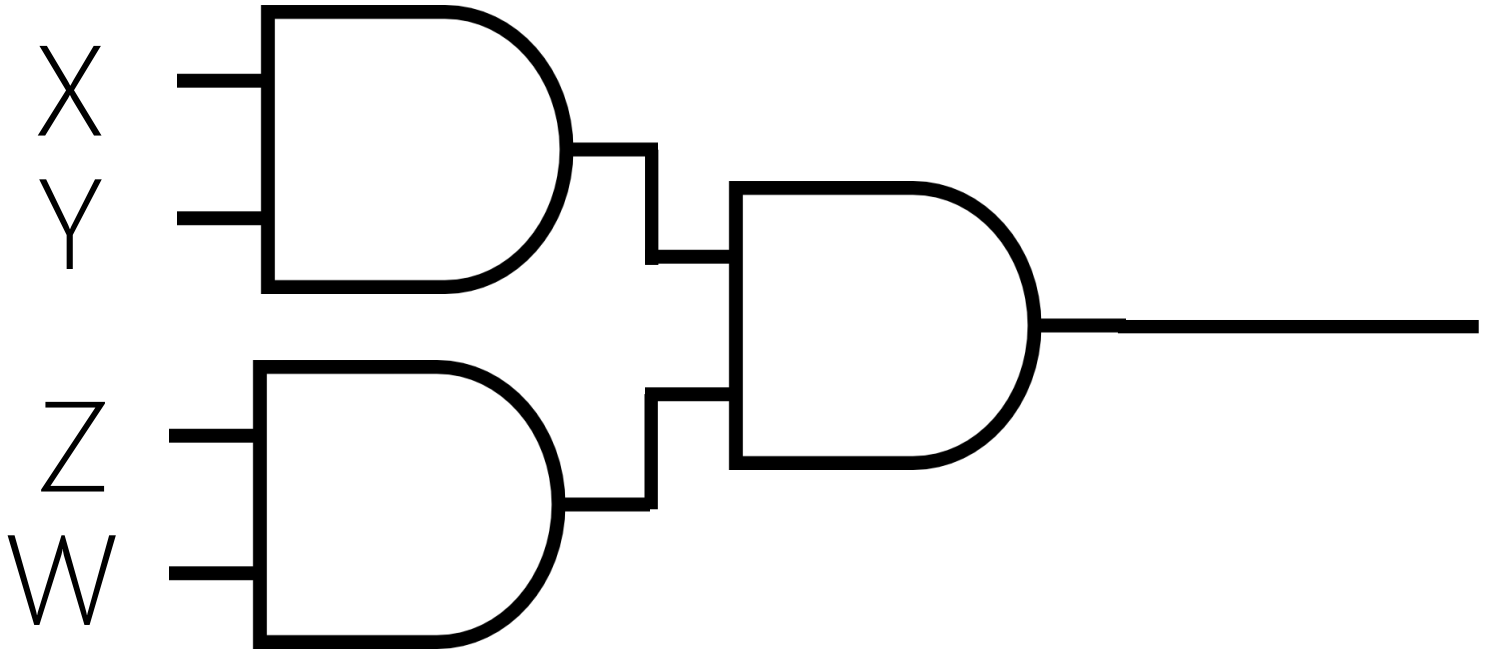
4-INPUT AND

DESIGN PATTERNS

Using Same or Similar Previous Designs for New Designs



$F_1 = W(Z(YX))$



$F_2 = (WZ)(YX)$

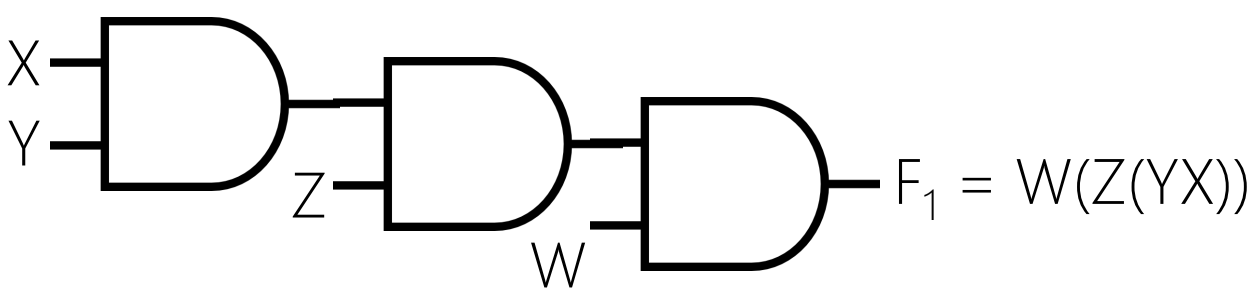
ANALYSIS

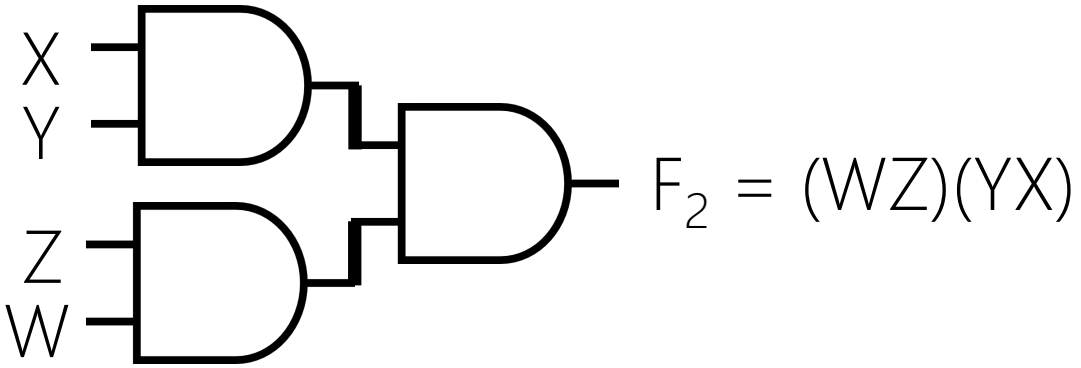
given the structure of a system, find its functionality.

determine the functionality exhibited by a structure.

EVALUATION

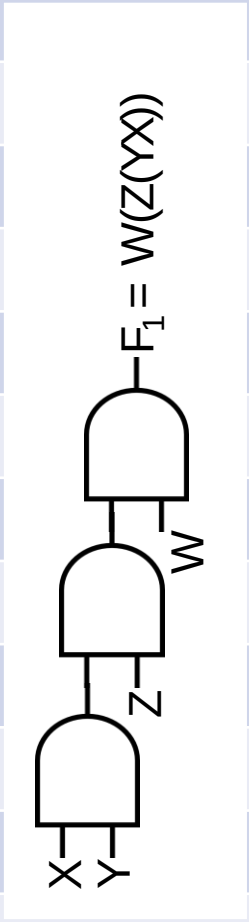
Correct vs. Wrong
Nice vs. Poor

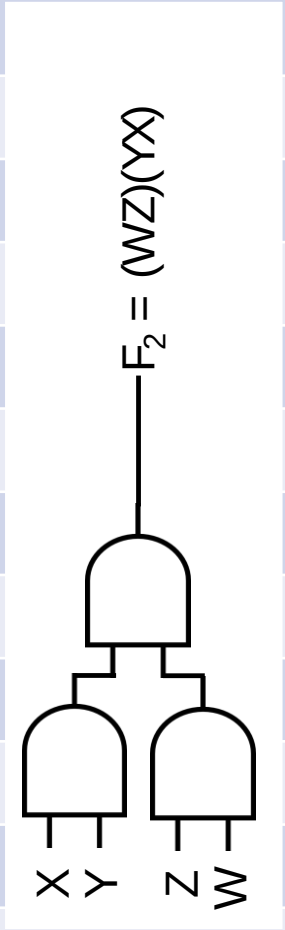


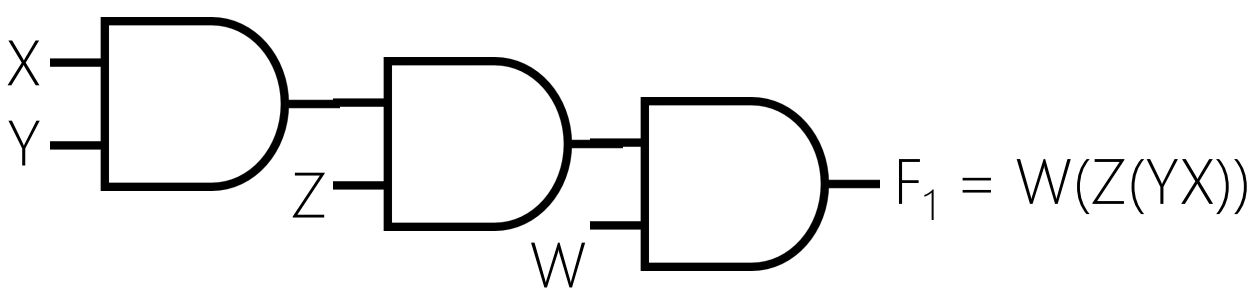


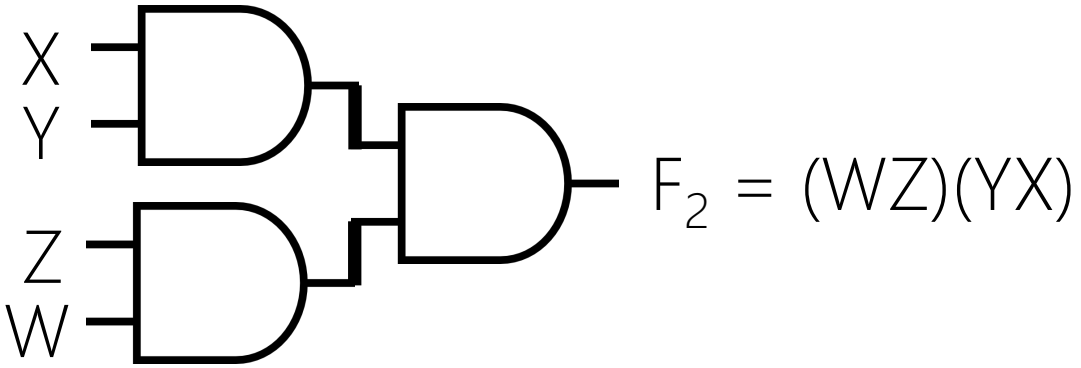
F = WZYX	F_1	F_2
Effective (True)		
Efficient (Fast)		
Min. Cost		

W	Z	Y	X	(W(Z(YX)))	(WZ)(YX)
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	1	1

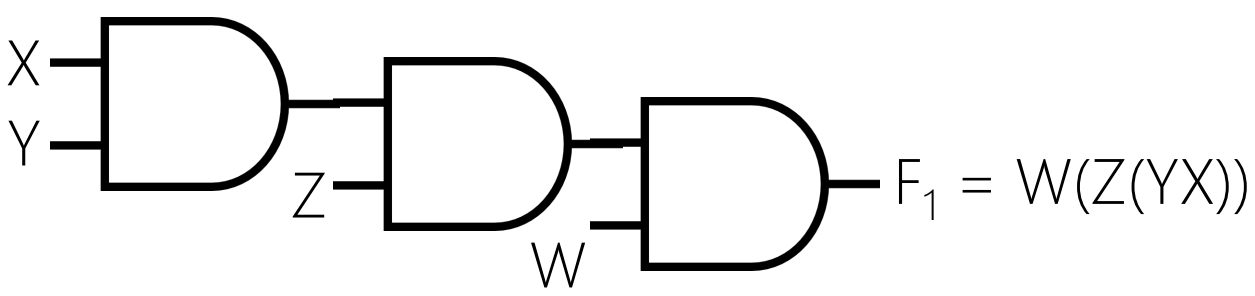


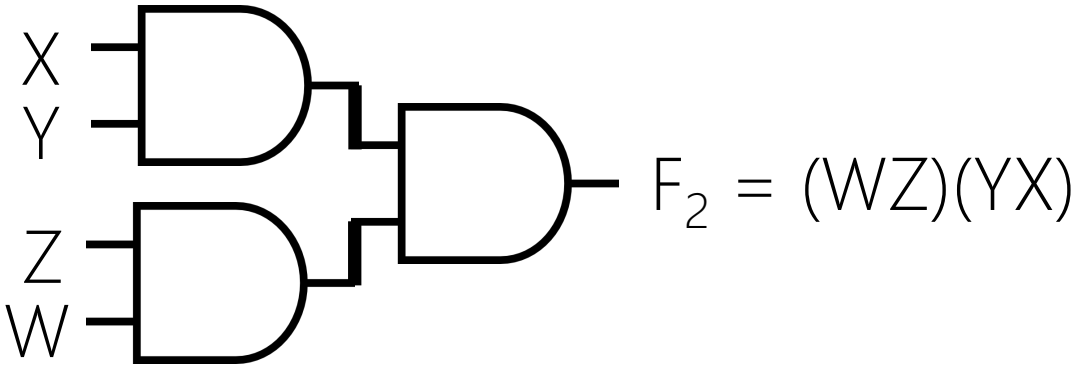




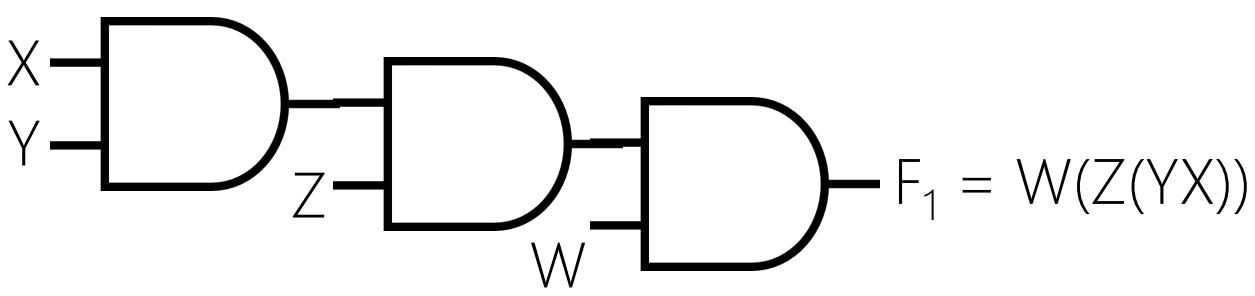


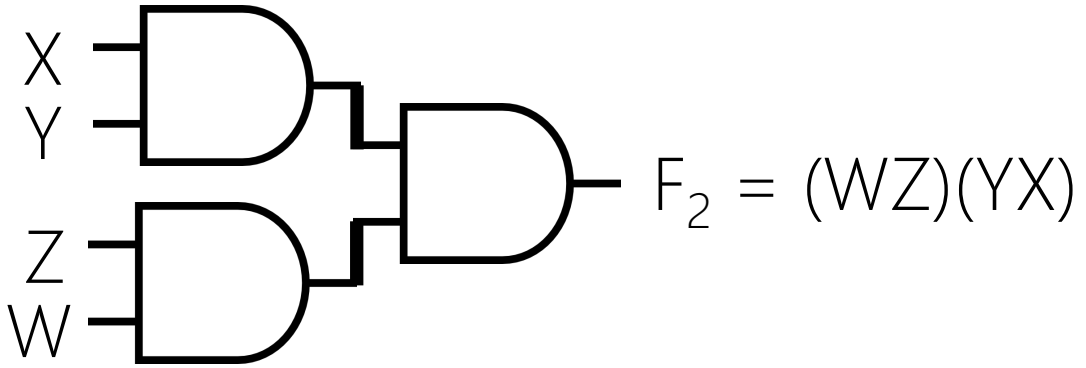
F = WZYX	F_1	F_2
Effective (True)	Yes	Yes
Efficient (Fast)		
Min. Cost		



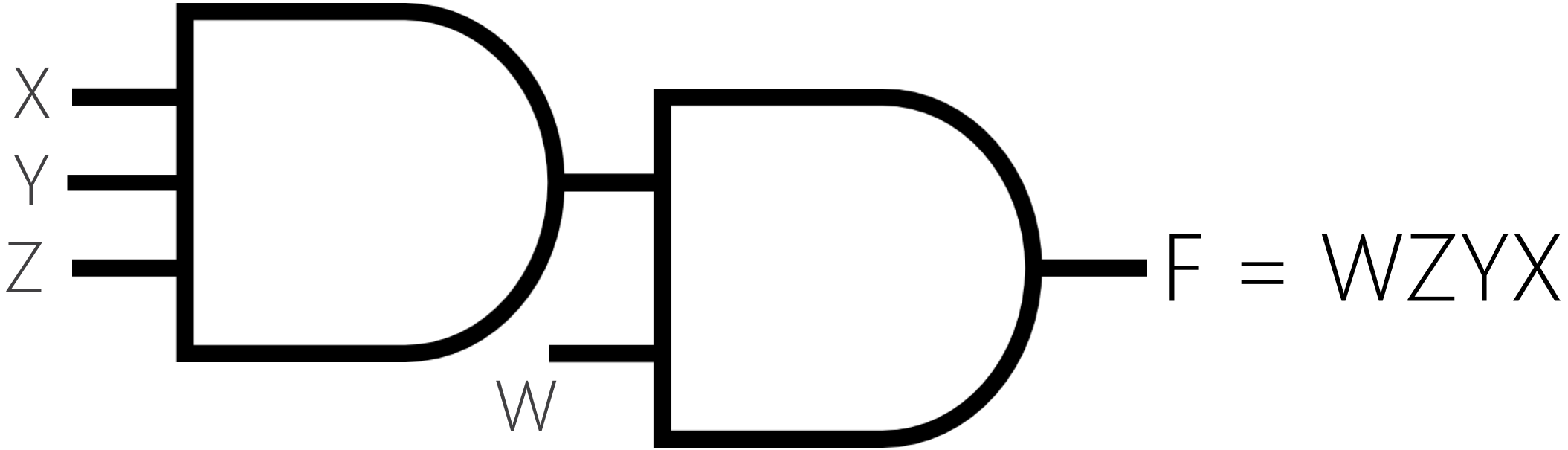


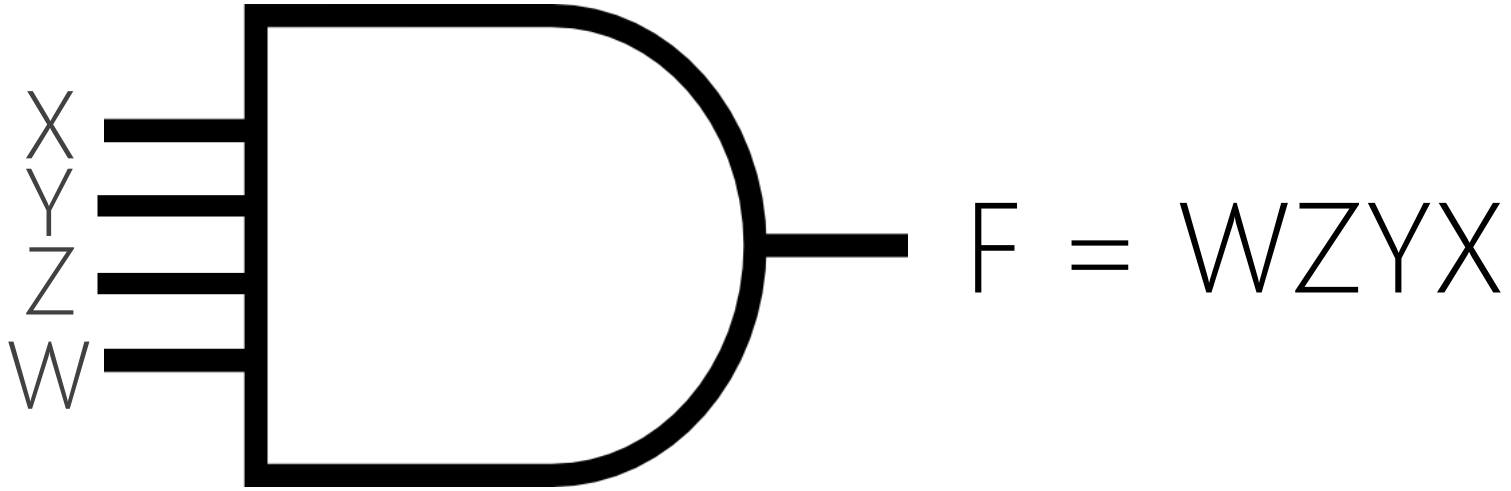
F = WZYX	F_1	F_2
Effective (True)	Yes	Yes
Efficient (Fast)		
Min. Cost	3 gates, Yes	3 gates, Yes





F = WZYX	F_1	F_2
Effective (True)	Yes	Yes
Efficient (Fast)	Hmm, 3 levels, No!	Yes! 2 levels
Min. Cost	3 gates, Yes	3 gates, Yes


X
Y
Z
W
F = WZYX

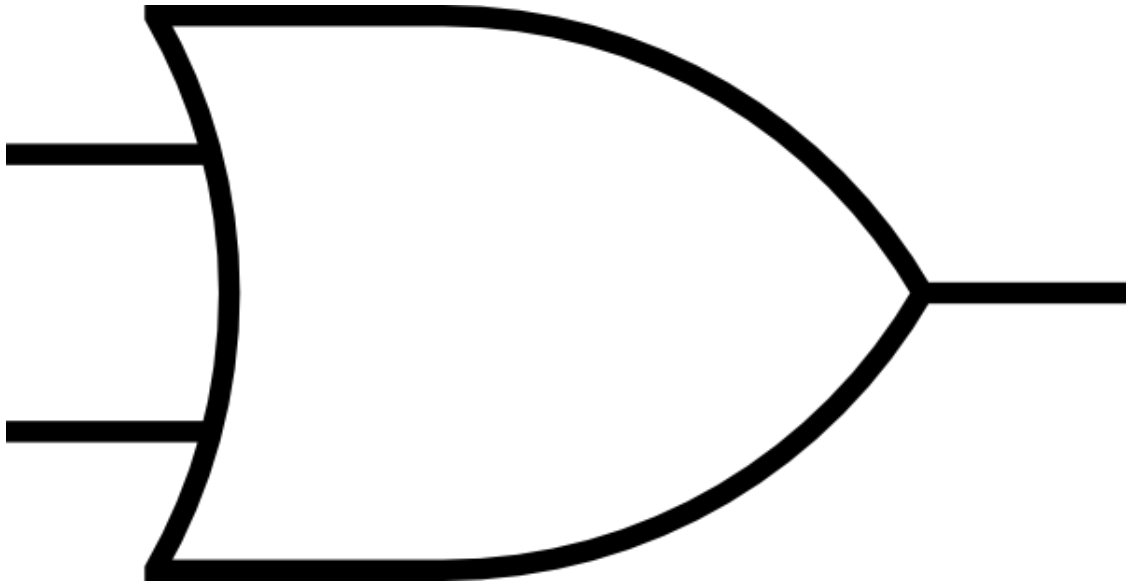

X
Y
Z
W
F = WZYX


Geöffnet
11-23 Uhr

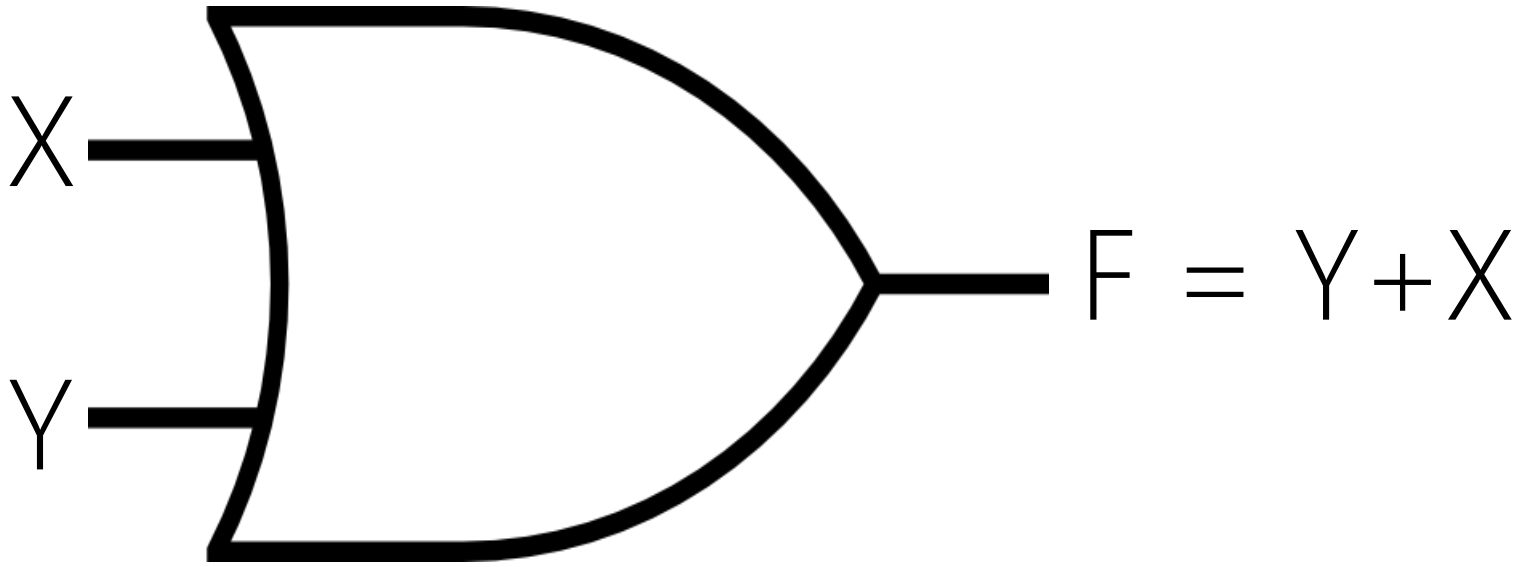

KARIM
RASHID
About Karim

Karim Rashid | I am a Design Pervert

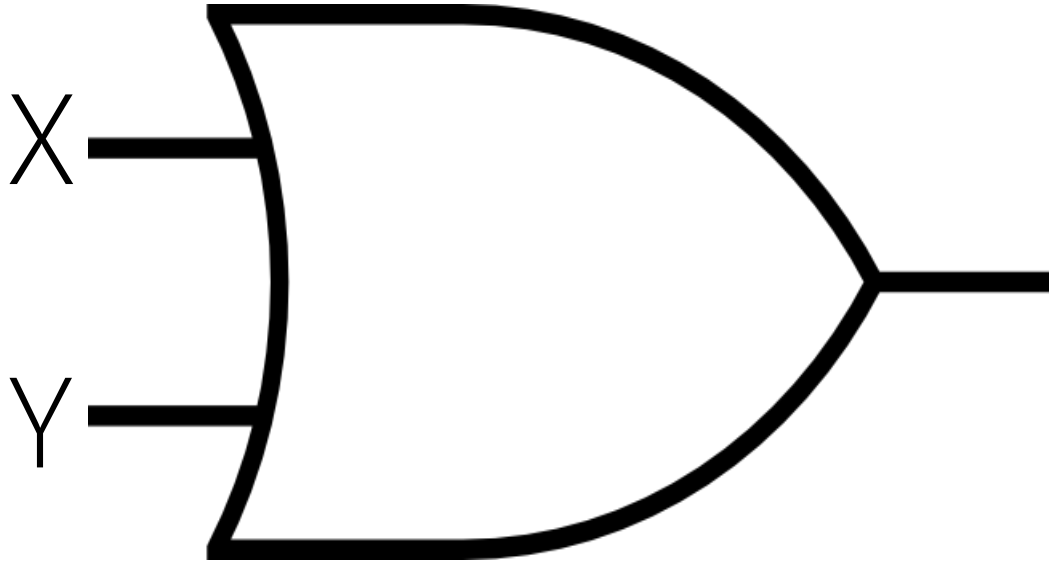
https://www.youtube.com/watch?v=ovjpszjQHN4



OR



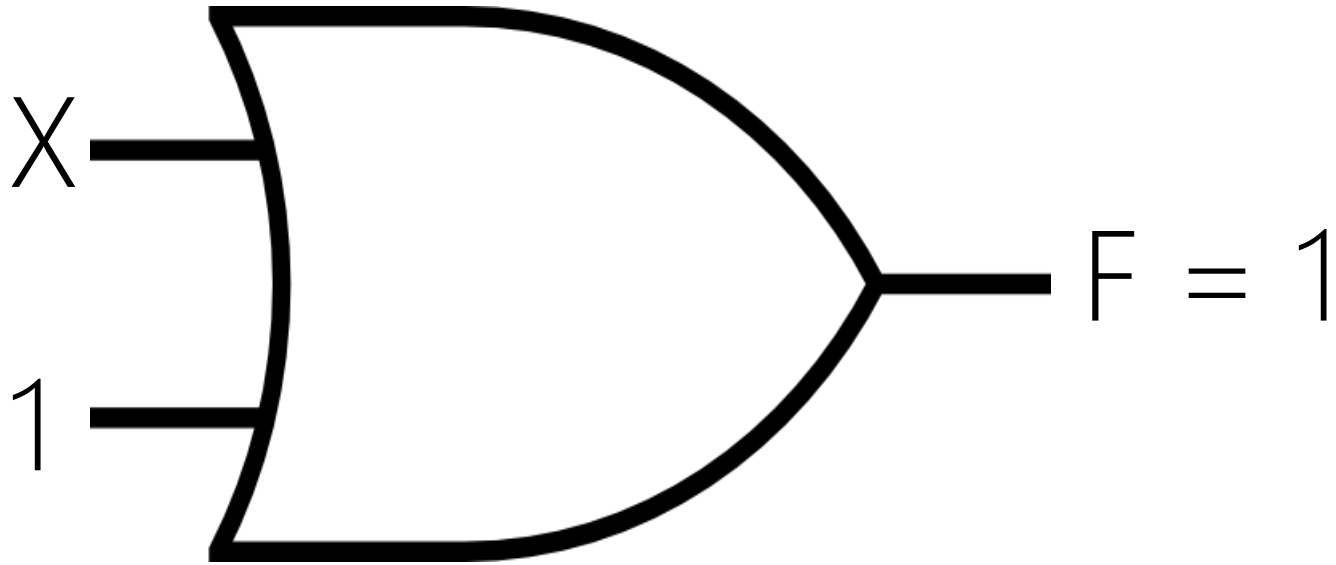
Y	X	Y OR X	Y+X
0	0	0	
0	1	1	
1	0	1	
1	1	1	



F=Y+X=X+Y

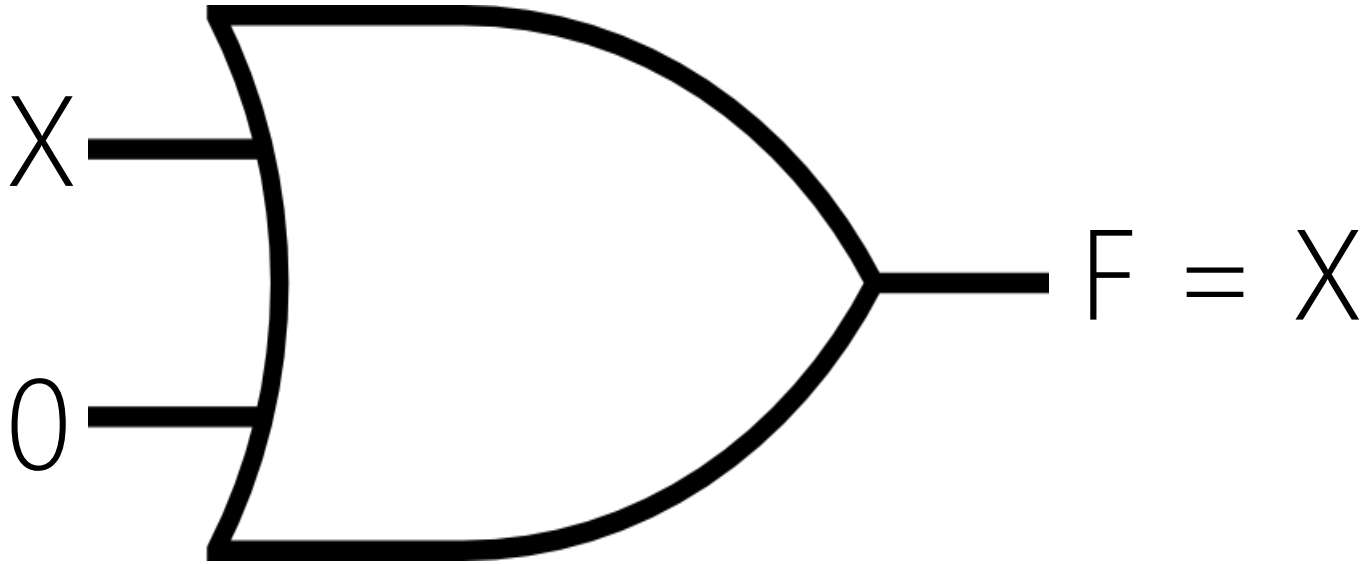
Commutative

X	Y	X OR Y	X+Y
0	0	0	
0	1	1	
1	0	1	
1	1	1	



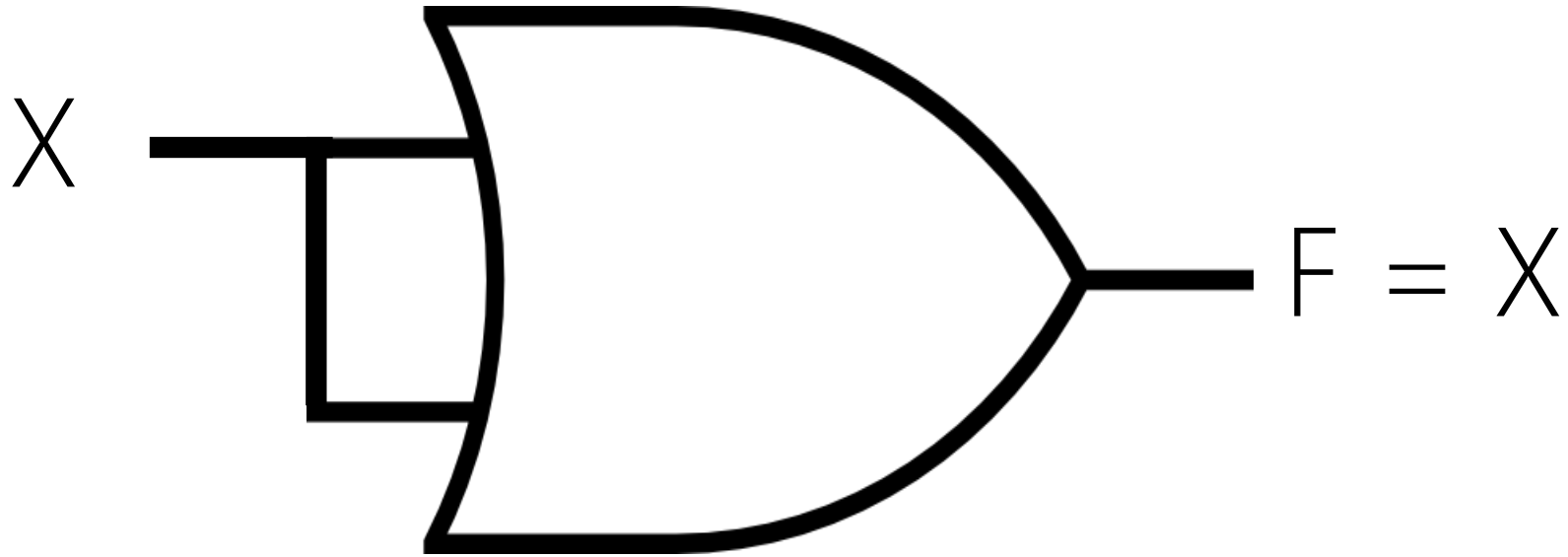
Y	X	Y+X
1	0	1
1	1	1

$$F = X+1 = 1$$



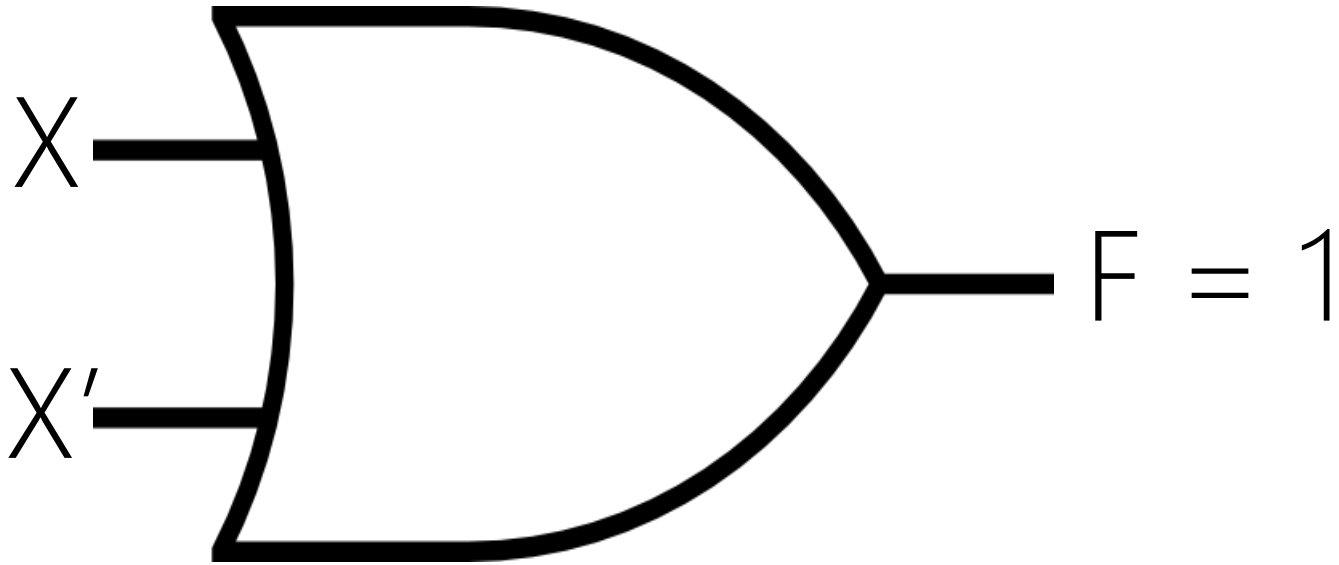
Y	X	Y+X
0	0	0
0	1	1

$$F = X + 0 = X$$



X	X	X+X
0	0	0
1	1	1

$$F = X+X = X$$



X′	X	X′+X
1	0	1
0	1	1

$$F = X + X' = 1$$

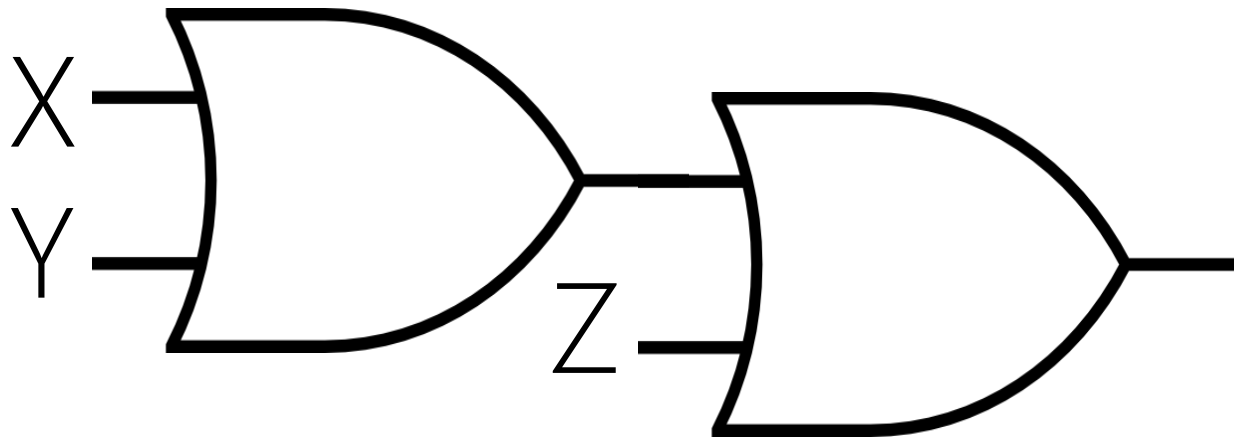
DESIGN

given the functionality, design the structure of a system

3-INPUT OR

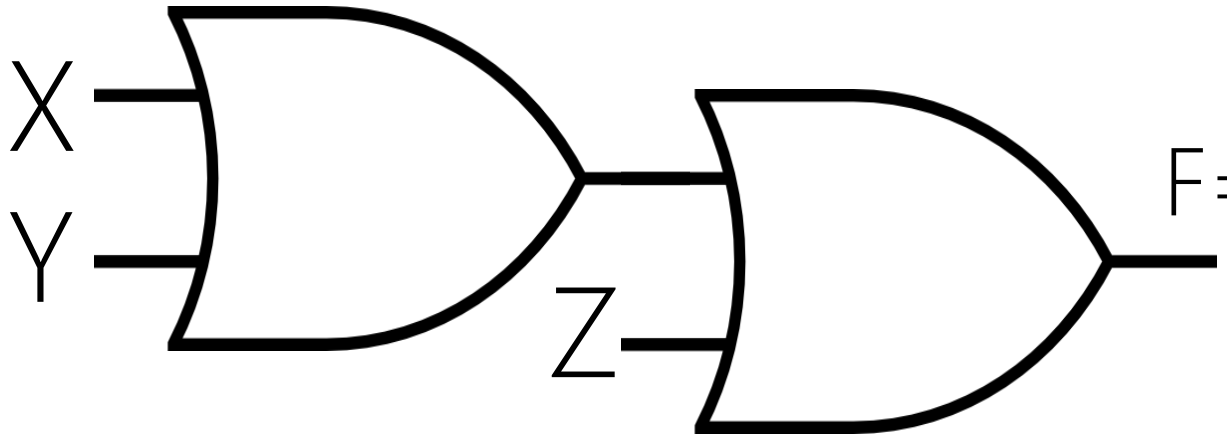
DESIGN PATTERNS

Using Same or Similar Previous Designs for New Designs



$$F = Z+(X+Y)$$

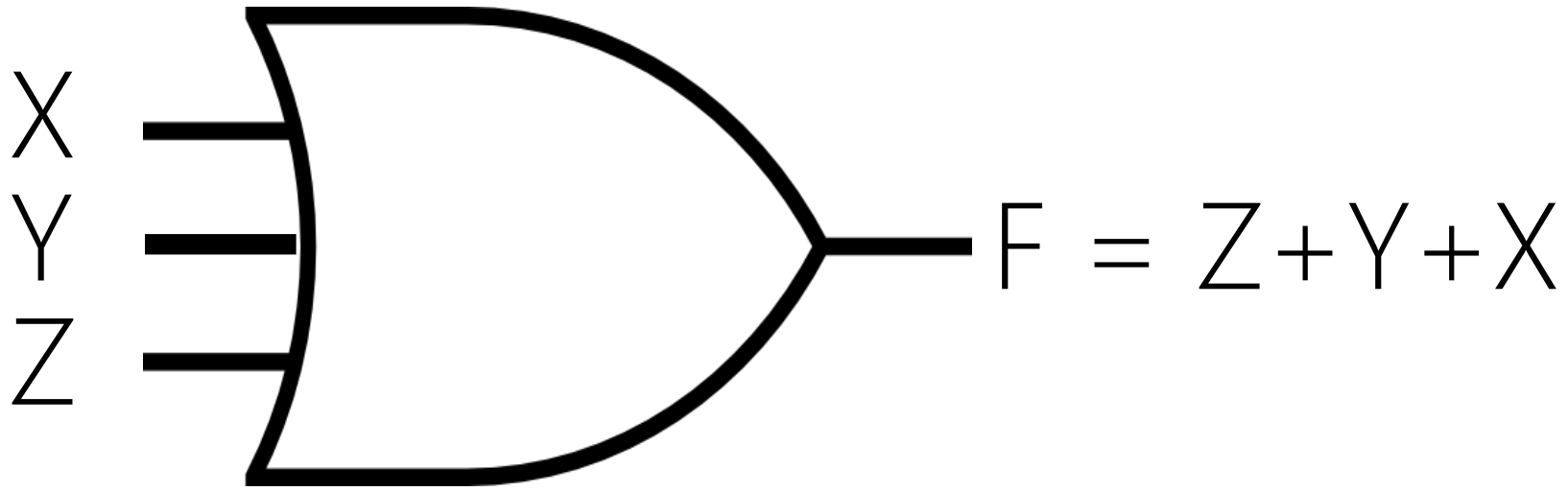
Z	Y	X	Z+(X+Y)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



$$F = Z + (Y + X) = Z + (X + Y) = (Z + X) + Y$$
$$= Z + Y + X$$

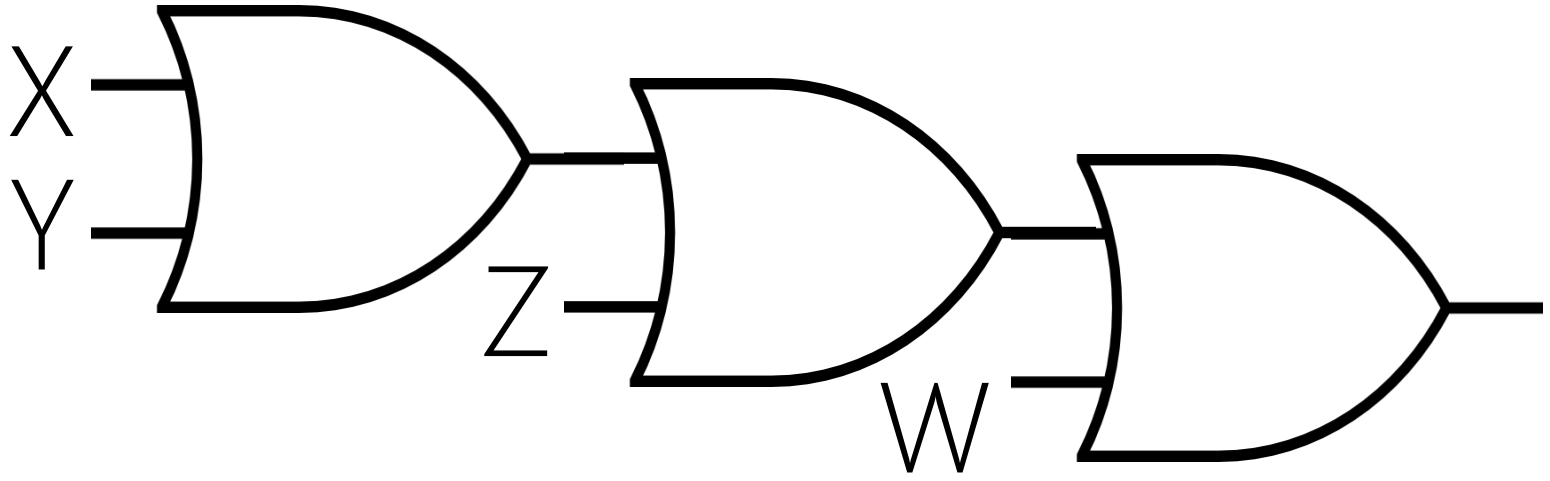
Associative

Z	Y	X	Z+(Y+X)	Z+(X+Y)	(Z+X)+Y	ZXY
0	0	0	0			
0	0	1	1			
0	1	0	1			
0	1	1	1			
1	0	0	1			
1	0	1	1			
1	1	0	1			
1	1	1	1			

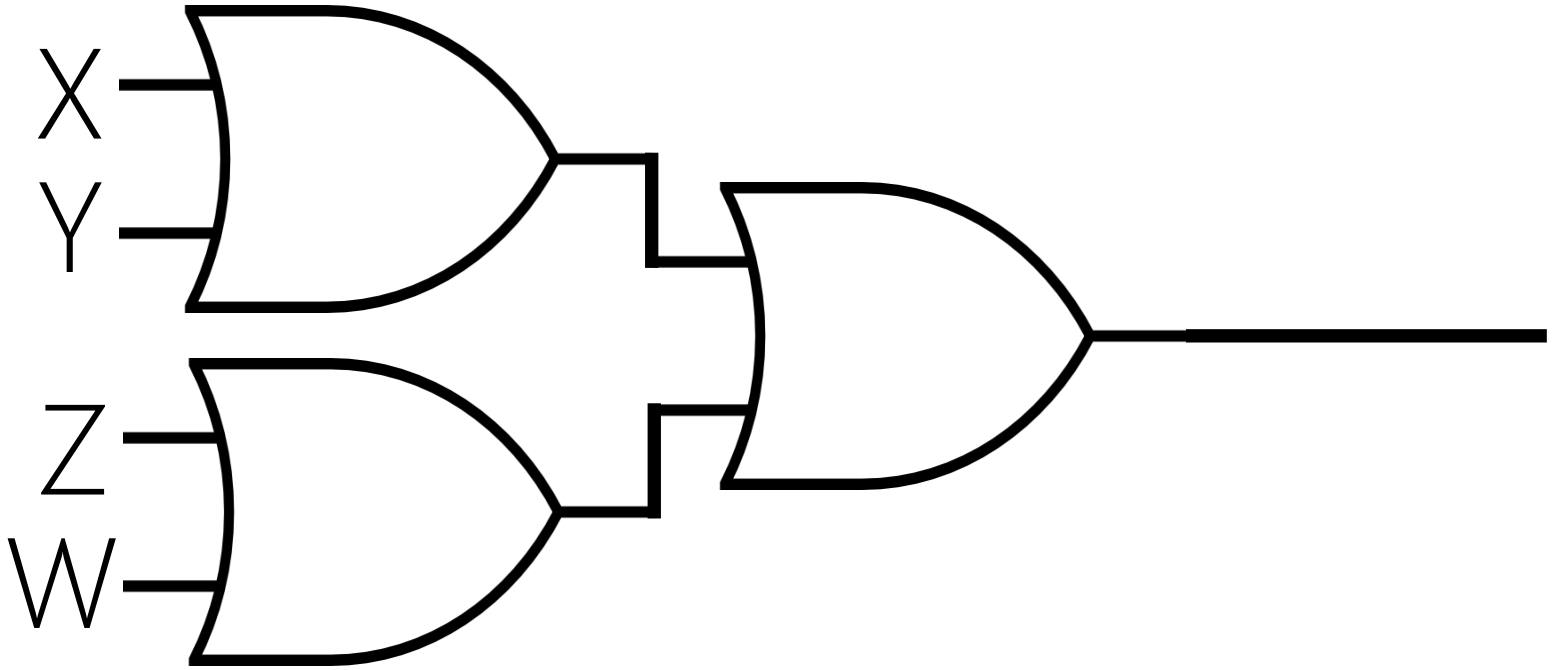


Z	Y	X	Z+(Y+X)	Z+(X+Y)	(Z+X)+Y	ZXY
0	0	0		0		
0	0	1		1		
0	1	0		1		
0	1	1		1		
1	0	0		1		
1	0	1		1		
1	1	0		1		
1	1	1		1		

4-INPUT OR

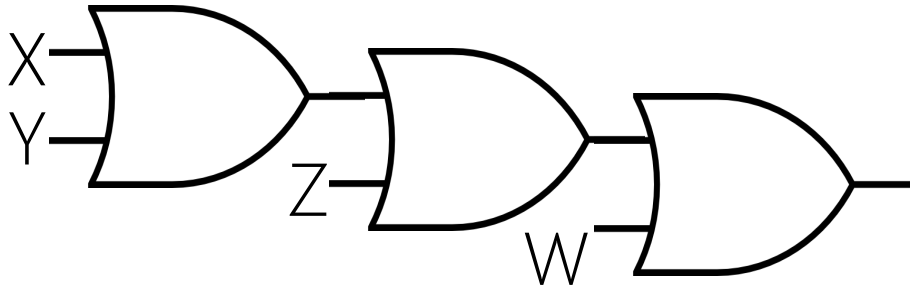


$F_1 = W+(Z+(Y+X))$

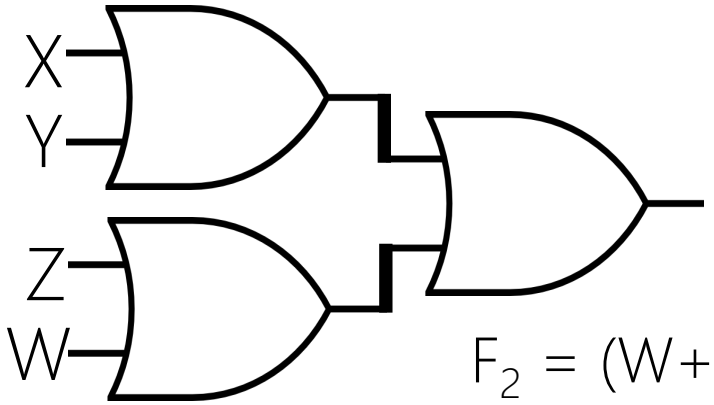


$F_2 = (W+Z)+(Y+X)$

DESIGN → ANALYSIS



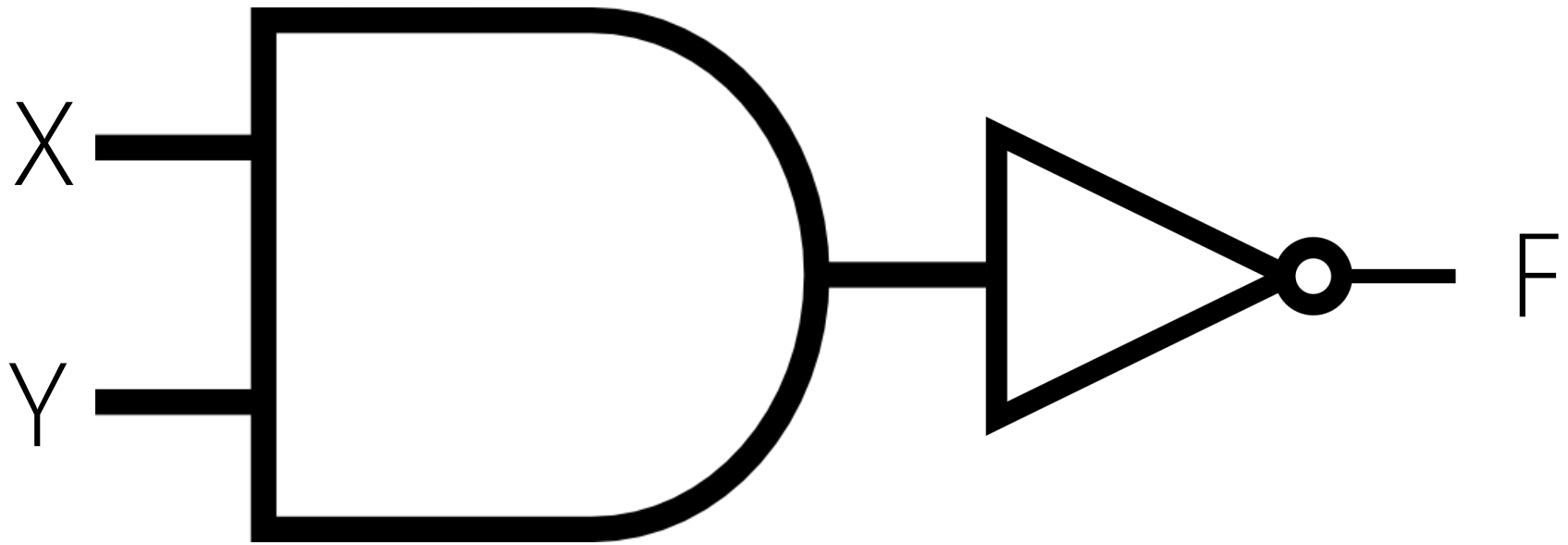
$F_1 = W+(Z+(Y+X))$

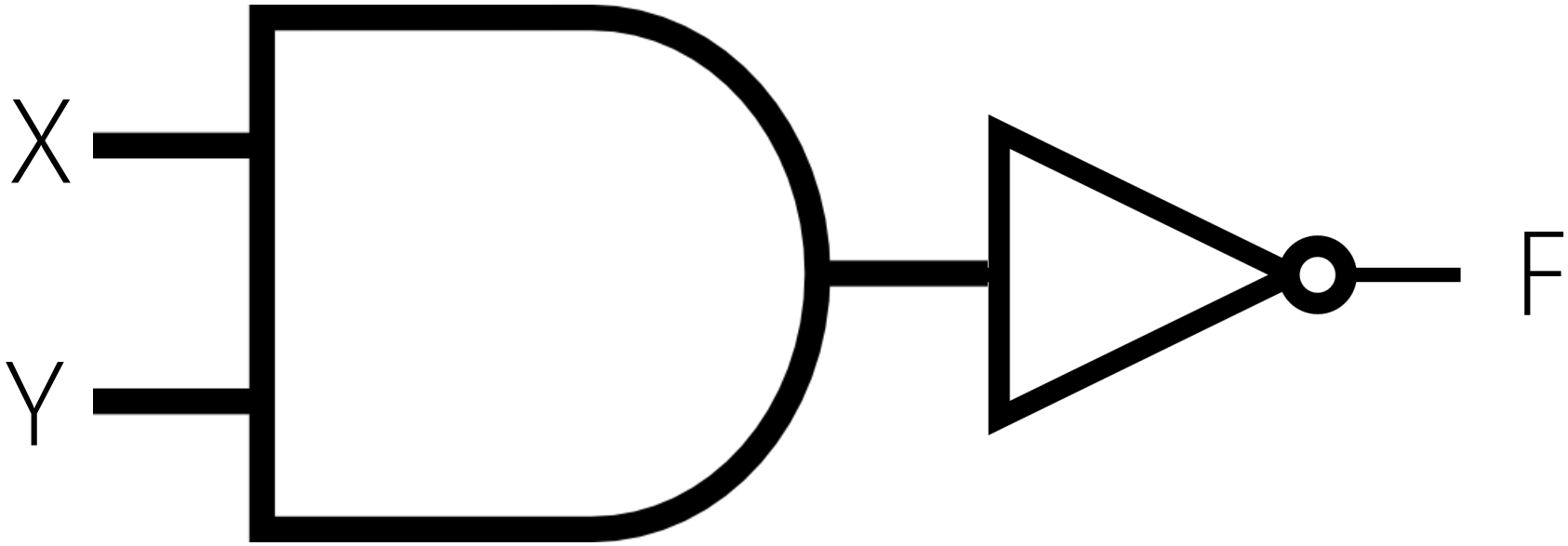


$F_2 = (W+Z)+(Y+X)$

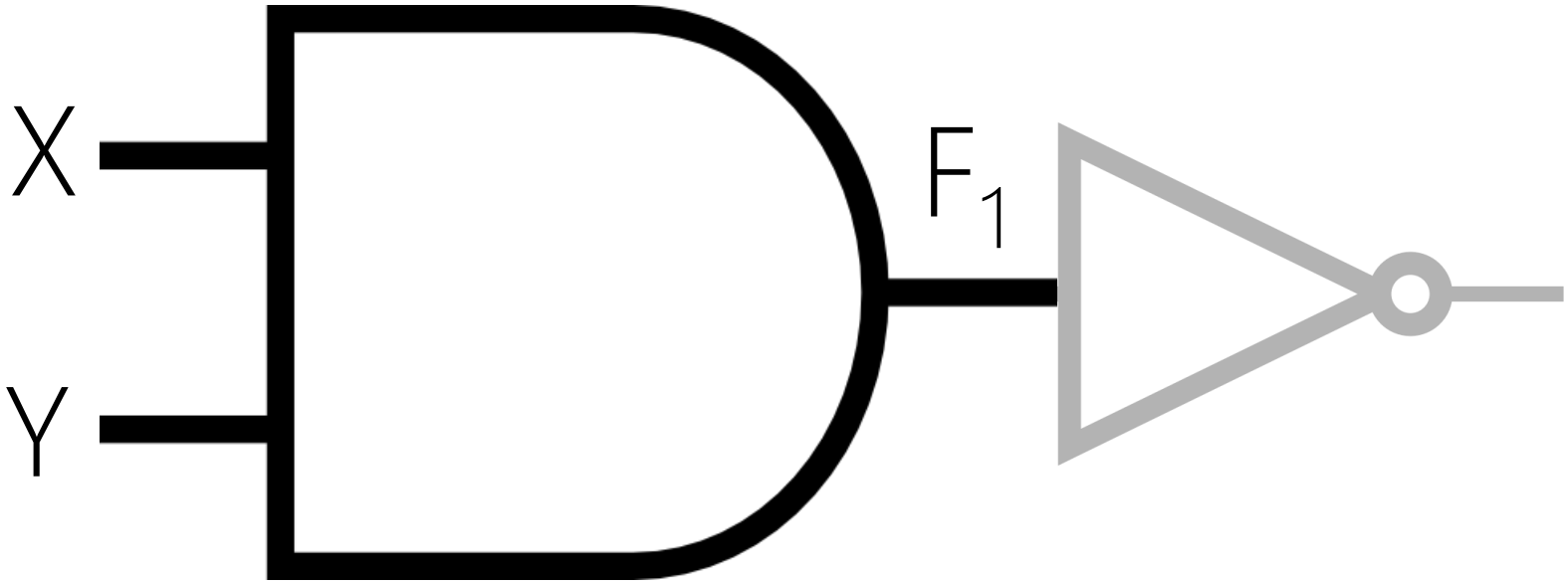
F = W+Z+Y+X	F_1	F_2
Effective (True)	Yes	Yes
Efficient (Fast)	Hmm, 3 levels, No!	Yes! 2 levels
Min. Cost	3 gates, Yes	3 gates, Yes

JUST ANALYSIS

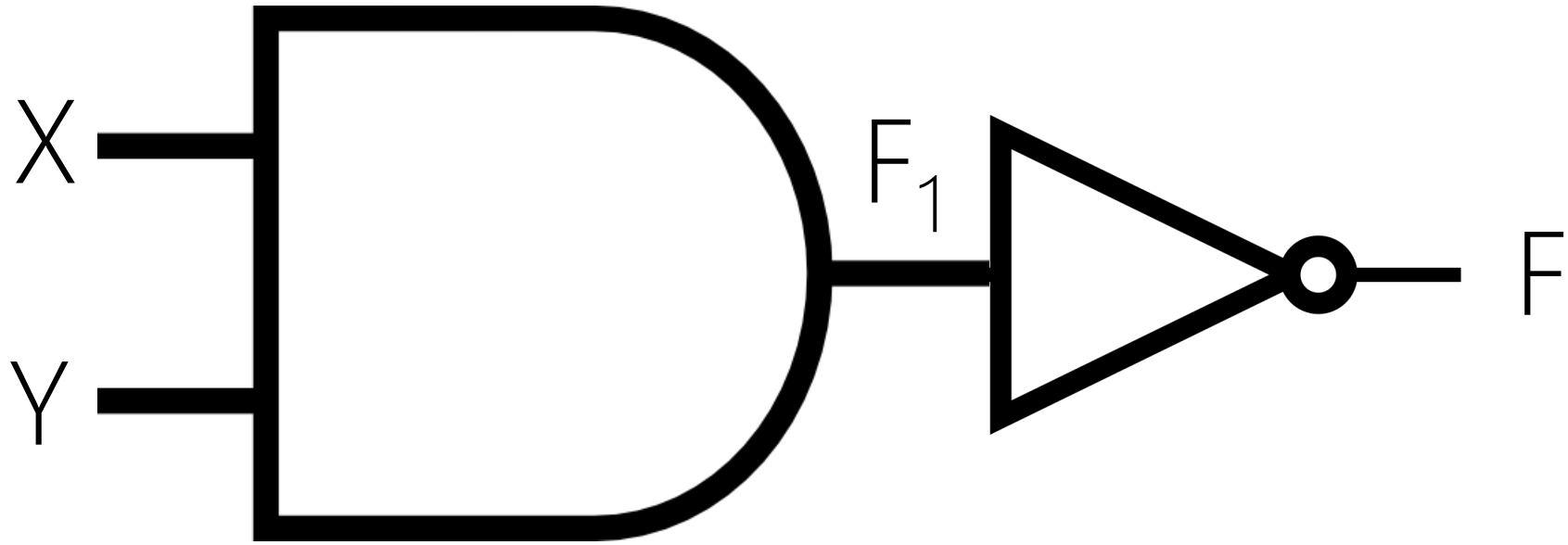

X
Y
F



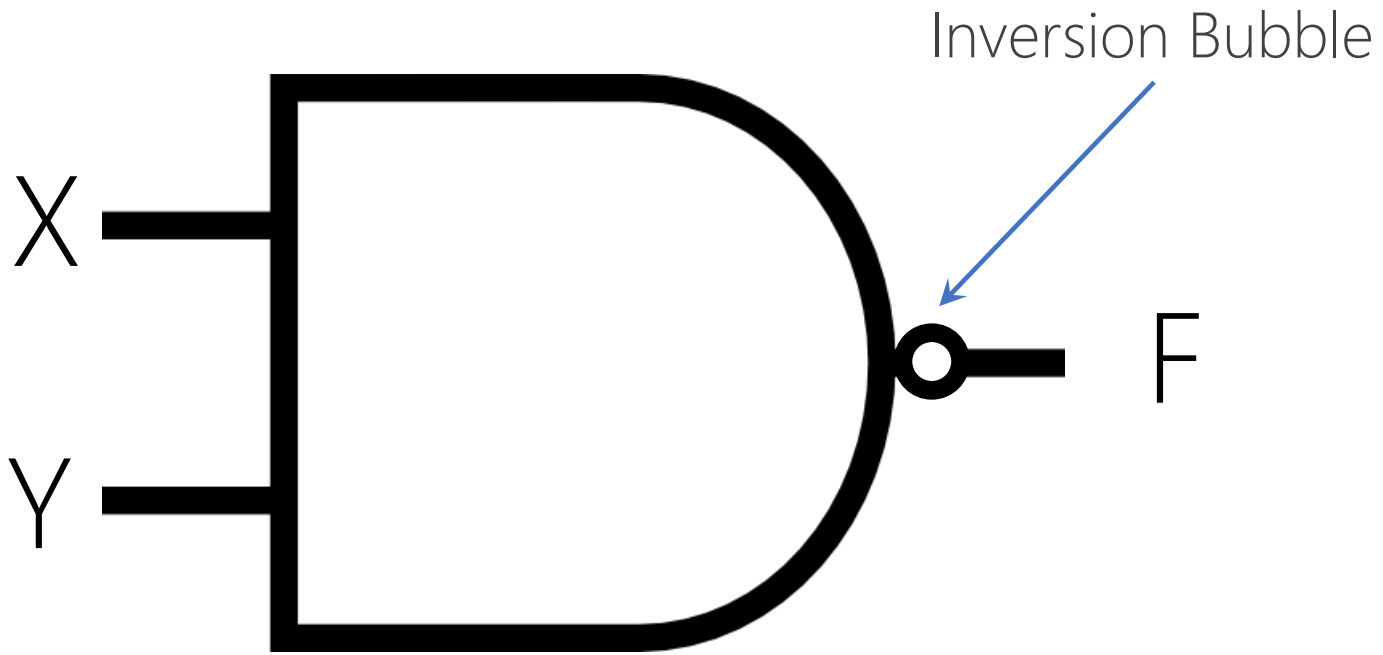
Y	X	F = ?
0	0	?
0	1	?
1	0	?
1	1	?



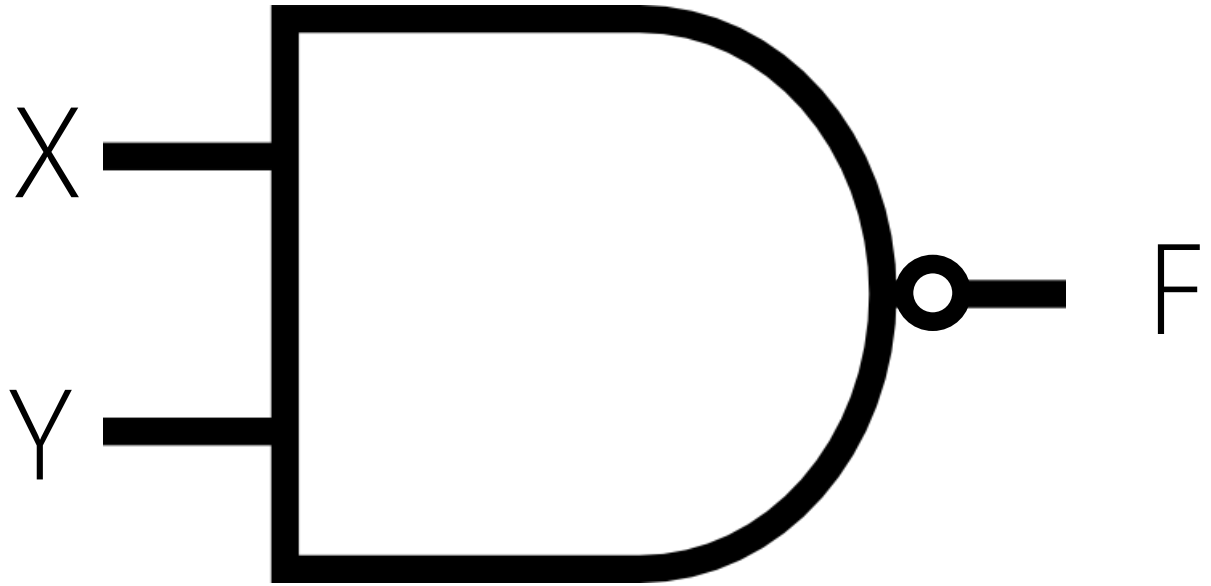
Y	X	$F_1 = YX$
0	0	0
0	1	0
1	0	0
1	1	1



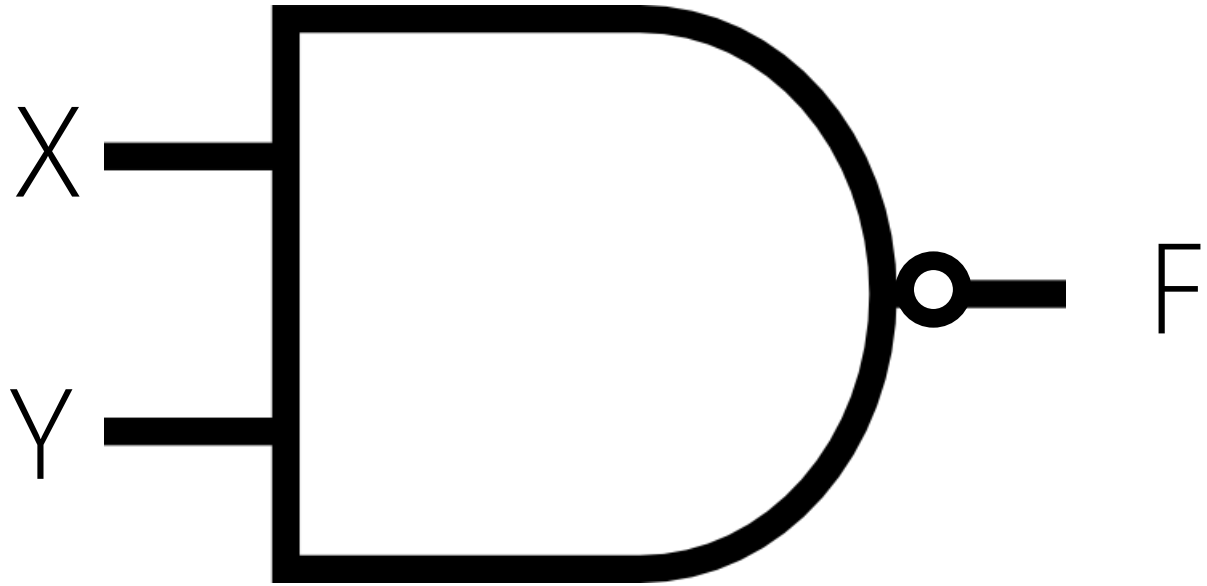
Y	X	$F_1 = YX$	$F = (YX)'$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0



NAND (Not – AND)

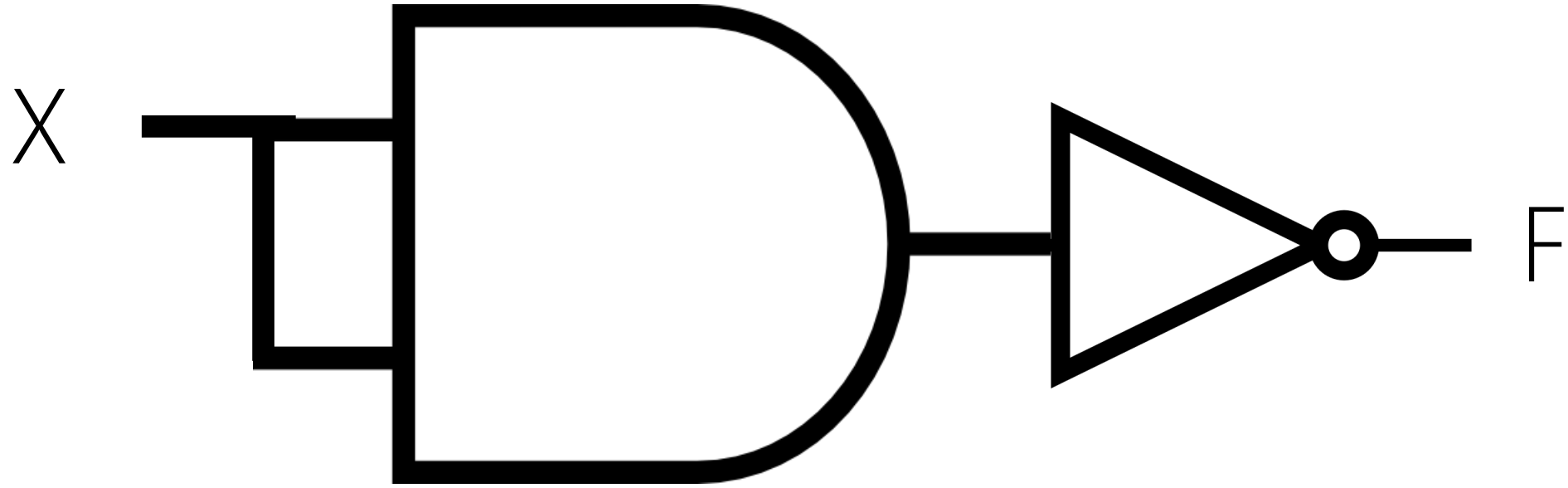


Y	X	F = (YX)′	F=Y ↑ X
0	0	1	
0	1	1	
1	0	1	
1	1	0	

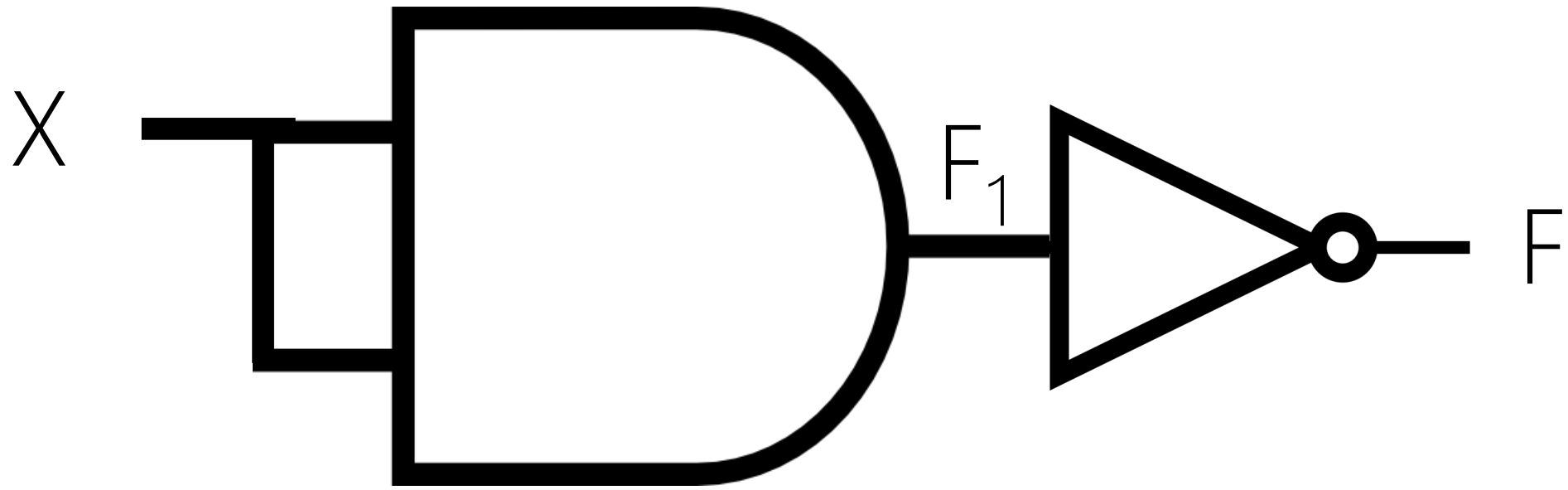


$$F = (YX)' = (XY)' = Y \uparrow X = X \uparrow Y$$

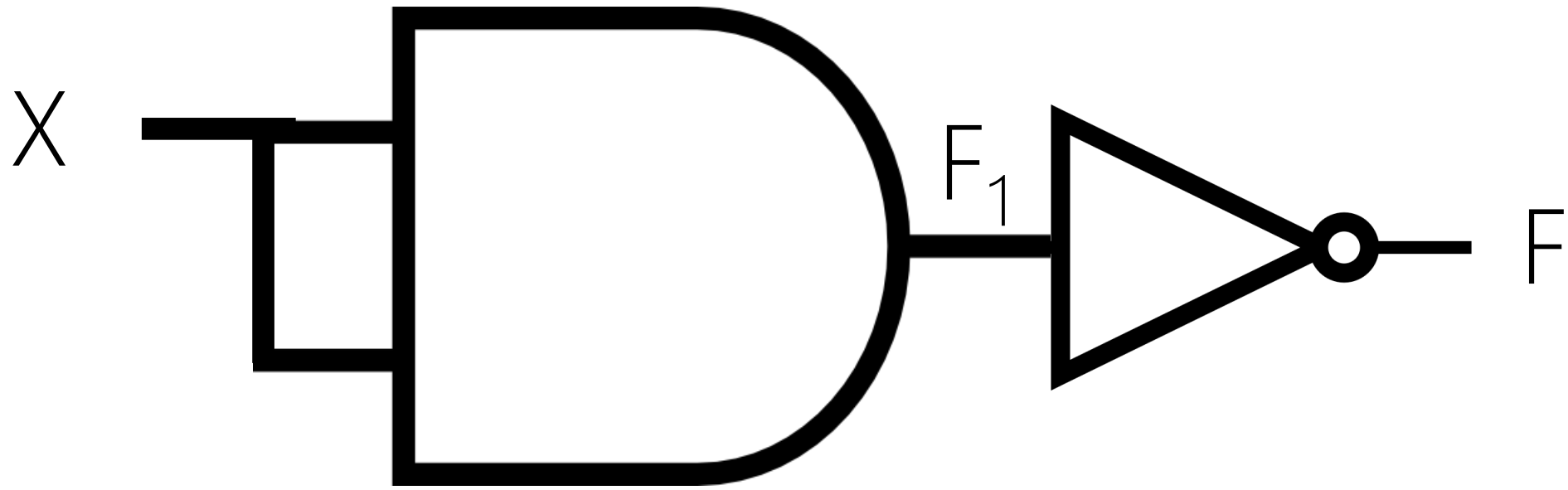
Commutative



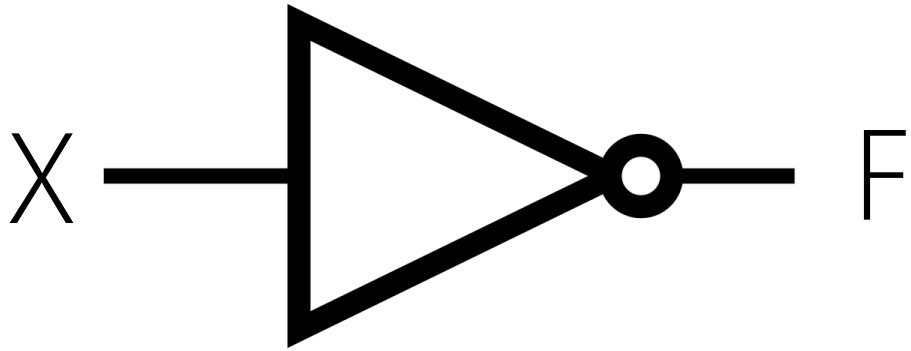
X	F = ?
0	
1	



X	$F_1 = XX$	F = ?
0	0	
1	1	



X	$F_1 = XX$	$F = (XX)'$
0	0	1
1	1	0



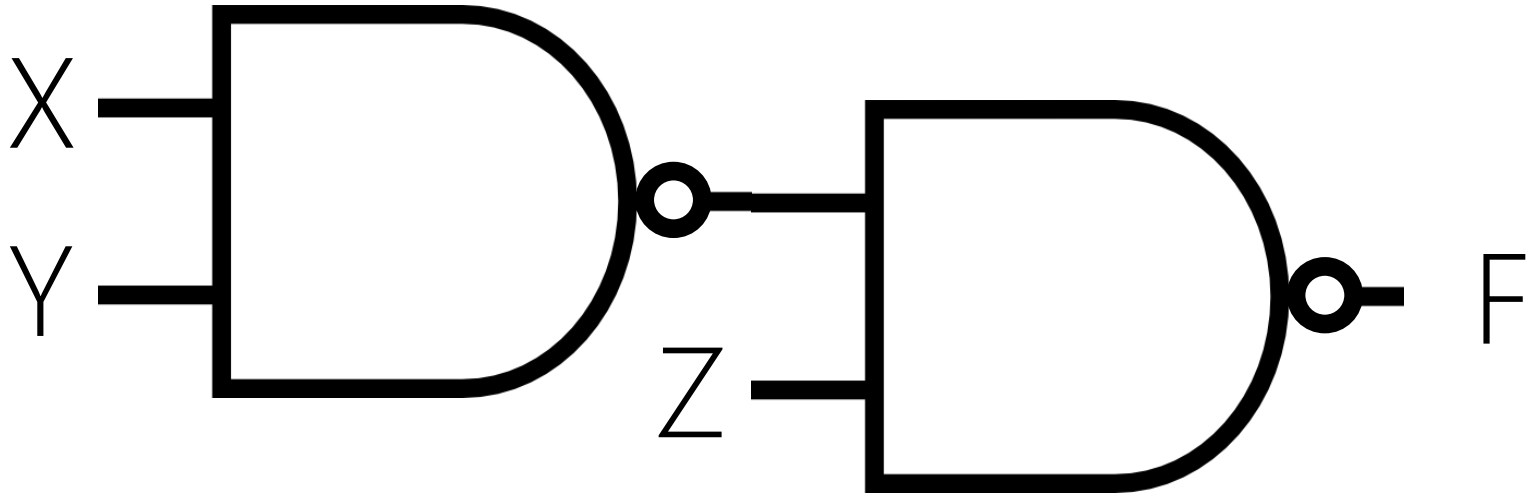
X	F = (XX)′=X↑X = X′
0	1
1	0

3-INPUT NAND

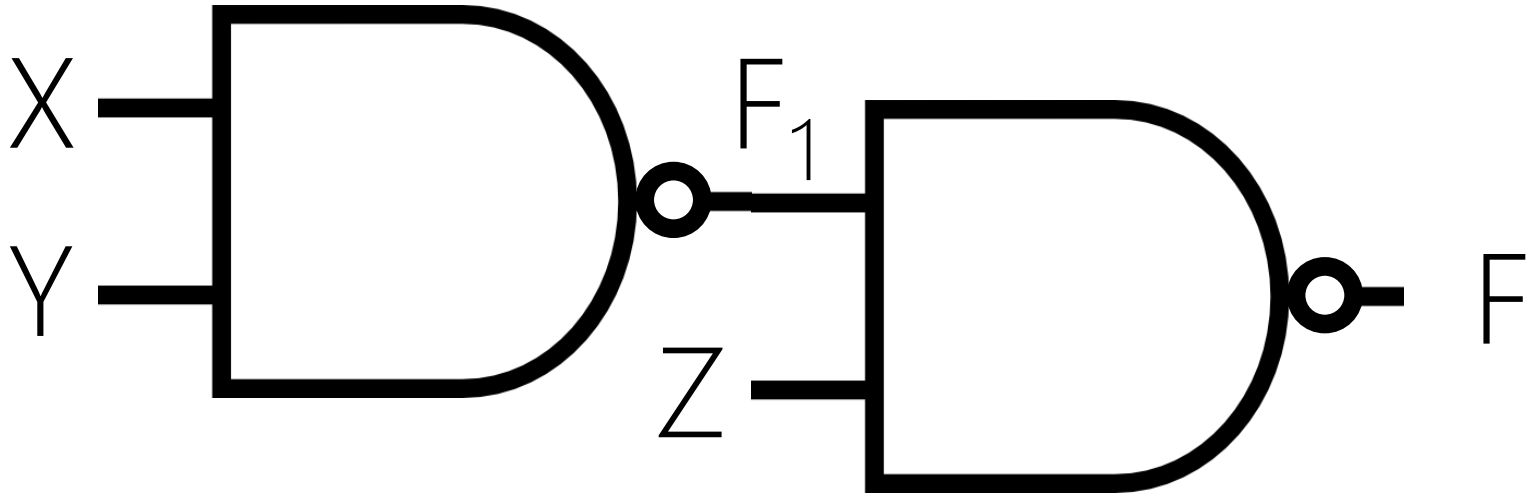
Z	Y	X	F=(ZYX)’
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

DESIGN PATTERNS

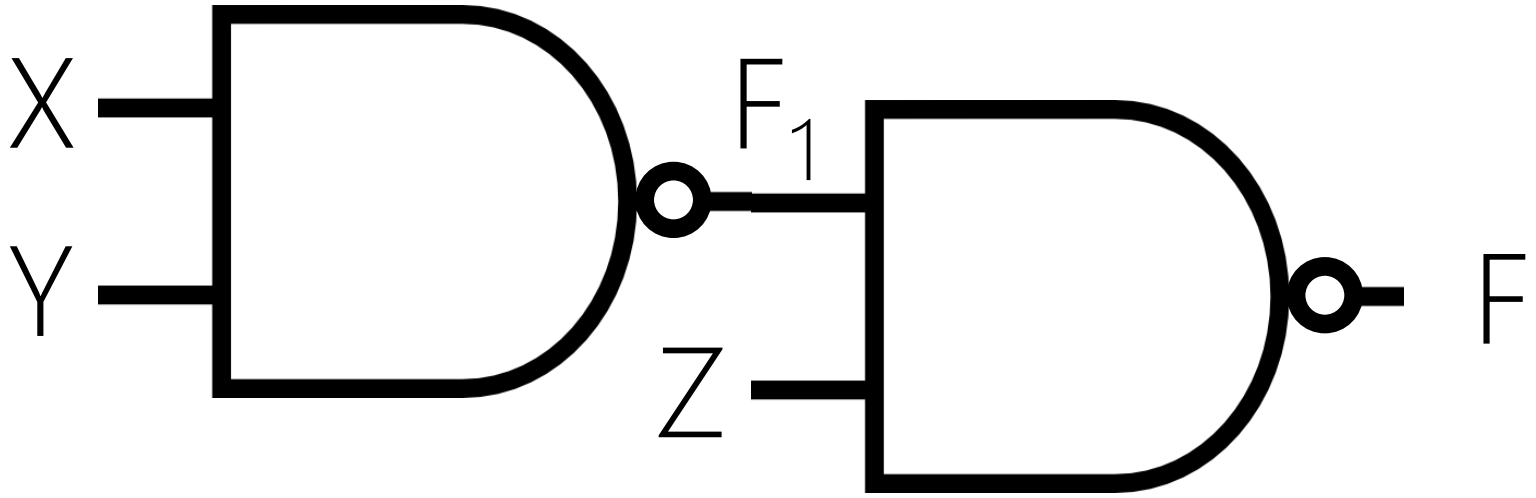
Using Same or Similar Previous Designs for New Designs



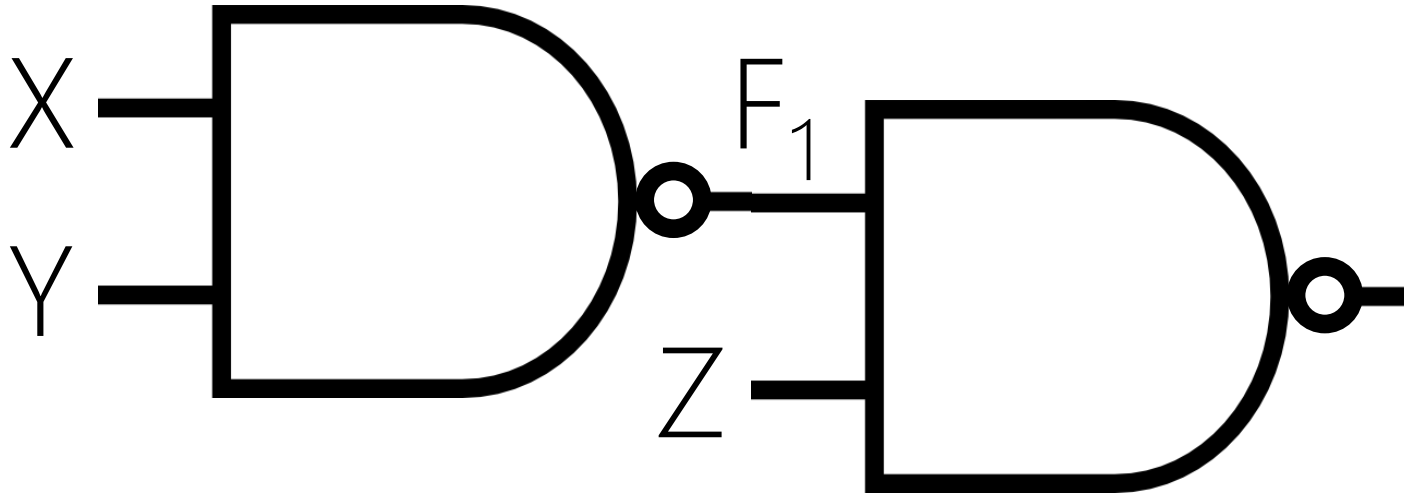
Z	Y	X	F = ?
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



Z	Y	X	$F_1 = (YX)'$	F = ?
0	0	0	1	
0	0	1	1	
0	1	0	1	
0	1	1	0	
1	0	0	1	
1	0	1	1	
1	1	0	1	
1	1	1	0	



Z	Y	X	$F_1 = (YX)'$	$F = (ZF_1)' = (Z(YX)')'$
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	0	1



$$F = (Z(YX)')' \neq (ZYX)'$$

Z	Y	X	$F_1 = (YX)'$	$F = (ZF_1)' = (Z(YX)')'$	$F=(ZYX)'$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	1	0	1
1	1	1	0	1	0

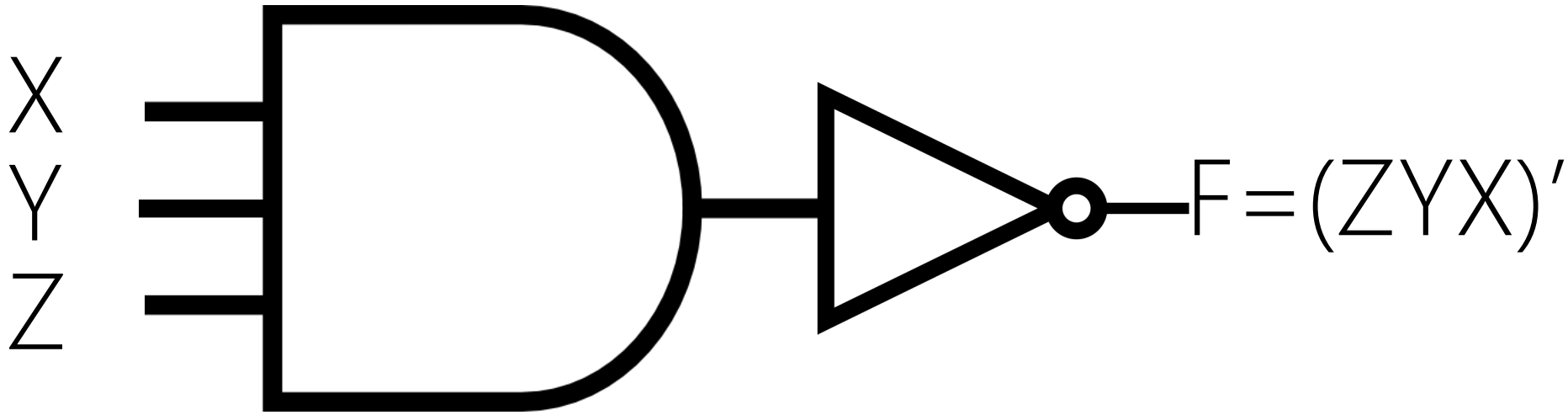
DESIGN PATTERNS

Using Same or Similar Previous Designs for New Designs

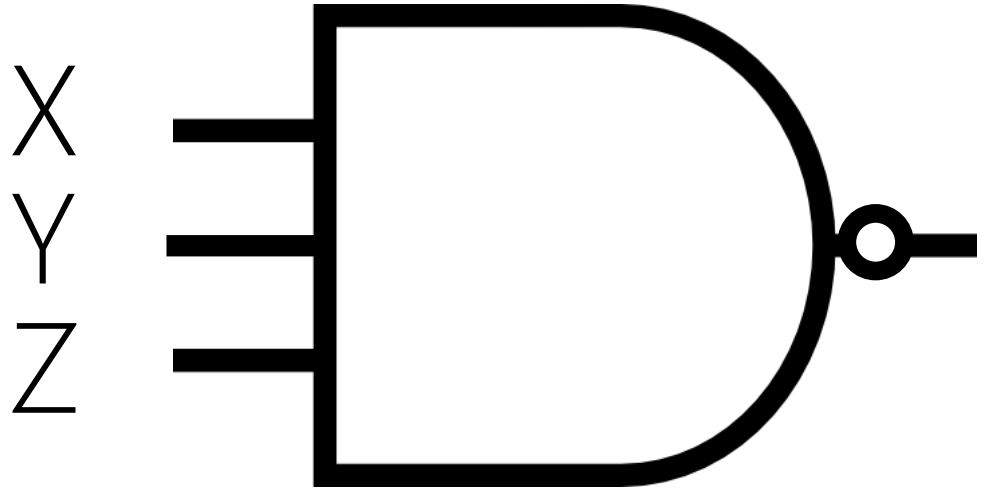
DESIGN → ANALYSIS → EVALUATION

Always Check

NOT (3-INPUT AND)



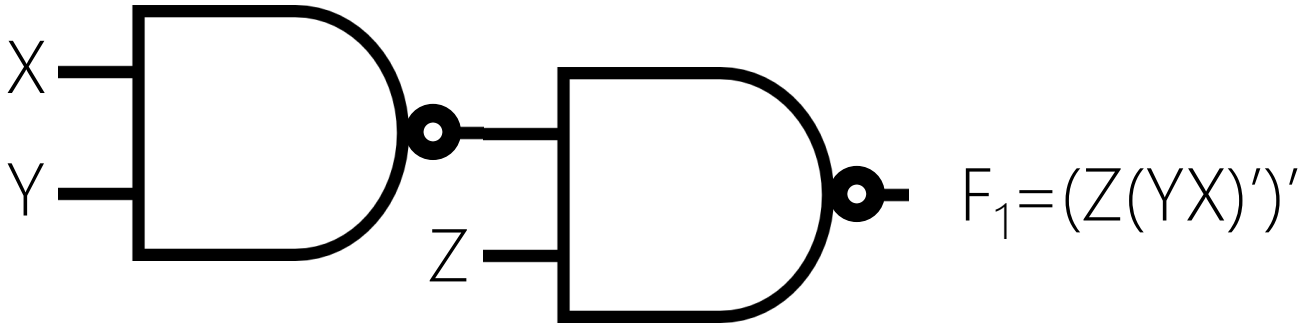
Z	Y	X	F=(ZYX)'	F=(ZYX)'
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0



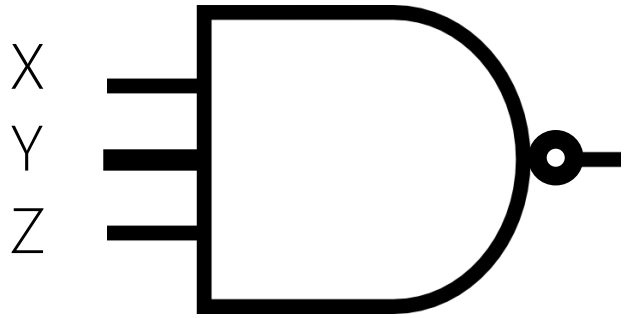
$F=(ZYX)'=(XZY)'$

Associative

Z	Y	X	F=(ZYX)'	F=(ZYX)'
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

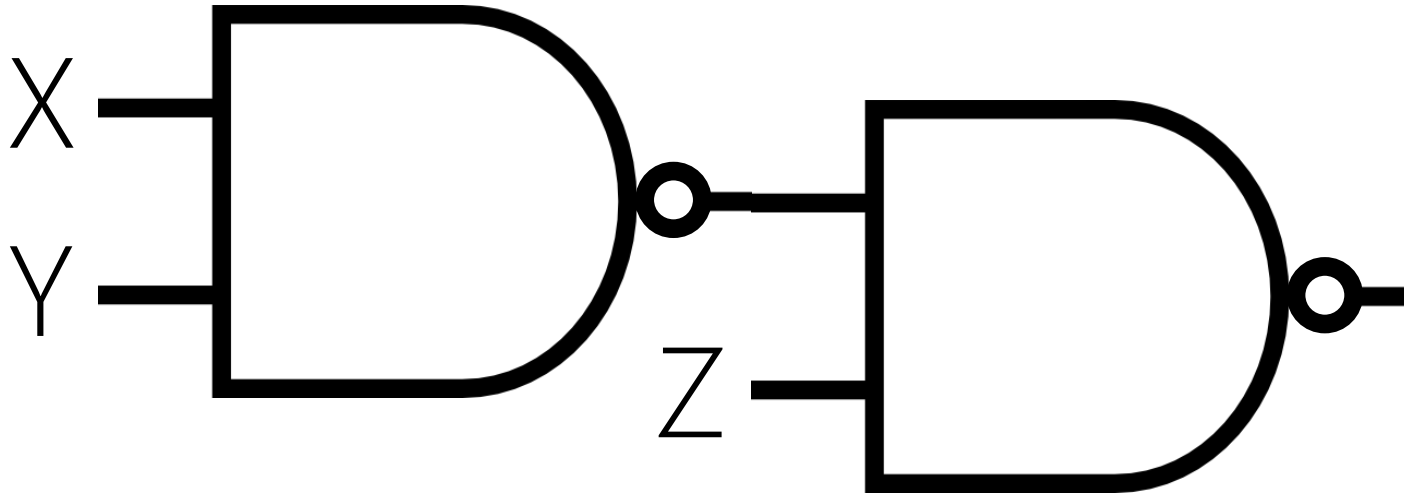


$F_1=(Z(YX)')'$



$F_2=(ZYX)'$

F = (ZYX)'	F_1	F_2
Effective (True)	No!	Yes
Efficient (Fast)	⋮	⋮
Min. Cost	⋮	⋮

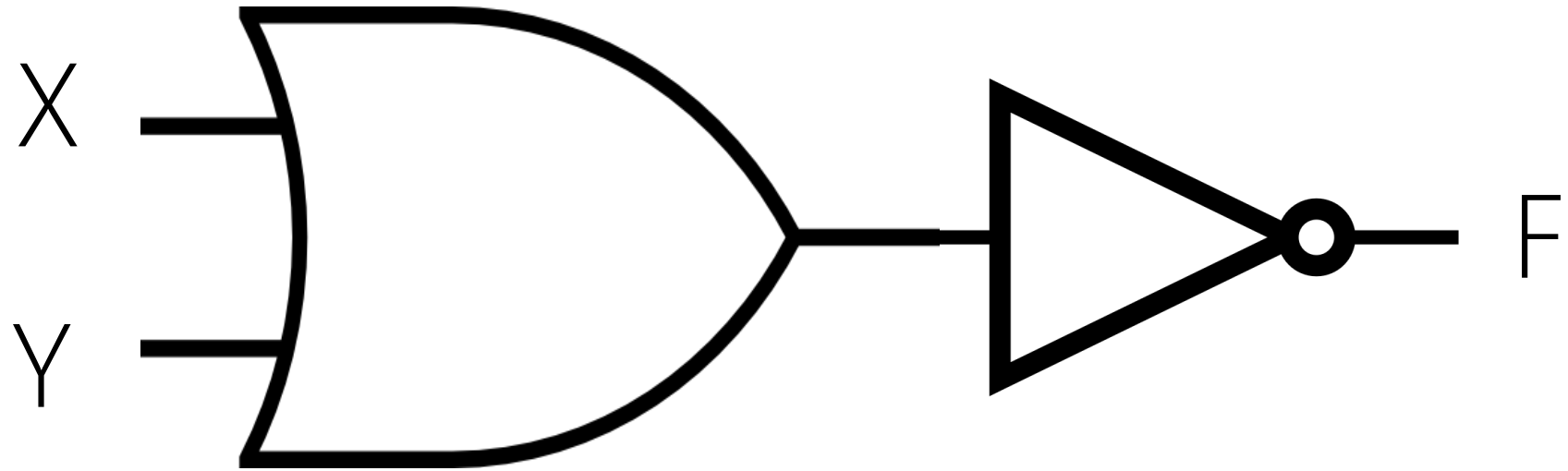


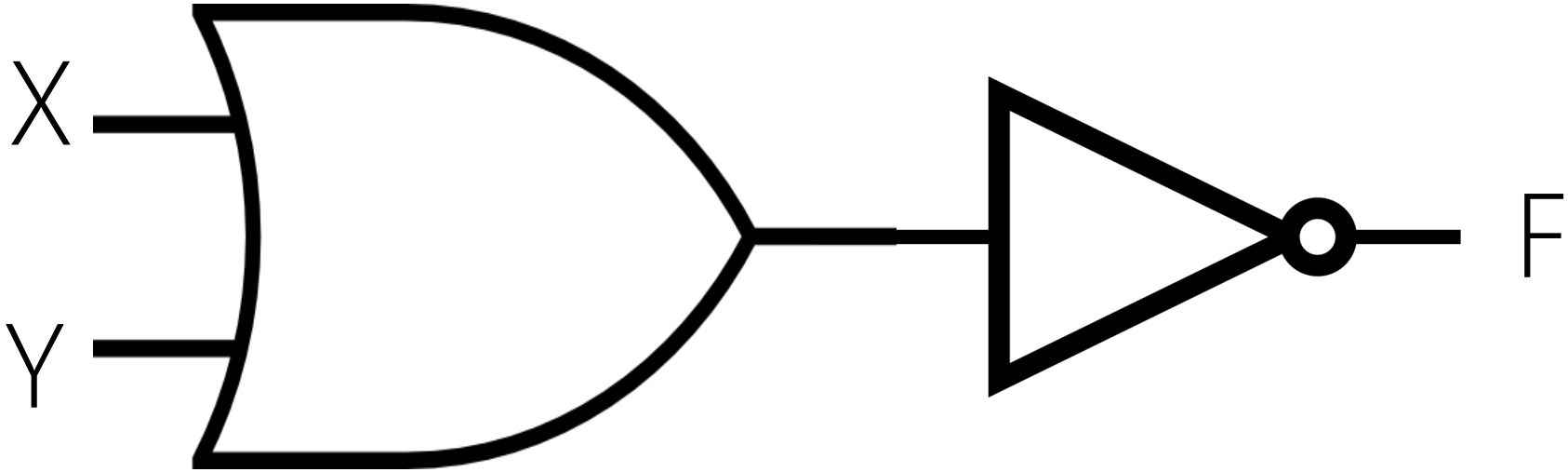
$F = Z \uparrow (Y \uparrow X)$

Associative?

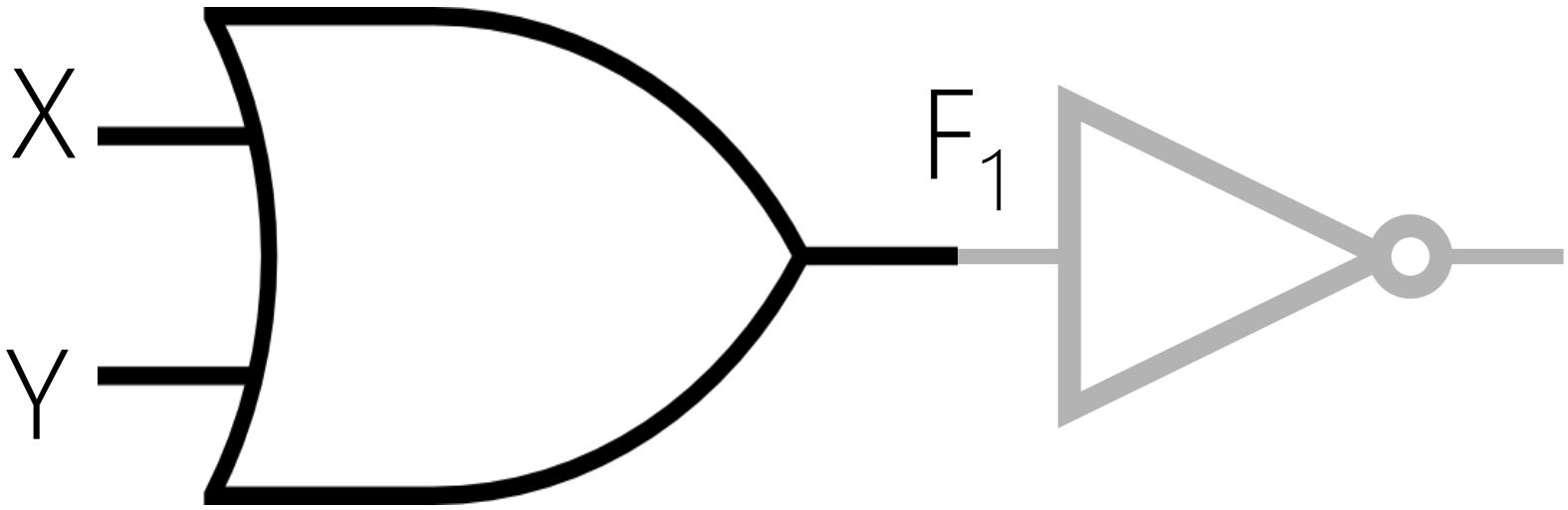
Z	Y	X	$F = (Z(YX)')' = Z \uparrow (Y \uparrow X)$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

ANALYSIS II

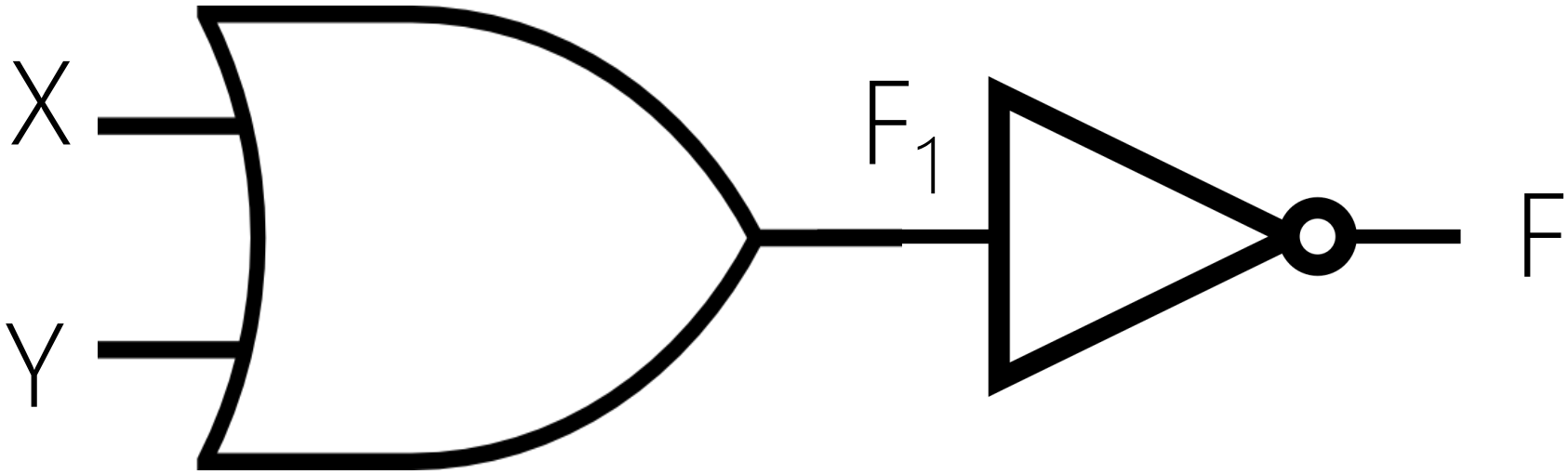

X
Y
F



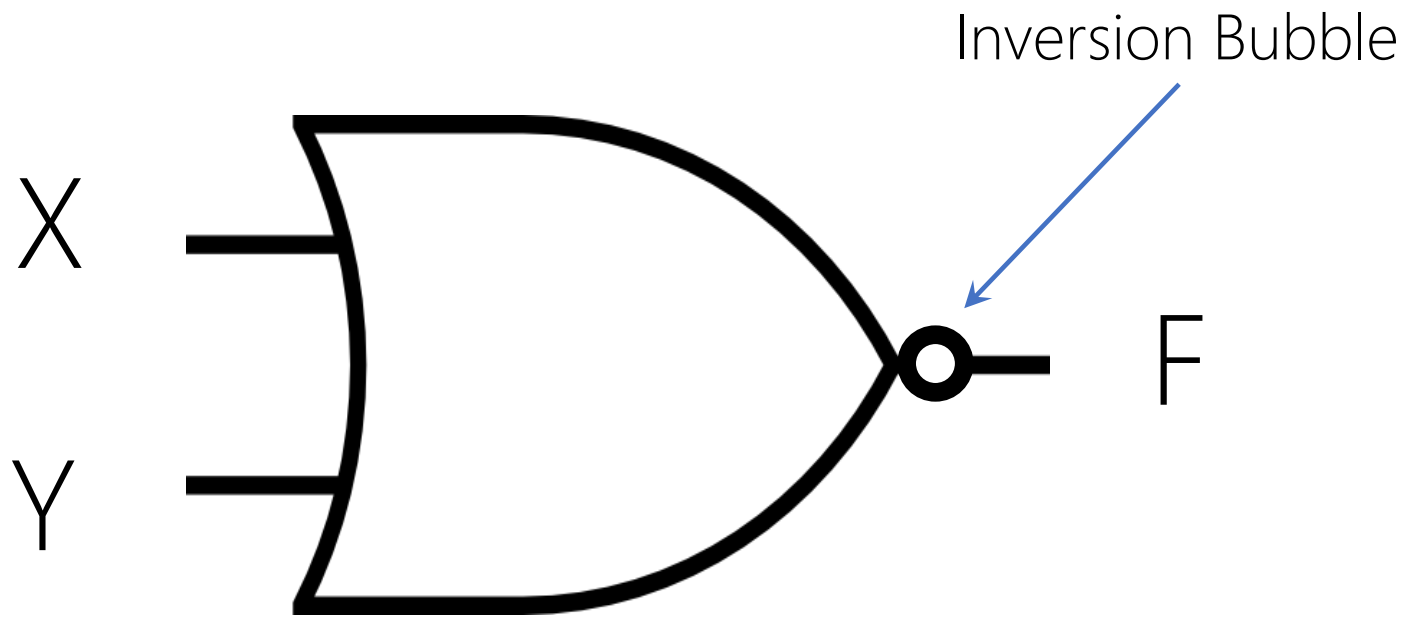
Y	X	F = ?
0	0	?
0	1	?
1	0	?
1	1	?



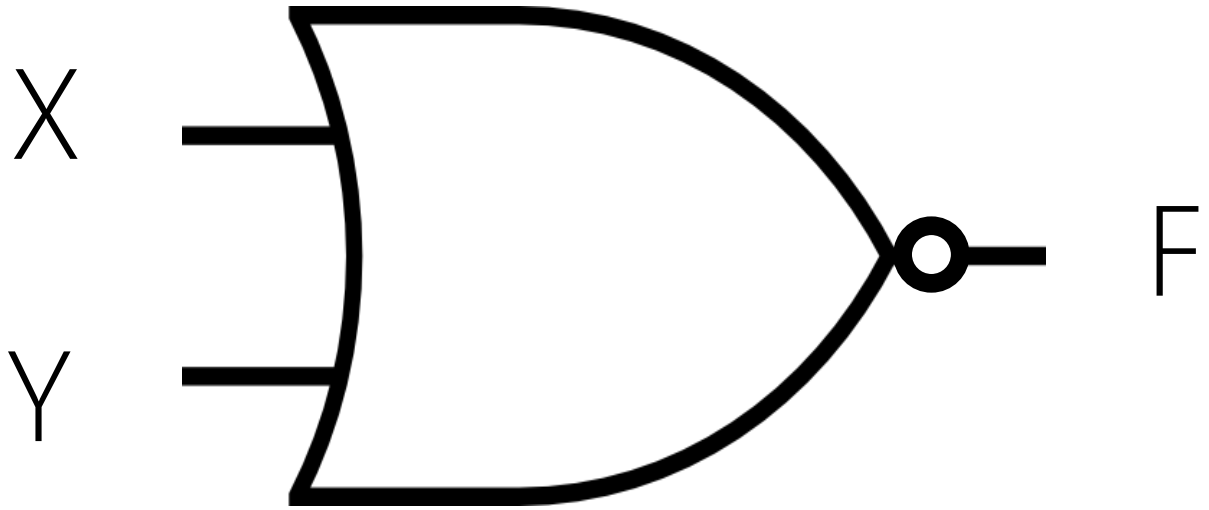
Y	X	$F_1 = Y+X$
0	0	0
0	1	1
1	0	1
1	1	1



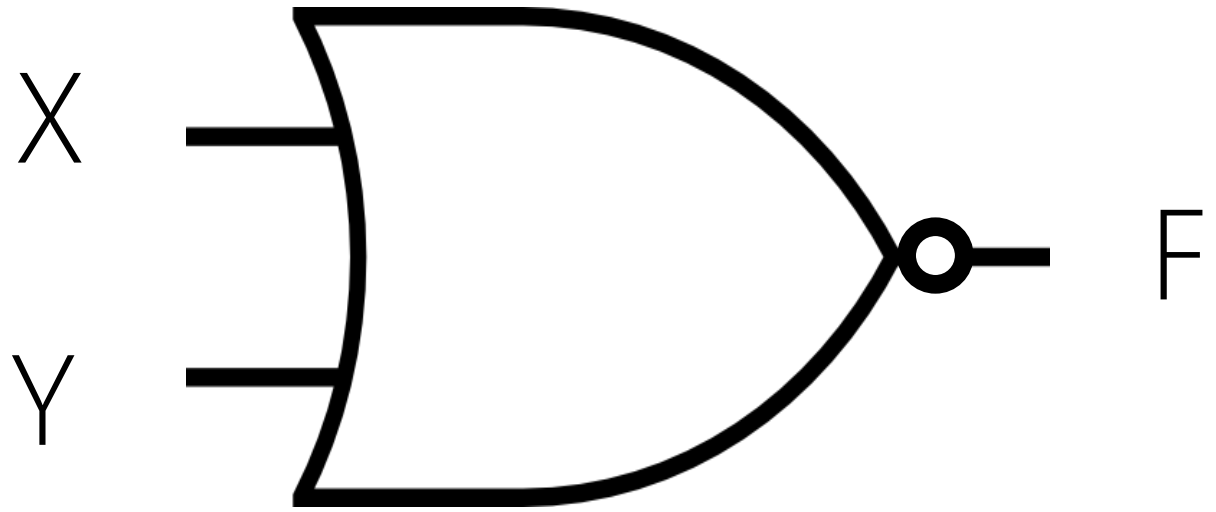
Y	X	$F_1 = Y+X$	$F = (Y+X)'$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0



NOR (Not – OR)

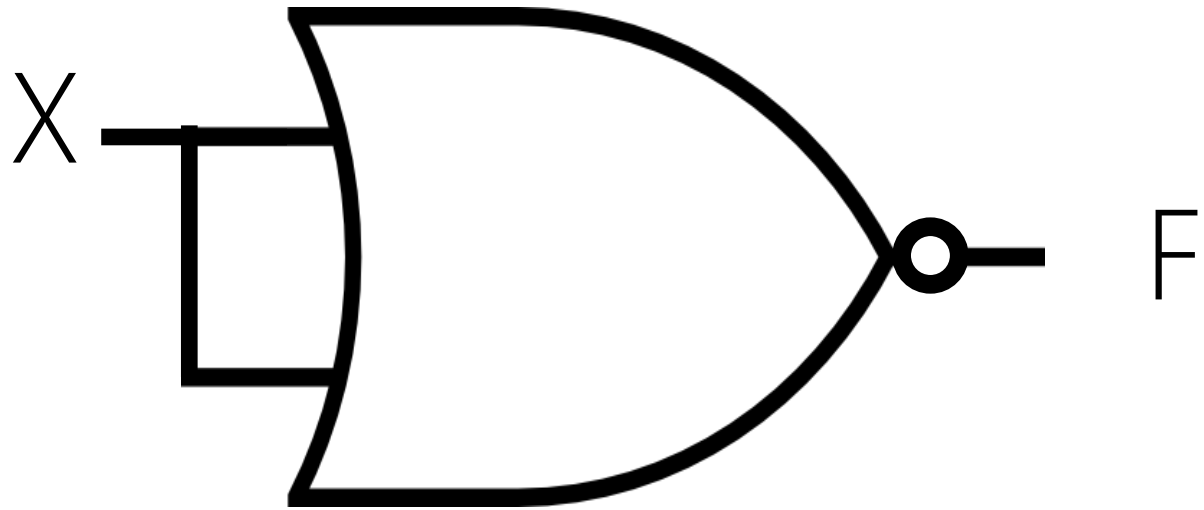


Y	X	F = (Y+X)'	F=Y ↓ X
0	0	1	
0	1	0	
1	0	0	
1	1	0	

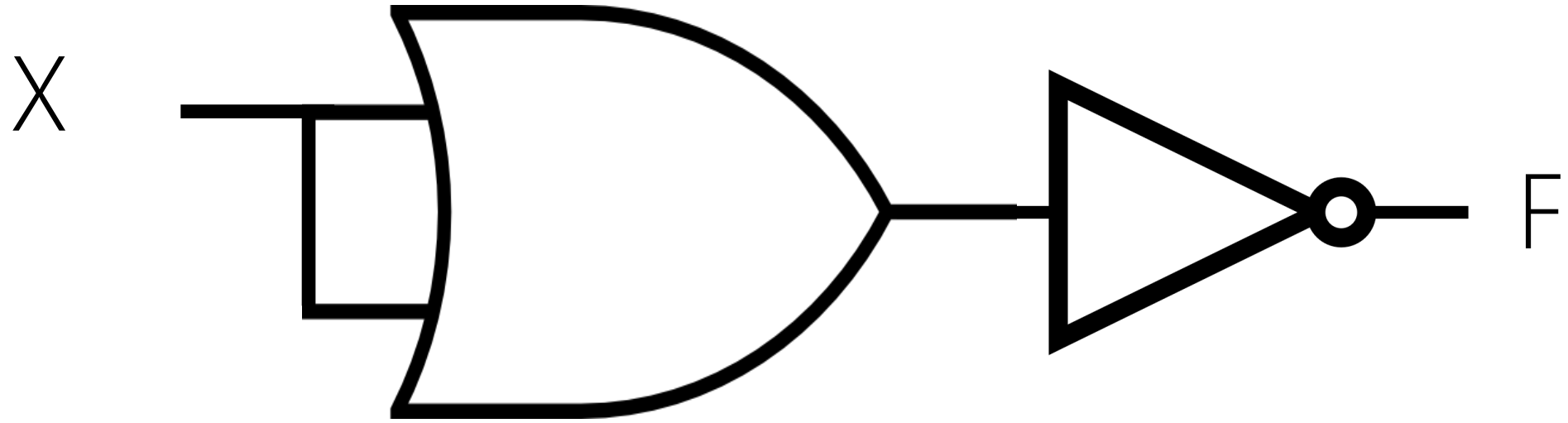


$$F = (Y + X)' = (X + Y)' = Y \downarrow X = X \downarrow Y$$

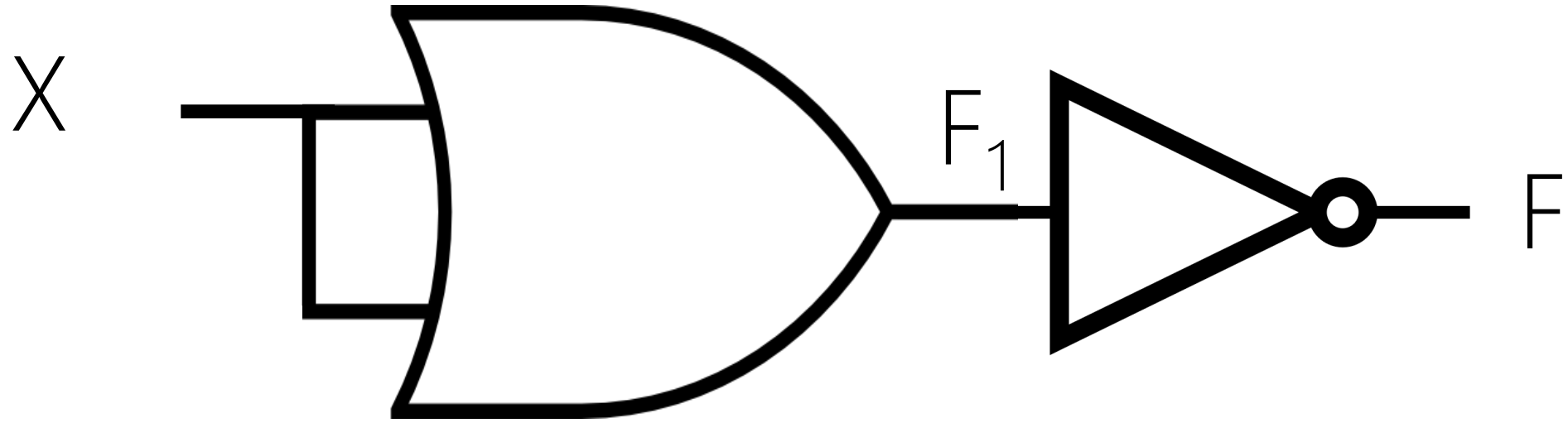
Commutative



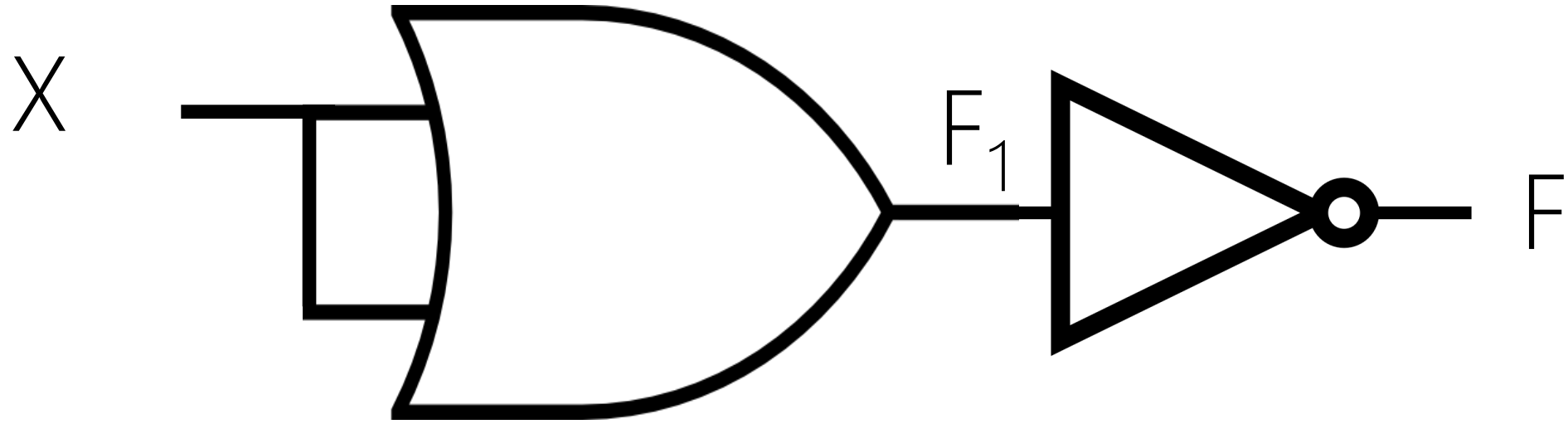
F=?



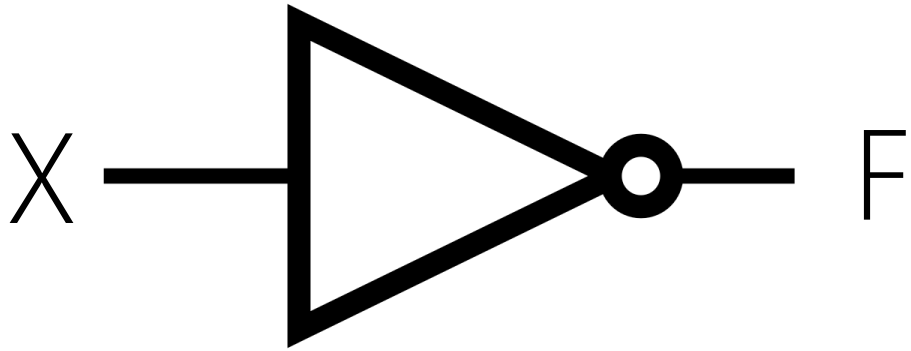
X	F = ?
0	
1	



X	$F_1 = X+X$	F = ?
0	0	
1	1	



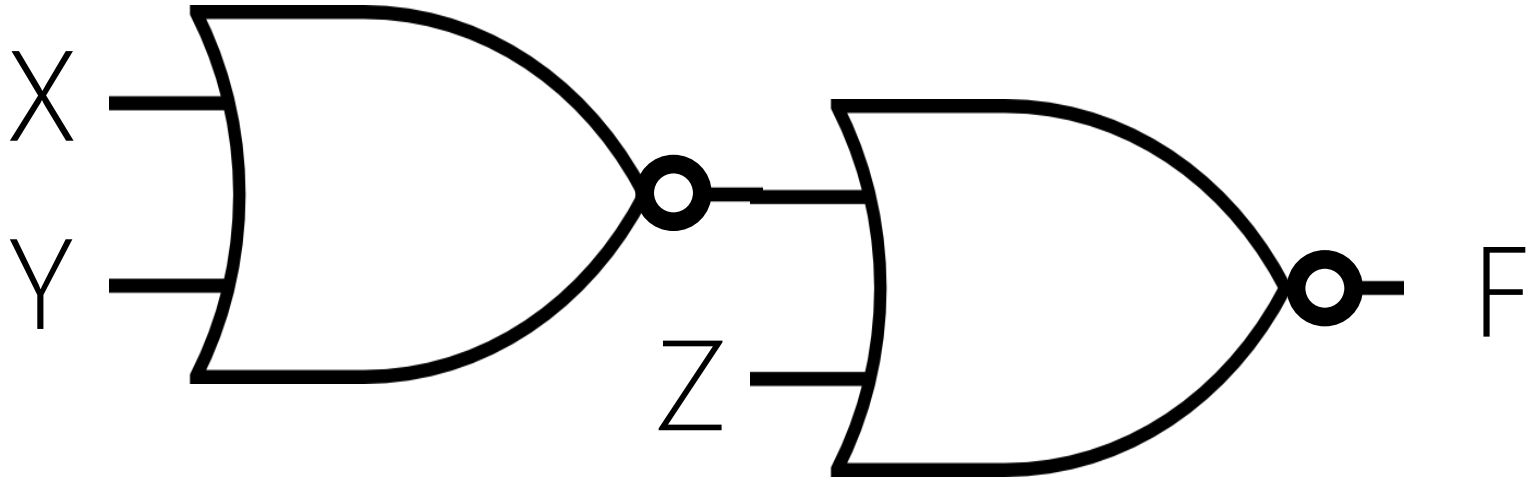
X	$F_1 = X+X$	$F = (X+X)'$
0	0	1
1	1	0



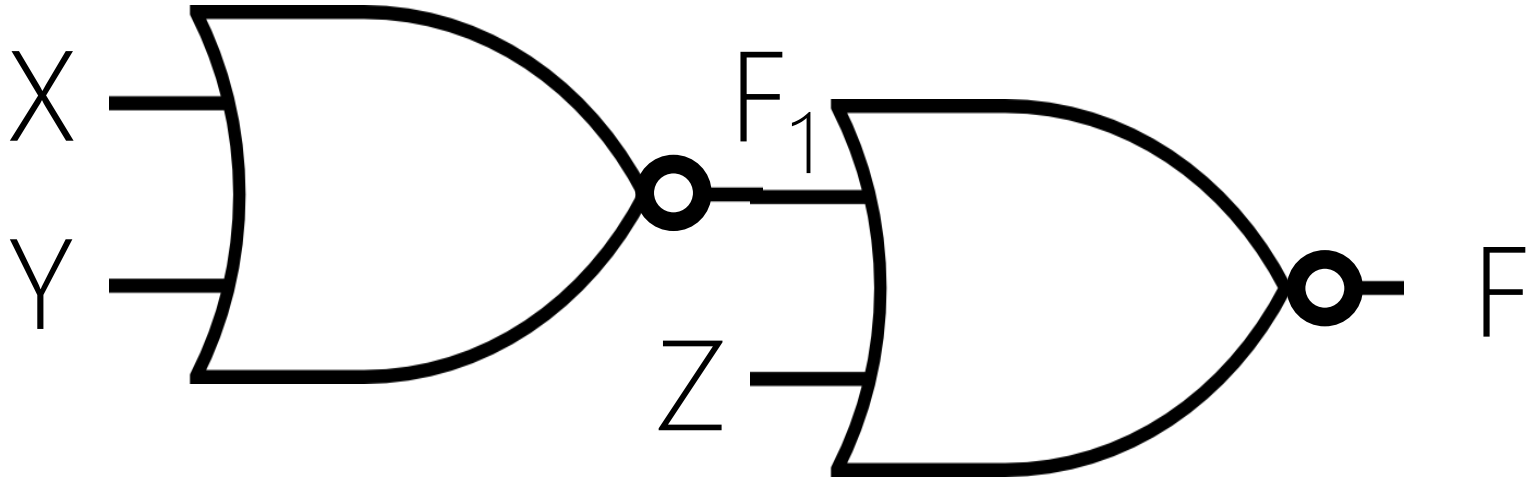
X	F = (X+X)′= X↓X =X′
0	1
1	0

3-INPUT NOR

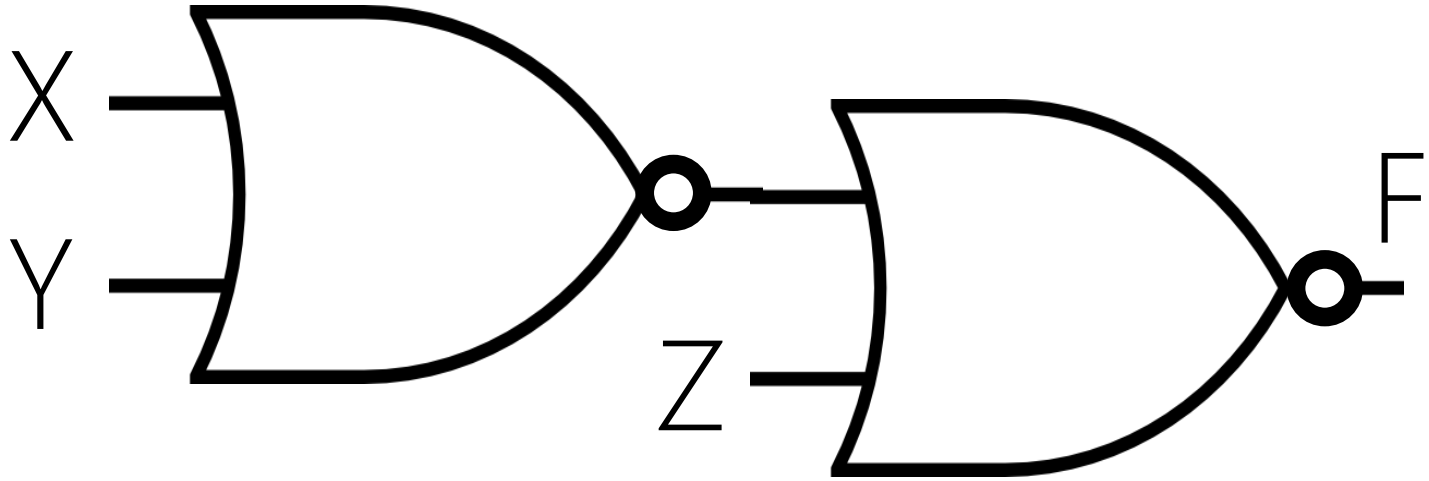
Z	Y	X	F=(Z+Y+X)’
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0



Z	Y	X	F = ?
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



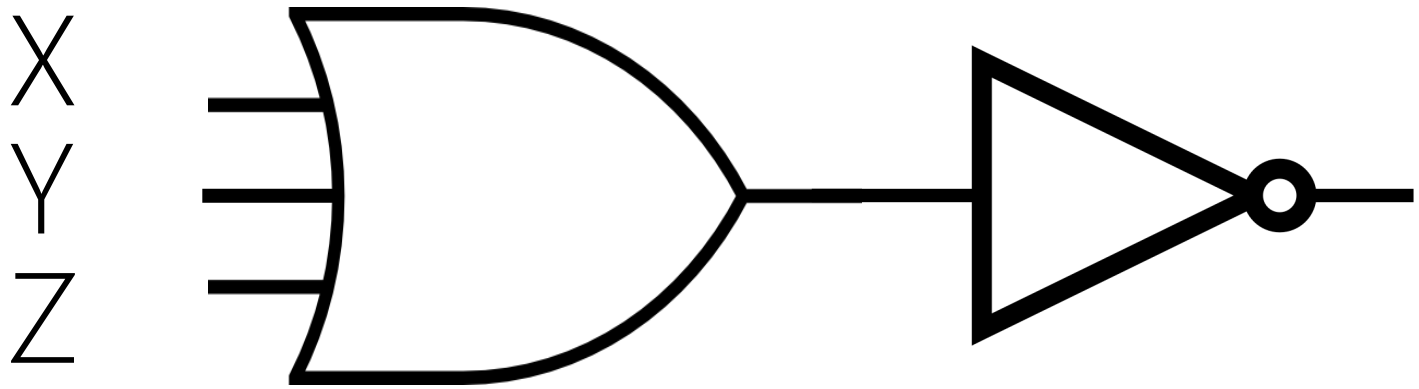
Z	Y	X	$F_1 = (Y+X)'$	$F = (Z+F_1)' = (Z+(Y+X)')'$
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0



$$F = (Z + (Y + X)')'$$
$$\neq (Z + Y + X)'$$

Z	Y	X	$F = (Z+F_1)' = (Z+(Y+X)')'$	$F=(Z+Y+X)'$
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

NOT (3-INPUT OR)

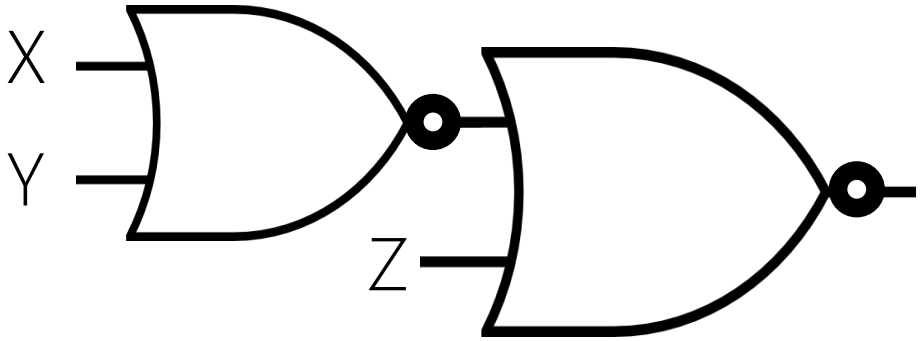


$F=(Z+Y+X)'$

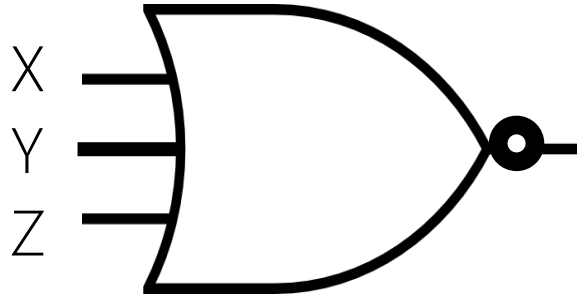
$F=(X+Y+Z)'$

Associative

Z	Y	X	F=(Z+Y+X)'	F=(X+Y+Z)'
0	0	0	1	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

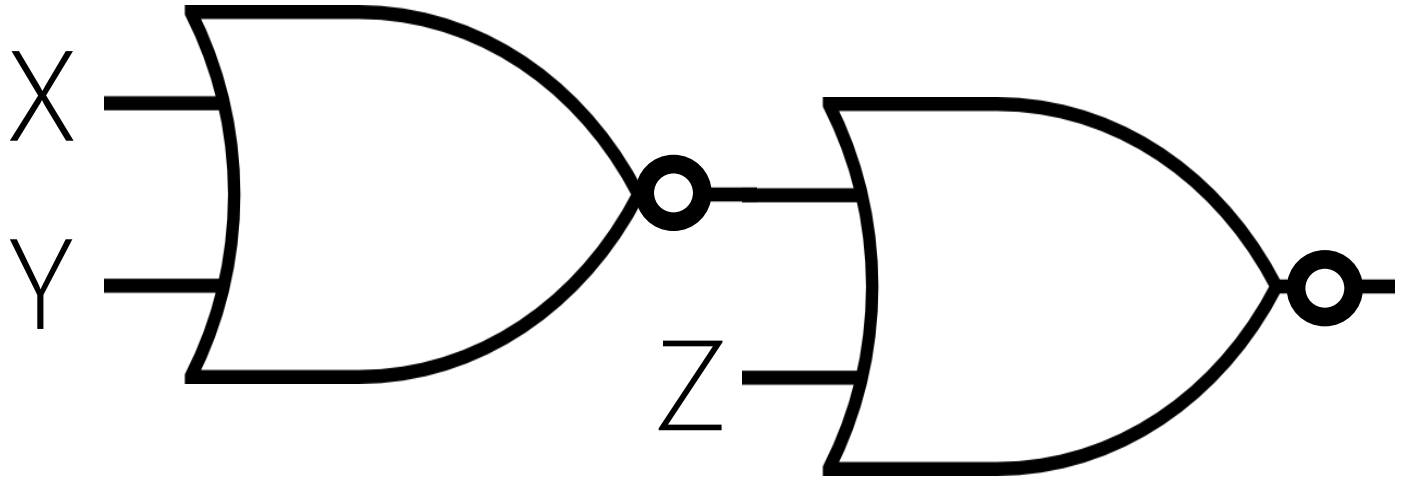


$F_1=(Z+(Y+X)')'$



$F_2=(Z+Y+X)'$

$F = (Z+Y+X)'$	F_1	F_2
Effective (True)	No!	Yes
Efficient (Fast)		
Min. Cost		



$F = Z \downarrow (Y \downarrow X)$

Associative?

Z	Y	X	F = (Z(YX)')' = Z ↓ (Y ↓ X)
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

RECAP

GATE	WHEN F=1
NOT	
AND	
OR	
NAND	
NOR	

GATE	WHEN F=1
NOT	The input is 0
AND	All the inputs are 1
OR	At least one input is 1
NAND	At least one input is 0
NOR	All the inputs are 0